

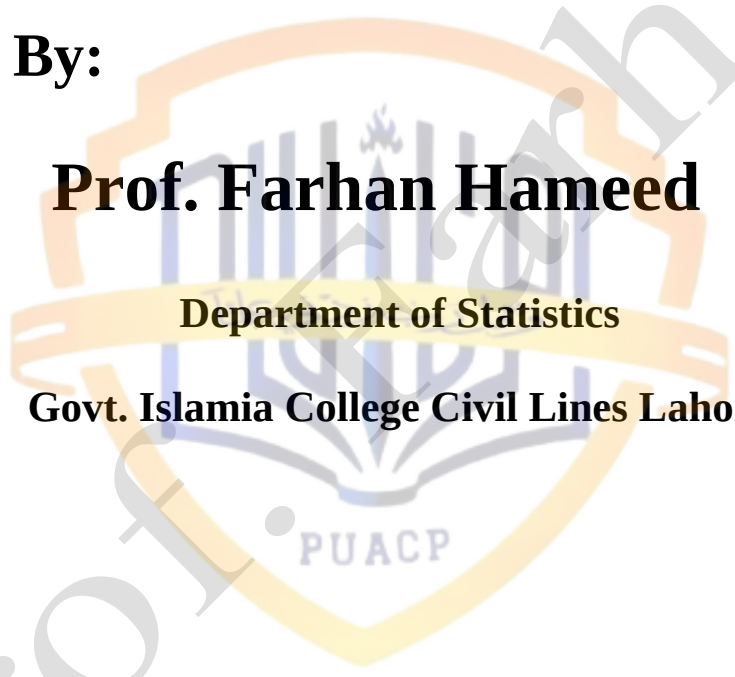
Methods of Estimation

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1.0 Methods of Estimating The Parameters

There are various methods of estimating parameters. They are listed below:-

- Method of Maximum Likelihood.
- Method of Moments.
- Least Square Method.
- Method of Minimum Chi-Square.
- Method of Modified Minimum chi-Square.
- Bayes' Estimation Procedure

1.1 Method of Maximum Likelihood

- Method of maximum likelihood is due to R.A fisher in 1921.

1.1.1 Likelihood Function

Let $X_1, X_2, \dots, X_n$ be n independent observations from $f(x; \theta)$, where, θ is a single unknown parameter. The joint probability density function of the sample is called likelihood function (LF) and is stated as

$$L(X|\theta) = f(x_1, \theta)f(x_2, \theta) \dots \dots \dots f(x_n, \theta)$$

1.1.2 Maximum Likelihood Principal

“To consider every possible value that the parameter might have, and for each value compute the probability that the given sample would have occurred if that were the true value of the parameter. That value of the parameter for which the probability of the given sample is greatest, is chosen as an estimate”

According to the maximum likelihood principal, one should take that value of the estimator θ within the admissible range of θ which makes $L(x|\theta)$ maximum. For this the method of maxima- minima is used.

If $L(x|\theta)$ is differentiable twice. Put first derivative of $L(x|\theta) = 0$, and solve for θ . Also for maximum check that the second derivative of $L(x|\theta)$ is negative for θ obtained by $L'(x|\theta)=0$.

If above two steps are verified, then the estimator obtained in such a manner is maximum likelihood estimator (MLE).

In practice it is better to take logarithm of $L(x|\theta)$ and then differentiate and solve it. This makes the estimation process easier.

If θ is a K dimensional parametric vector i.e. $\theta = (\theta_1, \theta_2, \dots, \theta_k)$

Then the estimator $(\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_k)$, which maximize the $L(x|\theta_1, \theta_2, \dots, \theta_k)$ can be obtained by differentiating $\log L(x|\theta_1, \theta_2, \dots, \theta_k)$ with respect to $\theta_1, \theta_2, \dots, \theta_k$ respectively and equating to zero. The solution of k equations provides the estimates of $\theta_1, \theta_2, \dots, \theta_k$.

Notationally,

$$\frac{\partial}{\partial \theta_1} [\log L(x|\theta_1, \theta_2, \dots, \theta_k)] = 0$$

$$\frac{\partial}{\partial \theta_2} [\log L(x|\theta_1, \theta_2, \dots, \theta_k)] = 0$$

$$\frac{\partial}{\partial \theta_k} [\log L(x|\theta_1, \theta_2, \dots, \theta_k)] = 0$$

In this way we get k equations in k unknowns. These equations are often called likelihood equations. Solving the equations, one gets the maximum likelihood estimators of $\theta_1, \theta_2, \dots, \theta_k$.

1.1.3 Properties of Maximum Likelihood Estimators

Uniqueness

- A MLE is not necessarily unique.

Unbiasedness

- A MLE is not necessarily unbiased
- They are asymptotically unbiased. i.e. when $n \rightarrow \infty$ then $bias \rightarrow 0$.

Consistency

- The MLE consistent which means that increasing the sample size the estimator becomes closer and closer to population parameter.
- A MLE may not be consistent in rare cases.

Asymptotic Variance

- The asymptotic variance of the MLE $\hat{\theta}$ for parameter θ can be obtained from the 2nd derivative of log likelihood function using

$$v(\hat{\theta}) = \frac{1}{-E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right]}$$

Asymptotic Normality

- Under certain regularity conditions a MLE has asymptotic normal distribution.

Efficiency

- If a minimum variance bound (MVB) unbiased estimator exists, MLE provides it
- When MLE exist it is most efficient for large samples under the assumption that the distribution of the estimator tends to normal.
- A MLE may not be uniformly minimum variance unbiased estimator (UMVUE).

Sufficiency

- If a sufficient statistic exists, The MLE of the parameter is a function of sufficient statistic.

Invariant

- The MLE is invariant under functional transformation.
- If $T = t(X_1, X_2, \dots, X_n)$ is a MLE of θ and $u(\theta)$ is a one-one function of θ , $u(T)$ is MLE of $u(\theta)$.
- For example, if $\hat{\sigma}^2$ is MLE of σ^2 , then $\hat{\sigma}$ will be MLE of σ .

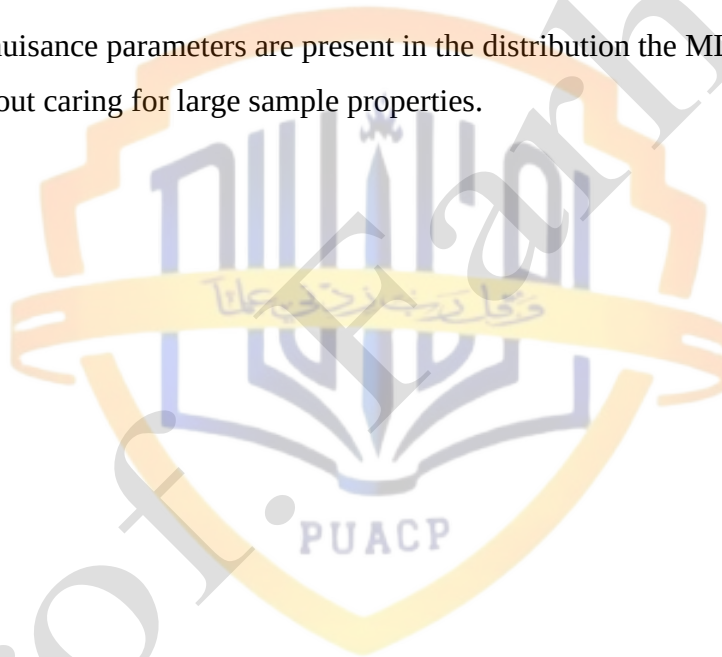
1.1.4 Merits of MLE

- It is a powerful method and precise for large samples.
- They become UMVE as the sample size increases.
- They have approximate normal distribution and approximate sample variance can be calculated, which can be used to construct confidence interval bounds.
- Likelihood functions can be used to test hypothesis about models and parameters.

- They are consistent.
- They are widely applicable.
- They are unaffected by monotonic transformation of the data.

1.1.5 Demerits of MLE

- They may be slightly biased.
- In case of small sample size MLE are not very precise and may even generate a line that lies above or below the data points.
- Computation of MLE is laborious and demanding.
- MLE can be failed to be estimated.
- Too many nuisance parameters are present in the distribution the MLE will be highly biased without caring for large sample properties.



2. Questions Related to Maximum Likelihood Estimator

2.1 MLE of Parameters and Variance-Covariance Matrix of Parameter of Normal Distribution

Question 1: A random sample of size n is drawn from $N(\mu, \sigma^2)$, find (i) MLE of μ when σ^2 is known. (ii) MLE for σ^2 when μ is known. (iii) MLE for μ and σ^2 when both are unknown. (iv) Also Find Variance Covariance Matrix

(i) MLE of μ

The pdf of normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

The likelihood function

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

$$L(\mu) = \frac{1}{\sigma^n (2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2\sigma^2} \sum (x-\mu)^2}$$

Taking ln both sides

$$\ln L(\mu) = \ln \left[\frac{1}{\sigma^n (2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x-\mu)^2} \right]$$

$$\ln L(\mu) = \ln 1 - \frac{n}{2} \ln(\sigma^2) - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x-\mu)^2 \ln e$$

$$\ln L(\mu) = -\frac{n}{2} \ln(\sigma^2) - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x-\mu)^2$$

Differentiate partially w.r.t μ

$$\frac{\partial}{\partial \mu} \ln L(\mu) = -\frac{2}{2\sigma^2} \sum_{i=1}^n (x-\mu) (-1)$$

$$\frac{\partial}{\partial \mu} \ln L(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (x-\mu) \dots \dots \dots (1)$$

Put

$$\frac{\partial}{\partial \mu} \ln L(\mu) = 0$$

$$\sum_{i=1}^n (x_i - \mu) = 0 \Rightarrow \sum_{i=1}^n x_i - n\mu = 0$$

$$\mu = \frac{\sum_{i=1}^n x_i}{n} = \bar{x}$$

Differentiate, (1) w.r.t μ

$$\frac{\partial^2}{\partial \mu^2} \ln L(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (0 - 1) = -\frac{n}{\sigma^2} < 0$$

So, the MLE of μ is $\bar{x}$.

(ii) MLE for σ^2

Now, to find MLE for σ^2

$$\begin{aligned} \frac{\partial}{\partial \sigma^2} \ln L(\sigma^2) &= -\frac{n}{2} \frac{\partial}{\partial \sigma^2} [\ln(\sigma^2)] - \frac{\partial}{\partial \sigma^2} \left[\frac{n}{2} \ln(2\pi) \right] - \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2 \frac{\partial}{\partial \sigma^2} (\sigma^2)^{-1} \\ \frac{\partial}{\partial \sigma^2} \ln L(\sigma^2) &= -\frac{n}{2} \frac{1}{\sigma^2} (1) - \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2 (-1) (\sigma^2)^{-2} (1) \end{aligned}$$

Putting

$$\begin{aligned} \frac{\partial}{\partial \sigma^2} \ln L(\sigma^2) &= 0 \\ -\frac{n}{2} \frac{1}{\sigma^2} + \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2 (\sigma^2)^{-2} &= 0 \\ \frac{n}{2\sigma^2} &= \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2 \Rightarrow n\sigma^2 = \sum_{i=1}^n (x_i - \mu)^2 \\ \hat{\sigma}^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 \end{aligned}$$

Again differentiate (2) w.r.t σ^2

$$\begin{aligned} \frac{\partial^2}{\partial (\sigma^2)^2} \ln L(\sigma^2) &= -\frac{n}{2} \frac{\partial}{\partial \sigma^2} (\sigma^2)^{-1} + \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2 \frac{\partial}{\partial \sigma^2} (\sigma^2)^{-2} \\ \frac{\partial^2}{\partial (\sigma^2)^2} \ln L(\sigma^2) &= \frac{n}{2} (\sigma^2)^{-2} - \frac{2}{2} \sum_{i=1}^n (x_i - \mu)^2 (\sigma^2)^{-3} \end{aligned}$$

$$\begin{aligned}\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) &= \frac{n}{2\sigma^4} - \frac{1}{\sigma^6} \sum_{i=1}^n (x - \mu)^2 \\ \frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) &= \frac{1}{2\sigma^6} \left[n\sigma^2 - 2 \sum_{i=1}^n (x - \mu)^2 \right] \\ \frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) &= \frac{1}{2\sigma^6} \left[\sum_{i=1}^n (x - \mu)^2 - 2 \sum_{i=1}^n (x - \mu)^2 \right] \\ \frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) &= \frac{-1}{2\sigma^6} \left[\sum_{i=1}^n (x - \mu)^2 \right] < 0\end{aligned}$$

Hence, MLE of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x - \mu)^2$$

(iii) MLE for μ and σ^2 when both are unknown

$$\frac{\partial}{\partial \mu} \ln L(\mu, \sigma^2) = 0 \text{ and } \frac{\partial}{\partial \sigma^2} \ln L(\mu, \sigma^2) = 0$$

Provides two equations one of them give MLE of μ and other given MLE of σ^2

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i \text{ and } \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x - \mu)^2$$

(iv) Variance -Covariance Matrix

$$E \left[\frac{\partial^2}{\partial \mu^2} \ln L(\mu, \sigma^2) \right] = E \left[-\frac{n}{\sigma^2} \right] = -\frac{n}{\sigma^2}$$

And

$$E \left[\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\mu, \sigma^2) \right] = E \left[\frac{-1}{2\sigma^6} \left[\sum_{i=1}^n (x - \mu)^2 \right] \right] = \frac{-1}{2\sigma^6} \left[\sum_{i=1}^n E(x - \mu)^2 \right] = -\frac{n\sigma^2}{2\sigma^6} = -\frac{n}{2\sigma^4}$$

$$E \left[\frac{\partial^2}{\partial \mu \partial \sigma^2} \ln L(\mu, \sigma^2) \right] = E \left[\sum_{i=1}^n (x - \mu) \frac{\partial}{\partial \sigma^2} \left(\frac{1}{\sigma^2} \right) \right] = E \left[\frac{-\sum x_i + n\mu}{\sigma^4} \right] = E \left[-\frac{-\sum x_i + \sum x_i}{\theta} \right] = 0$$

$$\begin{aligned}E \left[\frac{\partial^2}{\partial \sigma^2 \partial \mu} \ln L(\theta, p) \right] &= E \left[\frac{\partial}{\partial \sigma^2} \left\{ -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x - \mu)^2 \right\} \right] = E \left[\frac{-\sum x_i + n\mu}{\sigma^4} \right] \\ &= E \left[-\frac{-\sum x_i + \sum x_i}{\theta} \right] = 0\end{aligned}$$

$$V(\mu, \sigma^2) = - \begin{bmatrix} E \left[\frac{\partial^2}{\partial \mu^2} \ln L(\mu, \sigma^2) \right] & E \left[\frac{\partial^2}{\partial \mu \partial \sigma^2} \ln L(\mu, \sigma^2) \right] \\ E \left[\frac{\partial^2}{\partial \sigma^2 \partial \mu} \ln L(\theta, p) \right] & E \left[\frac{\partial^2}{\partial (\sigma^2)^2} \ln L(\mu, \sigma^2) \right] \end{bmatrix}^{-1}$$

$$V(\mu, \sigma^2) = - \begin{bmatrix} -\frac{n}{\sigma^2} & 0 \\ 0 & -\frac{n}{2\sigma^4} \end{bmatrix}^{-1}$$

$$V(\mu, \sigma^2) = \frac{1}{\left(\frac{n}{\sigma^2}\right) \left(\frac{n}{2\sigma^4}\right)} \begin{bmatrix} \frac{n}{2\sigma^4} & 0 \\ 0 & \frac{n}{\sigma^2} \end{bmatrix}$$

$$V(\mu, \sigma^2) = \frac{1}{\left(\frac{n}{2\sigma^6}\right)} \begin{bmatrix} \frac{n}{2\sigma^4} & 0 \\ 0 & \frac{n}{\sigma^2} \end{bmatrix}$$

$$V(\mu, \sigma^2) = \left(\frac{2\sigma^6}{n}\right) \begin{bmatrix} \frac{n}{2\sigma^4} & 0 \\ 0 & \frac{n}{\sigma^2} \end{bmatrix}$$

$$V(\mu, \sigma^2) = \begin{bmatrix} \frac{\sigma^2}{n} & 0 \\ 0 & \frac{2\sigma^4}{n} \end{bmatrix}$$

2.2 MLE of α

Question 2: Given the p.d.f of random sample of X , find MLE for α .

$$f(x; \alpha) = \alpha x^{\alpha-1}, 0 < \alpha < 1$$

Solution:

$$L(\alpha) = \prod_{i=1}^n \alpha x^{\alpha-1}$$

$$L(\alpha) = \alpha^n (x_1 \cdot x_2 \dots \dots x_n)^{\alpha-1}$$

Taking “ln” both sides

$$\ln L(\alpha) = n \ln \alpha + (\alpha - 1) \ln(x_1 \cdot x_2 \dots \dots x_n)$$

$$\ln L(\alpha) = n \ln \alpha + (\alpha - 1) \sum \ln x_i$$

Diff w.r.t α

$$\frac{\partial}{\partial \alpha} n \ln(\alpha) = n \frac{\partial}{\partial \alpha} \ln \alpha + \sum \ln x_i \frac{\partial}{\partial \alpha} (\alpha - 1)$$

$$\frac{\partial}{\partial \alpha} n \ln(\alpha) = \frac{n}{\alpha} + \sum \ln x_i \dots \dots \dots (1)$$

Put

$$\frac{\partial}{\partial \alpha} n \ln(\alpha) = 0$$

$$\frac{n}{\alpha} + \sum \ln x_i = 0$$

$$n = -\alpha \sum \ln x_i$$

$$\alpha = -\frac{n}{\sum \ln x_i}$$

Diff (1) w.r.t α

$$\frac{\partial^2}{\partial \alpha^2} n \ln(\alpha) = -\frac{n}{\alpha^2} < 0$$

So, The MLE of α is

$$\hat{\alpha} = -\frac{n}{\sum \ln x_i}$$

2.3 MLE of α as function of Sufficient Statistic

Question 2(a): Given the p.d.f of random sample of X , $f(x; \alpha) = \alpha x^{\alpha-1}$, $0 < \alpha < 1$

find MLE for α if sufficient statistic Y exists, express MLE as function of Y .

Solution:

$$L(\alpha) = \prod_{i=1}^n \alpha x_i^{\alpha-1} \dots \dots \dots (A)$$

$$L(\alpha) = \alpha^n (x_1 \cdot x_2 \dots \dots \dots x_n)^{\alpha-1}$$

Let $y = (x_1 \cdot x_2 \dots \dots \dots x_n)$

$$L(\alpha) = \alpha^n (y)^{\alpha-1}$$

$$L(\alpha) = \alpha^n y^{\alpha} \cdot \frac{1}{y}$$

$$L(\alpha) = g(y, \alpha) \cdot h(y)$$

By factorization theorem it is clear that y is sufficient statistic of
Taking "ln" both sides of (A)

$$\ln L(\alpha) = n \ln \alpha + (\alpha - 1) \ln y$$

$$\ln L(\alpha) = n \ln \alpha + (\alpha - 1) \ln y$$

Diff w.r.t α

$$\frac{\partial}{\partial \alpha} \ln L(\alpha) = n \frac{\partial}{\partial \alpha} \ln \alpha + \ln y \frac{\partial}{\partial \alpha} (\alpha - 1)$$

$$\frac{\partial}{\partial \alpha} \ln L(\alpha) = \frac{n}{\alpha} + \ln y \dots \dots \dots (1)$$

Put

$$\frac{\partial}{\partial \alpha} \ln L(\alpha) = 0$$

$$\frac{n}{\alpha} + \ln y = 0$$

$$n = -\alpha \ln y$$

$$\alpha = -\frac{n}{\ln y}$$

Diff (1) w.r.t α

$$\frac{\partial^2}{\partial \alpha^2} \ln L(\alpha) = -\frac{n}{\alpha^2} < 0$$

So, The MLE of α is

$$\hat{\alpha} = -\frac{n}{\sum \ln x_i}$$

where, y is sufficient statistic of α

2.4 MLE of Parameter of Exponential Distribution

Question 3: Suppose the random variable with p.d.f $f(x; \theta) = \theta e^{-\theta x}$, $x > 0$, find the MLE of the parameter θ .

Solution:-

$$L(\theta) = \prod_{i=1}^n \theta e^{-\theta x_i}$$

$$L(\theta) = \theta^n e^{-\theta \sum x_i}$$

Taking “ln” both sides

$$\ln L(\theta) = n \ln \theta - \theta \sum x_i$$

$$\ln L(\theta) = n \ln \theta - \theta \sum x_i$$

Diff w.r.t θ

$$\begin{aligned} \frac{\partial}{\partial \theta} \ln L(\theta) &= n \frac{\partial}{\partial \theta} \ln \theta - \sum x_i \frac{\partial}{\partial \theta} \theta \\ \frac{\partial}{\partial \theta} \ln L(\theta) &= \frac{n}{\theta} - \sum x_i \end{aligned} \quad (1)$$

Put

$$\begin{aligned} \frac{\partial}{\partial \theta} \ln L(\theta) &= 0 \\ \frac{n}{\theta} - \sum x_i &= 0 \\ \frac{n}{\theta} &= \sum x_i \\ n &= \theta \sum x_i \\ \theta &= \frac{n}{\sum x_i} \end{aligned}$$

Diff (1) w.r.t θ

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = -\frac{n}{\theta^2} < 0$$

So, MLE of θ is

$$\hat{\theta} = \frac{n}{\sum x_i} = \frac{1}{\frac{\sum x_i}{n}} = \frac{1}{\bar{x}}$$

2.5 MLE of Parameter “p” of Geometric Distribution

Question 4: Obtain the MLE of “p” in geometric distribution.

Solution:

Let $X_1, X_2, \dots, X_n$ be random samples of size “n” with Probability Mass Function.

$$f(x; p) = p(1 - p)^{x-1}, x = 1, 2, 3, \dots,$$

The likelihood function for parameter p is

$$L(p) = \prod_{i=1}^n p(1 - p)^{x-1}$$

$$L(p) = p^n (1 - p)^{\sum x - n}$$

Taking “ln” both sides

$$\ln L(p) = n \ln p + (\sum x - n) \ln (1 - p)$$

Diff w.r.t p

$$\frac{\partial}{\partial p} \ln L(p) = n \frac{\partial}{\partial p} \ln p + (\sum x - n) \frac{\partial}{\partial p} \ln (1 - p)$$

$$\frac{\partial}{\partial p} \ln L(p) = \frac{n}{p} - \frac{1}{1 - p} (\sum x - n) \dots \dots \dots (1)$$

Put

$$\frac{\partial}{\partial p} \ln L(p) = 0$$

$$\frac{n}{p} - \frac{1}{1 - p} (\sum x - n) = 0$$

$$\frac{n}{p} = \frac{1}{1 - p} (\sum x - n)$$

$$n(1 - p) = p (\sum x - n)$$

$$n - np = p \sum x - np$$

$$p = \frac{n}{\sum x}$$

Diff, (1) w.r.t p

$$\frac{\partial^2}{\partial p^2} \ln L(p) = n \frac{\partial}{\partial p} \left(\frac{1}{p} \right) - (\sum x - n) \frac{\partial}{\partial p} \left(\frac{1}{1 - p} \right)$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-n}{p^2} - \frac{(\sum x - n)}{(1 - p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-n(1 - p)^2 - p^2 (\sum x - n)}{p^2 (1 - p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-n - np^2 + 2np - p^2 \sum x + np^2}{p^2 (1 - p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-n + 2(p \sum x)p - p^2 \sum x}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-n + 2p^2 \sum x - p^2 \sum x}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-n + p^2 \sum x}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-n + p^2 \left(\frac{n}{p}\right)}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-n + np}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-n(1-p)}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-n}{p^2(1-p)} < 0, \text{ As } 0 < p < 1$$

So, MLE of p in Geometric distribution is

$$\hat{p} = \frac{n}{\sum x} = \frac{1}{\frac{\sum x}{n}} = \frac{1}{\bar{x}}$$

2.6 MLE of Parameter of “p” of Binomial Distribution

Question 5: Let $X_1, X_2, \dots, X_m$ be random sample from binomial distribution with Probability Mass Function

Solution:

$$f(x; p) = \binom{m}{x} p^x (1-p)^{m-x}, x = 0, 1, \dots, m$$

Find MLE for the parameter p

Solution:

The likelihood function

$$L(p) = \prod_{i=1}^m \binom{m}{x_i} p^{x_i} (1-p)^{m-x_i}$$

$$L(p) = \prod_{i=1}^m \binom{m}{x_i} p^{\sum x_i} (1-p)^{\sum m - \sum x_i}$$

$$L(p) = k p^{\sum x_i} (1-p)^{m^2 - \sum x_i}$$

Taking “ln” both sides

$$\ln L(p) = \ln k + \sum x_i \ln p + (m^2 - \sum x_i) \ln (1-p)$$

Differentiate w.r.t p

$$\frac{\partial}{\partial p} \ln L(p) = 0 + \sum x_i \frac{\partial}{\partial p} \ln p + (m^2 - \sum x_i) \frac{\partial}{\partial p} \ln (1-p)$$

$$\frac{\partial}{\partial p} \ln L(p) = \frac{\sum x_i}{p} - \frac{m^2 - \sum x_i}{1-p} \dots \dots \dots (1)$$

Put

$$\frac{\partial}{\partial p} \ln L(p) = 0$$

$$\frac{\sum x_i}{p} - \frac{m^2 - \sum x_i}{1-p} = 0$$

$$\frac{\sum x_i}{p} = \frac{m^2 - \sum x_i}{1-p}$$

$$(1-p) \sum x_i = p(m^2 - \sum x_i)$$

$$\sum x_i - p \sum x_i = pm^2 - p \sum x_i$$

$$\sum x_i = pm^2$$

$$p = \frac{\sum x_i}{m^2}$$

Diff (1) w.r.t p

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \sum x_i \frac{\partial}{\partial p} \left(\frac{1}{p} \right) - (m^2 - \sum x_i) \frac{\partial}{\partial p} \left(\frac{1}{1-p} \right)$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-\sum x}{p^2} - \left(\frac{m^2 - \sum x_i}{(1-p)^2} \right)$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-\sum x_i(1-p)^2 - p^2(m^2 - \sum x_i)}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-\sum x_i - p^2 \sum x_i + 2 \sum x_i p - p^2 m^2 + \sum x_i p^2}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-\sum x_i + 2 \sum x_i p - p^2 m^2}{p^2(1-p)^2}$$

Using $\sum x_i = m^2 p$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-m^2 p + 2m^2 p^2 - p^2 m^2}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-m^2 p + m^2 p^2}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-m^2 p(1-p)}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-m^2}{p(1-p)} < 0, \text{ As } 0 < p < 1$$

So, MLE of p of binomial distribution

$$p = \frac{\sum x_i}{m^2} = \frac{1}{m} \cdot \frac{\sum x_i}{m} = \frac{\bar{x}}{m}$$

2.7 MLE of Parameter of “p” Bernoulli Distribution

Question 6: let X be Bernoulli random variable with p.m.f $p(x) = p^x(1-p)^{1-x}; x = 0, 1$. Show that the MLE of p is $\hat{p} = \bar{x}$. Also using CRLB verify that

$$V(\hat{p}) \geq \frac{p(1-p)}{n}$$

Solution:-

$$L(p) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i}$$

$$L(p) = p^{\sum x_i} (1-p)^{n-\sum x_i}$$

Taking “ln” both sides

$$\ln L(p) = \sum x_i \ln p + (n - \sum x_i) \ln (1-p)$$

Differentiate w.r.t p

$$\frac{\partial}{\partial p} \ln L(p) = \sum x_i \frac{\partial}{\partial p} \ln p + (n - \sum x_i) \frac{\partial}{\partial p} \ln (1-p)$$

$$\frac{\partial}{\partial p} \ln L(p) = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1 - p} \dots \dots \dots (1)$$

Put

$$\frac{\partial}{\partial p} \ln L(p) = 0$$

$$\frac{\sum x_i}{p} - \frac{n - \sum x_i}{1 - p} = 0$$

$$\frac{\sum x_i}{p} = \frac{n - \sum x_i}{1 - p}$$

$$(1 - p) \sum x_i = p(n - \sum x_i)$$

$$\sum x_i - p \sum x_i = np - p \sum x_i$$

$$\sum x_i = np$$

$$p = \frac{\sum x_i}{n}$$

Diff (1) w.r.t p

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \sum x_i \frac{\partial}{\partial p} \left(\frac{1}{p} \right) - (n - \sum x_i) \frac{\partial}{\partial p} \left(\frac{1}{1 - p} \right)$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-\sum x_i}{p^2} - \left(\frac{n - \sum x_i}{(1 - p)^2} \right)$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-\sum x_i(1 - p)^2 - p^2(n - \sum x_i)}{p^2(1 - p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-\sum x_i - p^2 \sum x_i + 2 \sum x_i p - np^2 + \sum x_i p^2}{p^2(1 - p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-\sum x_i + 2 \sum x_i p - np^2}{p^2(1 - p)^2}$$

Using $\sum x_i = np$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-np + 2np^2 - np^2}{p^2(1 - p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-np + np^2}{p^2(1 - p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-np(1 - p)}{p^2(1 - p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-n}{p(1-p)} < 0, \text{ As } 0 < p < 1$$

So, MLE of p of Bernoulli Distribution

$$\hat{p} = \frac{\sum x_i}{n}$$

Now, to get CRLB

$$p(x) = p^x(1-p)^{1-x}$$

Taking “ln” both sides

$$\ln L[P(x)] = x \ln p + (1-x) \ln(1-p)$$

Differentiate w.r.t p

$$\frac{\partial}{\partial p} \ln L[P(x)] = x \frac{\partial}{\partial p} \ln p + (1-x) \frac{\partial}{\partial p} \ln(1-p)$$

$$\frac{\partial}{\partial p} \ln L[P(x)] = \frac{x}{p} - \frac{1-x}{1-p}$$

$$\frac{\partial}{\partial p} \ln L[P(x)] = \frac{x(1-p) - p(1-x)}{p(1-p)}$$

$$\frac{\partial}{\partial p} \ln L[P(x)] = \frac{x - xp - p + xp}{p(1-p)}$$

$$\frac{\partial}{\partial p} \ln L[P(x)] = \frac{x-p}{p(1-p)}$$

Squaring and taking Expectations

$$E \left[\frac{\partial}{\partial p} \ln L[P(x)] \right]^2 = E \left[\frac{x-p}{p(1-p)} \right]^2$$

$$E \left[\frac{\partial}{\partial p} \ln L[P(x)] \right]^2 = \frac{E(x^2) + E(p^2) - 2E(xp)}{p^2(1-p)^2}$$

$$E \left[\frac{\partial}{\partial p} \ln L[P(x)] \right]^2 = \frac{p + p^2 - 2p^2}{p^2(1-p)^2}$$

$$E \left[\frac{\partial}{\partial p} \ln L[P(x)] \right]^2 = \frac{p - p^2}{p^2(1-p)^2} = \frac{p(1-p)}{p^2(1-p)^2} = \frac{1}{p(1-p)}$$

Putting in

$$V(p) \geq \frac{1}{nE \left[\frac{\partial}{\partial p} \{ \ln p(x) \} \right]^2}$$

$$V(p) \geq \frac{1}{n \left[\frac{1}{p(1-p)} \right]}$$

$$V(p) \geq \frac{p(1-p)}{n}$$

2.8 MLE of Parameter λ of Poisson Distribution

Question 7: Let $X_1, X_2, \dots, X_n$ denote the random sample from poisson distribution with parameter λ . Find MLE of λ .

Solution:-

The probability Mass Function of Poisson Distribution is

$$f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$$

The likelihood function is

$$L(\lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$L(\lambda) = \frac{e^{-\lambda n} \lambda^{\sum x_i}}{x_1! \cdot x_2! \cdot \dots \cdot x_n!}$$

Taking "ln" Both sides

$$\ln L(\lambda) = -\lambda n \ln e + \sum x_i \ln \lambda - \ln (x_1! \cdot x_2! \cdot \dots \cdot x_n!)$$

$$\ln L(\lambda) = -\lambda n + \sum x_i \ln \lambda - \sum \ln (x_i!)$$

Diff w.r.t λ

$$\begin{aligned} \frac{\partial}{\partial \lambda} \ln L(\lambda) &= -n \frac{\partial}{\partial \lambda} \lambda + \sum x_i \frac{\partial}{\partial \lambda} \ln \lambda - \frac{\partial}{\partial \lambda} \sum \ln (x_i!) \\ \frac{\partial}{\partial \lambda} \ln L(\lambda) &= -n + \frac{\sum x_i}{\lambda} \dots \dots \dots (1) \end{aligned}$$

Put

$$\begin{aligned} \frac{\partial}{\partial \lambda} \ln L(\lambda) &= 0 \\ -n + \frac{\sum x_i}{\lambda} &= 0 \\ n &= \frac{\sum x_i}{\lambda} \end{aligned}$$

$$\lambda = \frac{\sum x_i}{n}$$

Diff (1) w.r.t λ

$$\frac{\partial^2}{\partial \lambda^2} \ln L(\lambda) = -\frac{\sum x_i}{\lambda^2} < 0$$

So, MLE for λ

$$\hat{\lambda} = \frac{\sum x_i}{n}$$

2.9 MLE of Parameter p of Negative Binomial Distribution

Question 8: Let $X_1, X_2, \dots, X_n$ be random sample from negative binomial distribution with Probability Mass Function

$$f(x; p) = \binom{r+x-1}{x} p^r (1-p)^x$$

Find MLE for the parameter p

Solution:

The likelihood function

$$L(p) = \prod_{i=1}^n \binom{r+x-1}{x} p^r (1-p)^x$$

$$L(p) = \prod_{i=1}^n \binom{r+x-1}{x} p^{nr} (1-p)^{\sum x_i}$$

$$L(p) = k p^{nr} (1-p)^{\sum x_i}$$

Taking “ln” both sides

$$\ln L(p) = \ln k + nr \ln p + \sum x_i \ln (1-p)$$

Differentiate w.r.t p

$$\frac{\partial}{\partial p} \ln L(p) = 0 + nr \frac{\partial}{\partial p} \ln p + \sum x_i \frac{\partial}{\partial p} \ln (1-p)$$

$$\frac{\partial}{\partial p} \ln L(p) = \frac{nr}{p} - \frac{\sum x_i}{1-p} \dots \dots \dots (1)$$

Put

$$\frac{\partial}{\partial p} \ln L(p) = 0$$

$$\frac{nr}{p} - \frac{\sum x_i}{1-p} = 0$$

$$\frac{nr}{p} = \frac{\sum x_i}{1-p}$$

$$(1-p)nr = p(\sum x_i)$$

$$nr - npr = p \sum x_i$$

$$nr = p(\sum x_i + nr)$$

$$p = \frac{nr}{\sum x_i + nr}$$

Diff (1) w.r.t p

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{\partial}{\partial p} \left(\frac{nr}{p} \right) - \frac{\partial}{\partial p} \left(\frac{\sum x_i}{1-p} \right)$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-nr}{p^2} - \left(\frac{\sum x_i}{(1-p)^2} \right)$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-nr(1-p)^2 - p^2 \sum x_i}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-nr(1-p)(1-p) - p^2 \sum x_i}{p^2(1-p)^2}$$

Using $nr(1-p) = p \sum x_i$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-p \sum x_i (1-p) - p^2 \sum x_i}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-p \sum x_i + p^2 \sum x_i - p^2 \sum x_i}{p^2(1-p)^2}$$

$$\frac{\partial^2}{\partial p^2} \ln L(p) = \frac{-p \sum x_i}{p^2(1-p)^2} < 0, \text{ As } 0 < p < 1$$

Hence MLE of p

$$\hat{p} = \frac{nr}{\sum x_i + nr}$$

2.10 MLE of Parameters λ^2 and λ from the Rayleigh Distribution

Question 9 : Obtain the MLE of λ^2 and λ from the Rayleigh Distribution which has the p.d.f

$$f(x, \lambda) = \frac{2x}{\lambda^2} e^{-\frac{x^2}{\lambda^2}}; \quad 0 < x < \infty$$

Solution:

The likelihood function:-

$$L(\lambda) = \prod_{i=1}^n \frac{2x_i}{\lambda^2} e^{-\frac{x_i^2}{\lambda^2}}$$

$$L(\lambda) = \prod_{i=1}^n x_i \left(\frac{2}{\lambda^2}\right)^n e^{-\frac{\sum x_i^2}{\lambda^2}}$$

Taking “ln” Both sides

$$\ln L(\lambda) = \ln(x_1 \cdot x_2 \dots x_n) + n \ln 2 - n \ln \lambda^2 - \frac{\sum x_i^2}{\lambda^2} \ln e$$

$$\ln L(\lambda) = \sum \ln x_i + n \ln 2 - 2n \ln \lambda - \frac{\sum x_i^2}{\lambda^2}$$

Diff w.r.t λ

$$\frac{\partial}{\partial \lambda} \ln L(\lambda) = -2n \frac{\partial}{\partial \lambda} \ln \lambda - \sum x_i^2 \frac{\partial}{\partial \lambda} \left(\frac{1}{\lambda^2}\right)$$

$$\frac{\partial}{\partial \lambda} \ln L(\lambda) = -\frac{2n}{\lambda} + \frac{2 \sum x_i^2}{\lambda^3} \dots \dots \dots (1)$$

put

$$\frac{\partial}{\partial \lambda} \ln L(\lambda) = 0$$

$$-\frac{2n}{\lambda} + \frac{2 \sum x_i^2}{\lambda^3} = 0$$

$$\frac{n}{\lambda} = \frac{\sum x_i^2}{\lambda^3}$$

$$n\lambda^2 = \sum x_i^2$$

$$\lambda^2 = \frac{\sum x_i^2}{n}$$

Diff (1) w.r.t λ

$$\frac{\partial^2}{\partial \lambda^2} \ln L(\lambda) = \frac{\partial}{\partial \lambda} \left(-\frac{2n}{\lambda}\right) + \frac{\partial}{\partial \lambda} \left(\frac{2 \sum x_i^2}{\lambda^3}\right)$$

$$\frac{\partial^2}{\partial \lambda^2} \ln L(\lambda) = \frac{2n}{\lambda^2} - \frac{6 \sum x_i^2}{\lambda^4}$$

$$\frac{\partial^2}{\partial \lambda^2} \ln L(\lambda) = \frac{2n\lambda^2 - 6 \sum x_i^2}{\lambda^4}$$

Using $n\lambda^2 = \sum x_i^2$

$$\frac{\partial^2}{\partial \lambda^2} \ln L(\lambda) = \frac{2\sum x_i^2 - 6\sum x_i^2}{\lambda^4}$$

$$\frac{\partial^2}{\partial \lambda^2} \ln L(\lambda) = \frac{-4\sum x_i^2}{\lambda^4} < 0$$

So, MLE for λ^2 is

$$\hat{\lambda}^2 = \frac{\sum x_i^2}{n}$$

The MLE for λ will be obtained as

$$\hat{\lambda} = \left[\frac{\sum x_i^2}{n} \right]^{\frac{1}{2}}$$

Since λ is in one-one relationship of λ^2 , and its invariant property of MLE.

2.11 MLE of Parameter σ of the $N(0, \sigma^2)$, Also Variance.

Question 10: Find the asymptotic variance of the MLE of σ of the $N(0, \sigma^2)$

Solution:-

The pdf of normal distribution $N(0, \sigma^2)$.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}}$$

The likelihood function

$$L(\sigma) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x_i^2}{2\sigma^2}}$$

$$L(\sigma) = \frac{1}{\sigma^n (2\pi)^{\frac{n}{2}}} e^{-\frac{\sum x_i^2}{2\sigma^2}}$$

Taking ln both sides

$$\ln L(\sigma) = \ln 1 - n \ln \sigma - \frac{n}{2} \ln (2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2 \ln e$$

$$\ln L(\sigma) = -n \ln \sigma - \frac{n}{2} \ln (2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2$$

Diff w.r.t σ

$$\frac{\partial}{\partial \sigma} \ln L(\sigma) = -\frac{\partial}{\partial \sigma} n \ln \sigma - \frac{\partial}{\partial \sigma} \frac{n}{2} \ln (2\pi) - \frac{\partial}{\partial \sigma} \frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2$$

$$\frac{\partial}{\partial \sigma} \ln L(\sigma) = -\frac{n}{\sigma} + \frac{\sum x_i^2}{\sigma^3} \dots \dots \dots (1)$$

Put

$$\frac{\partial}{\partial \sigma} \ln L(\sigma) = 0$$

$$-\frac{n}{\sigma} + \frac{\sum x_i^2}{\sigma^3} = 0$$

$$\frac{n}{\sigma} = \frac{\sum x_i^2}{\sigma^3}$$

$$\sigma^2 = \frac{\sum x_i^2}{n}$$

Diff (1) w.r.t σ

$$\frac{\partial^2}{\partial \sigma^2} \ln L(\sigma) = -\frac{\partial}{\partial \sigma} \frac{n}{\sigma} + \frac{\partial}{\partial \sigma} \frac{\sum x_i^2}{\sigma^3}$$

$$\frac{\partial^2}{\partial \sigma^2} \ln L(\sigma) = \frac{n}{\sigma^2} - \frac{3 \sum x_i^2}{\sigma^4}$$

$$\frac{\partial^2}{\partial \sigma^2} \ln L(\sigma) = \frac{n\sigma^2 - 3 \sum x_i^2}{\sigma^4}$$

Using $n\sigma^2 = \sum x_i^2$

$$\frac{\partial^2}{\partial \sigma^2} \ln L(\sigma) = \frac{\sum x_i^2 - 3 \sum x_i^2}{\sigma^4}$$

$$\frac{\partial^2}{\partial \sigma^2} \ln L(\sigma) = \frac{-2 \sum x_i^2}{\sigma^4} < 0$$

Hence MLE for σ is

$$\sigma = \left(\frac{\sum x_i^2}{n} \right)^{\frac{1}{2}}$$

Now, to find the asymptotic variance of σ

$$E \left[\frac{\partial^2}{\partial \sigma^2} \ln L(\sigma) \right] = \frac{-2 \sum E[x_i^2]}{\sigma^4}$$

$$E \left[\frac{\partial^2}{\partial \sigma^2} \ln L(\sigma) \right] = \frac{-2n\sigma^2}{\sigma^4}$$

$$-E \left[\frac{\partial^2}{\partial \sigma^2} \ln L(\sigma) \right] = \frac{2n}{\sigma^2}$$

Now, putting in

$$V(\hat{\sigma}) = \frac{1}{-E \left[\frac{\partial^2}{\partial \sigma^2} \ln L(\sigma) \right]} = \frac{1}{\frac{2n}{\sigma^2}} = \frac{\sigma^2}{2n}$$

2.12 MLE of Parameter θ of $N(o, \theta)$, as Function of Sufficient Statistic

Question11: let X_n be random sample from normal distribution.

$$f(x) = \frac{1}{\sqrt{2\pi\theta}} e^{-\frac{x^2}{2\theta}} \quad -\infty < x < \infty$$

Show that $\sum x_i^2$ is a sufficient statistic for θ and MLE of θ is a function of sufficient statistic

Solution:-

The likelihood function

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\theta}} e^{-\frac{x_i^2}{2\theta}} \\ L(\theta) &= \frac{1}{(2\pi\theta)^{\frac{n}{2}}} e^{-\frac{\sum x_i^2}{2\theta}} \quad \text{--- (1)} \\ L(\theta) &= [e^{-\frac{\sum x_i^2}{2\theta}} \theta^{-\frac{n}{2}} (2\pi)^{-\frac{n}{2}}] [1] \\ L(\theta) &= g(\theta, \sum x_i^2) h(x) \end{aligned}$$

By factorization theorem it is clear that $g(\theta, \sum x_i^2)$ is function of parameter and statistic and $h(x)$ is independent of parameter and statistic.

Taking ln both sides of (1)

$$\begin{aligned} \ln L(\theta) &= \ln 1 - \frac{n}{2} \ln \theta - \frac{n}{2} \ln (2\pi) - \frac{1}{2\theta} \sum_{i=1}^n x_i^2 \ln e \\ \ln L(\sigma) &= -\frac{n}{2} \ln \theta - \frac{n}{2} \ln (2\pi) - \frac{1}{2\theta} \sum_{i=1}^n x_i^2 \end{aligned}$$

Diff w.r.t θ

$$\frac{\partial}{\partial \theta} \ln L(\theta) = -\frac{\partial}{\partial \theta} \frac{n}{2} \ln \theta - \frac{\partial}{\partial \theta} \frac{n}{2} \ln (2\pi) - \frac{\partial}{\partial \theta} \frac{1}{2\theta} \sum_{i=1}^n x_i^2$$

$$\frac{\partial}{\partial \theta} \ln L(\theta) = -\frac{n}{2\theta} + \frac{\sum x_i^2}{2\theta^2} \dots \dots \dots (2)$$

Put

$$\frac{\partial}{\partial \theta} \ln L(\theta) = 0$$

$$-\frac{n}{2\theta} + \frac{\sum x_i^2}{2\theta^2} = 0$$

$$\frac{n}{2\theta} = \frac{\sum x_i^2}{2\theta^2}$$

$$\theta = \frac{\sum x_i^2}{n}$$

Diff (2) w.r.t σ

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = -\frac{\partial}{\partial \theta} \frac{n}{2\theta} + \frac{\partial}{\partial \theta} \frac{\sum x_i^2}{2\theta^2}$$

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = \frac{n}{2\theta^2} - \frac{\sum x_i^2}{\theta^3}$$

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = \frac{n\theta - 2\sum x_i^2}{2\theta^3}$$

Using

$$n\theta = \sum x_i^2$$

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = \frac{\sum x_i^2 - 2\sum x_i^2}{2\theta^3}$$

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = \frac{-\sum x_i^2}{2\theta^3} < 0$$

Hence MLE for θ is

$$\theta = \frac{\sum x_i^2}{n}$$

2.13 MLE for μ and σ^2 of Log Normal Distribution

Question 12: Given the p.d.f of log normal distribution, find MLE for μ and σ^2 .

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma x}} e^{-\frac{1}{2\sigma^2}(\ln x - \mu)^2}$$

Solution:-

The likelihood function

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma x_i}} e^{-\frac{1}{2\sigma^2}(\ln x_i - \mu)^2}$$

$$L(\mu) = \frac{1}{(2\pi)^{\frac{n}{2}}} \frac{1}{(\sigma^2)^{\frac{n}{2}}} \prod_{i=1}^n \left(\frac{1}{x_i} \right) e^{-\frac{1}{2\sigma^2} \sum (\ln x_i - \mu)^2}$$

Taking “ln” Both sides

$$\ln L(\mu, \sigma^2) = \ln 1 - \frac{n}{2} \ln \pi + \ln 1 - \frac{n}{2} \ln \sigma^2 + \ln (x_1 \cdot x_2 \dots x_n) - \frac{1}{2\sigma^2} \sum (\ln x_i - \mu)^2 \ln e$$

$$\ln L(\mu, \sigma^2) = -\frac{n}{2} \ln \pi - \frac{n}{2} \ln \sigma^2 + \sum \ln x_i - \frac{1}{2\sigma^2} \sum (\ln x_i - \mu)^2 \quad (1)$$

Diff (1) w.r.t μ

$$\frac{\partial}{\partial \mu} \ln L(\mu, \sigma^2) = -\frac{2}{2\sigma^2} \sum (\ln x_i - \mu)(0 - 1)$$

$$\frac{\partial}{\partial \mu} \ln L(\mu, \sigma^2) = \frac{1}{\sigma^2} \sum (\ln x_i - \mu) \dots \dots \dots (2)$$

Put

$$\frac{\partial}{\partial \mu} \ln L(\mu, \sigma^2) = 0$$

$$\frac{1}{\sigma^2} \sum (\ln x_i - \mu) = 0$$

$$\sum (\ln x_i - \mu) = 0$$

$$\sum \ln x - n\mu = 0$$

$$\sum \ln x = n\mu$$

$$\mu = \frac{\sum \ln x}{n}$$

Diff (2) w.r.t μ

$$\frac{\partial^2}{\partial \mu^2} \ln L(\mu, \sigma^2) = \frac{1}{\sigma^2} \sum (0 - 1) = -\frac{n}{\sigma^2} < 0$$

MLE of μ

$$\hat{\mu} = \frac{\sum \ln x}{n}$$

Diff (1) w.r.t σ^2

$$\frac{\partial}{\partial \sigma^2} \ln L(\mu, \sigma^2) = -\frac{n}{2} \frac{1}{\sigma^2} - \frac{1}{2} \sum (\ln x_i - \mu)^2 (-1)(\sigma^2)^{-2}$$

$$\frac{\partial}{\partial \sigma^2} \ln L(\mu, \sigma^2) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (\ln x_i - \mu)^2 \dots \dots \dots (3)$$

Put

$$\begin{aligned} \frac{\partial}{\partial \sigma^2} \ln L(\mu, \sigma^2) &= 0 \\ -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (\ln x_i - \mu)^2 &= 0 \end{aligned}$$

$$\frac{n}{2\sigma^2} = \frac{1}{2\sigma^4} \sum (\ln x_i - \mu)^2$$

$$n = \frac{1}{\sigma^2} \sum (\ln x_i - \mu)^2$$

$$\sigma^2 = \frac{\sum (\ln x_i - \mu)^2}{n}$$

Diff (3) w.r.t σ^2

$$\frac{\partial^2}{\partial (\sigma^2)^2} \ln L(\mu, \sigma^2) = -\frac{n}{2} (-1)(\sigma^2)^{-2} + \frac{1}{2} \sum (\ln x_i - \mu)^2 (-2)(\sigma^2)^{-3}$$

$$\frac{\partial^2}{\partial (\sigma^2)^2} \ln L(\mu, \sigma^2) = \frac{n}{2\sigma^4} - \frac{\sum (\ln x_i - \mu)^2}{\sigma^6}$$

$$\frac{\partial^2}{\partial (\sigma^2)^2} \ln L(\mu, \sigma^2) = \frac{n\sigma^2 - 2\sum (\ln x_i - \mu)^2}{2\sigma^6}$$

$$\frac{\partial^2}{\partial (\sigma^2)^2} \ln L(\mu, \sigma^2) = \frac{\sum (\ln x_i - \mu)^2 - 2\sum (\ln x_i - \mu)^2}{2\sigma^6}$$

$$\frac{\partial^2}{\partial (\sigma^2)^2} \ln L(\mu, \sigma^2) = \frac{-\sum (\ln x_i - \mu)^2}{2\sigma^6} < 0$$

Hence, MLE of σ^2

$$\hat{\sigma}^2 = \frac{\sum (\ln x_i - \mu)^2}{n}$$

2.14 MLE of the Parameter α

Question 13: prove that the MLE of the parameter α for population with density

$$f(x) = \frac{2}{\alpha^2}(\alpha - x), \quad 0 < x < \alpha$$

For a sample of unit size is $2x$, x being a sample value. Show that estimate is biased.

Solution:

Since the sample size is unit so the likelihood function is

$$L(\alpha) = \frac{2}{\alpha^2}(\alpha - x)$$

Taking ln both sides

$$\ln L(\alpha) = \ln 2 - 2\ln \alpha + \ln(\alpha - x)$$

Diff w.r.t α

$$\begin{aligned} \frac{\partial}{\partial \alpha} \ln L(\alpha) &= \frac{\partial}{\partial \alpha} \ln 2 - \frac{\partial}{\partial \alpha} 2\ln \alpha + \frac{\partial}{\partial \alpha} \ln(\alpha - x) \\ \frac{\partial}{\partial \alpha} \ln L(\alpha) &= -\frac{2}{\alpha} + \frac{1}{\alpha - x} \dots \dots \dots (1) \end{aligned}$$

Put

$$\frac{\partial}{\partial \alpha} \ln L(\alpha) = 0$$

$$-\frac{2}{\alpha} + \frac{1}{\alpha - x} = 0$$

$$\frac{2}{\alpha} = \frac{1}{\alpha - x}$$

$$2(\alpha - x) = \alpha$$

$$2\alpha - 2x = \alpha$$

$$2\alpha - \alpha = 2x$$

$$\alpha = 2x$$

Diff (1) w.r.t α

$$\frac{\partial^2}{\partial \alpha^2} \ln L(\alpha) = -\frac{\partial}{\partial \alpha} \frac{2}{\alpha} + \frac{\partial}{\partial \alpha} \frac{1}{\alpha - x}$$

$$\frac{\partial^2}{\partial \alpha^2} \ln L(\alpha) = \frac{2}{\alpha^2} - \frac{1}{(\alpha - x)^2}$$

$$\frac{\partial^2}{\partial \alpha^2} \ln L(\alpha) = \frac{2(\alpha - x)^2 - \alpha^2}{\alpha^2(\alpha - x)^2}$$

$$\frac{\partial^2}{\partial \alpha^2} \ln L(\alpha) = \frac{2(\alpha - x)(\alpha - x) - \alpha^2}{\alpha^2(\alpha - x)^2}$$

Using $2(\alpha - x) = \alpha$

$$\frac{\partial^2}{\partial \alpha^2} \ln L(\alpha) = \frac{\alpha(\alpha - x) - \alpha^2}{\alpha^2(\alpha - x)^2}$$

$$\frac{\partial^2}{\partial \alpha^2} \ln L(\alpha) = \frac{\alpha^2 - \alpha x - \alpha^2}{\alpha^2(\alpha - x)^2}$$

$$\frac{\partial^2}{\partial \alpha^2} \ln L(\alpha) = \frac{-\alpha x}{\alpha^2(\alpha - x)^2} < 0, \text{ As } 0 < x < \alpha$$

Hence, the MLE for α is

$$\hat{\alpha} = 2x$$

Now, we find expectation

$$E(\hat{\alpha}) = \int_0^\alpha 2x \frac{2}{\alpha^2} (\alpha - x) dx$$

$$E(\hat{\alpha}) = \frac{4}{\alpha^2} \int_0^\alpha (x\alpha - x^2) dx$$

$$E(\hat{\alpha}) = \frac{4\alpha}{\alpha^2} \int_0^\alpha x dx - \frac{4}{\alpha^2} \int_0^\alpha x^2 dx$$

$$E(\hat{\alpha}) = \frac{4}{\alpha} \cdot \frac{x^2}{2} \Big|_0^\alpha - \frac{4}{\alpha^2} \frac{x^3}{3} \Big|_0^\alpha$$

$$E(\hat{\alpha}) = \frac{2}{\alpha} \cdot (\alpha^2 - 0) - \frac{4}{3\alpha^2} (\alpha^3 - 0)$$

$$E(\hat{\alpha}) = 2\alpha - \frac{4\alpha}{3}$$

$$E(\hat{\alpha}) = \frac{6\alpha - 4\alpha}{3} = \frac{2\alpha}{3}$$

Here,

$$E(\hat{\alpha}) \neq \alpha$$

So, it can be noted that MLE may not be unbiased.

2.15 MLE of parameter “c” and “a”

Question 14: Find MLE of parameter “c” when “a” is known from the following distribution . what would be your answers when it is asked to find the MLE of “a” when “c” is known.

$$f(x) = \frac{ca^c}{x^{c+1}}, a > 0, c > 0, a < x < \infty$$

Solution:-

$$L(c) = \prod_{i=1}^n \frac{ca^c}{x_i^{c+1}}$$

$$L(c) = \prod_{i=1}^n \left(\frac{1}{x_i^{c+1}} \right) c^n a^{nc}$$

Taking Ln both sides

$$\ln L(c) = \ln 1 - \ln \left[\prod_{i=1}^n x_i^{c+1} \right] + n \ln c + n \ln a$$

$$\ln L(c) = -(c+1) (\sum \ln x_i) + n \ln c + n \ln a$$

Diff w.r.t c

$$\frac{\partial}{\partial c} \ln L(c) = - (\sum \ln x_i) \frac{\partial}{\partial c} (c+1) + n \frac{\partial}{\partial c} \ln c + n \ln a \frac{\partial}{\partial c} c$$

$$\frac{\partial}{\partial c} \ln L(c) = - (\sum \ln x_i) (0+1) + \frac{n}{c} + n \ln a (1)$$

$$\frac{\partial}{\partial c} \ln L(c) = - (\sum \ln x_i) + \frac{n}{c} + n \ln a \quad \text{--- (1)}$$

Put

$$\frac{\partial}{\partial c} \ln L(c) = 0$$

$$- (\sum \ln x_i) + \frac{n}{c} + n \ln a = 0$$

$$\frac{n}{c} = \sum \ln x_i - n \ln a$$

$$c = \frac{n}{\sum \ln x_i - n \ln a}$$

Diff (1) w.r.t c

$$\frac{\partial^2}{\partial c^2} \ln L(c) = \frac{-n}{c^2} < 0$$

Hence, MLE of c

$$\hat{c} = \frac{n}{\sum \ln x_i - n \ln a}$$

It is interesting to note that the MLE of “c” requires the known value of the parameter “a”.

Now, to find MLE of “a” the likelihood function is

$$\ln L(a) = -(c+1) (\sum \ln x_i) + n \ln c + n \ln a$$

Diff w.r.t a

$$\frac{\partial}{\partial a} \ln L(a) = -\frac{\partial}{\partial a} (\sum \ln x_i) (c+1) + \frac{\partial}{\partial a} n \ln c + nc \frac{\partial}{\partial a} \ln a$$

$$\frac{\partial}{\partial a} \ln L(a) = \frac{nc}{a}$$

Put

$$\frac{\partial}{\partial a} \ln L(a) = 0$$

$$\frac{nc}{a} = 0$$

$$nc = 0$$

Here, the estimator is vanished so no MLE for “a” exists.

We cannot find the MLE, of “a” because it is in the range of the distribution. In such case the variance of the estimator cannot be evaluated

2.16 MLE of $P(X \leq 2)$ for Exponential Distribution

Question 15: Let $X_1, X_2, \dots, X_n$ be IID each with p.d.f $f(x; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}}, 0 < x < \infty$, find the MLE of $P(X \leq 2)$.

Solution:

$$L(\theta) = \prod_{i=1}^n \frac{1}{\theta} e^{-\frac{x_i}{\theta}}$$

$$L(\theta) = \left(\frac{1}{\theta}\right)^n e^{-\frac{\sum x_i}{\theta}}$$

Taking ln both sides

$$\ln L(\theta) = n \ln 1 - n \ln \theta - \frac{\sum x_i}{\theta} \ln e$$

$$\ln L(\theta) = -n \ln \theta - \frac{\sum x_i}{\theta}$$

Diff w.r.t

$$\frac{\partial}{\partial \theta} \ln L(\theta) = -n \frac{\partial}{\partial \theta} \ln \theta - \sum x_i \frac{\partial}{\partial \theta} \left(\frac{1}{\theta}\right)$$

$$\frac{\partial}{\partial \theta} \ln L(\theta) = -\frac{n}{\theta} + \frac{\sum x_i}{\theta^2} \dots \dots \dots (1)$$

Put

$$\frac{\partial}{\partial \theta} \ln L(\theta) = 0$$

$$-\frac{n}{\theta} + \frac{\sum x_i}{\theta^2} = 0$$

$$\frac{n}{\theta} = \frac{\sum x_i}{\theta^2}$$

$$\theta = \frac{\sum x_i}{n} = \bar{x}$$

Diff (1) w.r.t θ

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = -\frac{\partial}{\partial \theta} \frac{n}{\theta} + \frac{\partial}{\partial \theta} \frac{\sum x_i}{\theta^2}$$

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = \frac{n}{\theta^2} - \frac{2\sum x_i}{\theta^3}$$

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = \frac{n\theta - 2\sum x_i}{\theta^3}$$

Using $n\theta = \sum x_i$

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = \frac{\sum x_i - 2\sum x_i}{\theta^3}$$

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = \frac{-\sum x_i}{\theta^3} < 0$$

As $X_1, X_2, \dots, X_n$ are I.I.D

$$X_1 = X_2 = \dots = X_n$$

$$\theta = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{nX}{n} = X$$

$$P(X \leq 2) = \int_0^2 \frac{1}{\theta} e^{-\frac{x}{\theta}} dx$$

$$P(X \leq 2) = - \int_0^2 e^{-\frac{x}{\theta}} \left(-\frac{1}{\theta} dx \right)$$

$$P(X \leq 2) = -e^{-\frac{x}{\theta}} \Big|_0^2$$

$$P(X \leq 2) = - \left(e^{-\frac{2}{\theta}} - e^0 \right)$$

$$P(X \leq 2) = 1 - e^{-\frac{2}{\theta}}$$

Putting value of MLE θ

$$P(X \leq 2) = 1 - e^{-\frac{2}{\bar{X}}}$$

2.17 MLE of θ (Non-Existence of MLE)

Question 16: The probability density function of r.v is given as

$$f(x) = \frac{1}{2} e^{-|x-\theta|}; -\infty < x < \infty$$

Whether, MLE of θ can be obtained

Solution:-

The likelihood function is

$$L(\theta) = \prod_{i=1}^n \frac{1}{2} e^{-|x_i - \theta|}$$

$$L(\theta) = \left(\frac{1}{2}\right)^n e^{-\sum |x_i - \theta|}$$

Taking ln both sides

$$\ln L(\theta) = \ln \left(\frac{1}{2}\right)^n - \sum |x_i - \theta| \ln e$$

$$\ln L(\theta) = \ln \left(\frac{1}{2}\right)^n - \sum |x_i - \theta|$$

Arranging x_i in ascending order such that first k -values are less than θ and from $(k+1)$ to n are greater than θ

$$\ln L(\theta) = \ln \left(\frac{1}{2}\right)^n - [-(x_1 - \theta) - (x_2 - \theta) \dots \dots \dots - (x_k - \theta) + (x_{k+1} - \theta) \dots \dots (x_n - \theta)]$$

$$\ln L(\theta) = \ln \left(\frac{1}{2}\right)^n + [(x_1 - \theta) + (x_2 - \theta) \dots \dots \dots + (x_k - \theta)] - [(x_{k+1} - \theta) \dots \dots (x_n - \theta)]$$

Diff w.r.t θ

$$\frac{\partial}{\partial \theta} \ln L(\theta) = [-1 - 1 \dots \dots K - \text{times}] - [-1 - 1 \dots \dots (n - k) - \text{times}]$$

$$\frac{\partial}{\partial \theta} \ln L(\theta) = -k + (n - k)$$

$$\frac{\partial}{\partial \theta} \ln L(\theta) = n - 2k \dots \dots \dots (1)$$

Put

$$\frac{\partial}{\partial \theta} \ln L(\theta) = 0$$

$$n - 2k = 0$$

$$n = 2k$$

It is independent of θ

Diff (1) w.r.t θ

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = 0$$

The 2nd derivative is zero, which means the MLE does not exist .

However, according to ML principal $\ln L$ has to be maximized . $\ln L$ is maximum if $\sum |x_i - \theta|$ is minimum. We know $\sum |x_i - \theta|$ will be minimum when $\theta = \text{median}$.

So, when n is odd the MLE will be

$$\hat{\theta} = \text{med}(x_1, x_2, \dots, x_n)$$

When n is even and $Y_1, Y_2, \dots, Y_n$ taken as order statistic

$$Y_{\frac{n}{2}} \leq \theta \leq Y_{\frac{n}{2}+1}$$

this shows that MLE are not necessarily unique

2.18 MLE of μ and Large Sample Variance (Sample from Two normal Populations)

Question 17: independent samples of size n_1 and n_2 are taken from two normal populations with equal mean μ and variances $\lambda\sigma^2$ and σ^2 respectively. Find MLE of μ and its large sample variance. Hence, show that unbiased estimator

$$t = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

Has efficiency

$$\frac{\lambda(n_1 + n_2)^2}{(n_1\lambda + n_2)(n_1 + n_2\lambda)}$$

(a) Which attains the value 1 for $\lambda = 1$. (b) Find the value of λ for which MLE and estimator t will be equally efficient.

Solution:

Let $X_{1i}, i = 1, 2, \dots, n_1$ and $X_{2j}, j = 1, 2, \dots, n_2$ be two independent samples from given normal populations respectively. As the samples are independent, the joint likelihood function will be given as

$$L(\mu) = \prod_{i=1}^{n_1} \frac{1}{\sqrt{\lambda}\sigma\sqrt{2\pi}} e^{-\frac{1}{2\lambda\sigma^2}(x_{1i}-\mu)^2} \prod_{j=1}^{n_2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x_{2j}-\mu)^2}$$

$$L(\mu) = \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^{n_2} \left(\frac{1}{\sqrt{\lambda}\sigma\sqrt{2\pi}}\right)^{n_1} e^{-\frac{1}{2\lambda\sigma^2}\sum(x_{1i}-\mu)^2} e^{-\frac{1}{2\sigma^2}\sum(x_{2j}-\mu)^2}$$

$$L(\mu) = \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^{n_2} \left(\frac{1}{\sqrt{\lambda}\sigma\sqrt{2\pi}}\right)^{n_1} e^{-\frac{1}{2\lambda\sigma^2}\sum(x_{1i}-\mu)^2 - \frac{1}{2\sigma^2}\sum(x_{2j}-\mu)^2}$$

$$k = \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^{n_2} \left(\frac{1}{\sqrt{\lambda}\sigma\sqrt{2\pi}}\right)^{n_1}$$

$$L(\mu) = K e^{-\frac{1}{2\lambda\sigma^2}[\sum(x_{1i}-\mu)^2 + \lambda\sum(x_{2j}-\mu)^2]}$$

Taking ln both sides

$$\ln L(\mu) = \ln K - \frac{1}{2\lambda\sigma^2} [\sum(x_{1i} - \mu)^2 + \lambda\sum(x_{2j} - \mu)^2]$$

Diff w.r.t μ

$$\frac{\partial}{\partial \mu} \ln L(\mu) = -\frac{1}{2\lambda\sigma^2} [2\sum(x_{1i} - \mu)(-1) + 2\lambda\sum(x_{2j} - \mu)(-1)]$$

$$\frac{\partial}{\partial \mu} \ln L(\mu) = -\frac{2}{2\lambda\sigma^2} [\sum x_{1i} - n_1\mu + \lambda\sum x_{2j} - n_2\lambda\mu]$$

$$\frac{\partial}{\partial \mu} \ln L(\mu) = \frac{\sum x_{1i} + \lambda\sum x_{2j} - \mu(n_1 + \lambda n_2)}{\lambda\sigma^2} \text{----- (1)}$$

Put

$$\frac{\partial}{\partial \mu} \ln L(\mu) = 0$$

$$\frac{\sum x_{1i} + \lambda\sum x_{2j} - \mu(n_1 + \lambda n_2)}{\lambda\sigma^2} = 0$$

$$\sum x_{1i} + \lambda\sum x_{2j} - \mu(n_1 + \lambda n_2) = 0$$

$$\mu = \frac{\sum x_{1i} + \lambda\sum x_{2j}}{n_1 + \lambda n_2}$$

$$\mu = \frac{n_1\bar{x}_1 + n_2\lambda\bar{x}_2}{n_1 + \lambda n_2}$$

Diff (1) w.r.t μ

$$\frac{\partial^2}{\partial \mu^2} \ln L(\mu) = \frac{-(n_1 + \lambda n_2)}{\lambda\sigma^2} < 0$$

Therefore, the MLE of μ is

$$\hat{\mu} = \frac{n_1\bar{x}_1 + n_2\lambda\bar{x}_2}{n_1 + \lambda n_2}$$

Now to obtain variance of MLE

$$V(\hat{\mu}) = V\left[\frac{n_1\bar{x}_1 + n_2\lambda\bar{x}_2}{n_1 + \lambda n_2}\right]$$

$$V(\hat{\mu}) = \frac{n_1^2 V(\bar{x}_1) + n_2^2 \lambda^2 V(\bar{x}_2)}{(n_1 + \lambda n_2)^2}$$

$$V(\hat{\mu}) = \frac{n_1^2 \left(\frac{\lambda\sigma^2}{n_1}\right) + \lambda^2 n_2^2 \left(\frac{\sigma^2}{n_2}\right)}{(n_1 + \lambda n_2)^2}$$

$$V(\hat{\mu}) = \frac{n_1\lambda\sigma^2 + \lambda^2 n_2\sigma^2}{(n_1 + \lambda n_2)^2}$$

$$V(\hat{\mu}) = \frac{\lambda\sigma^2(n_1 + \lambda n_2)}{(n_1 + \lambda n_2)^2} = \frac{\lambda\sigma^2}{(n_1 + \lambda n_2)}$$

Now, consider

$$t = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

Now to obtain variance of t

$$V(t) = V\left[\frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}\right]$$

$$V(t) = \frac{n_1^2 V(\bar{x}_1) + n_2^2 V(\bar{x}_2)}{(n_1 + n_2)^2}$$

$$V(t) = \frac{n_1^2 \left(\frac{\lambda \sigma^2}{n_1}\right) + n_2^2 \left(\frac{\sigma^2}{n_2}\right)}{(n_1 + n_2)^2}$$

$$V(t) = \frac{n_1 \lambda \sigma^2 + n_2 \sigma^2}{(n_1 + n_2)^2}$$

$$V(t) = \frac{(n_1 \lambda + n_2) \sigma^2}{(n_1 + n_2)^2}$$

Now we obtain relative Efficiency of MLE of μ and unbiased estimator t .

$$\frac{V(\hat{\mu})}{V(t)} = \frac{\lambda \sigma^2}{(n_1 + \lambda n_2)} \times \frac{(n_1 + n_2)^2}{(n_1 \lambda + n_2) \sigma^2}$$

$$\frac{V(\hat{\mu})}{V(t)} = \frac{\lambda (n_1 + n_2)^2}{(n_1 + \lambda n_2)(n_1 \lambda + n_2)}$$

(a) Putting $\lambda = 1$

$$\frac{V(\hat{\mu})}{V(t)} = \frac{(n_1 + n_2)^2}{(n_1 + n_2)^2} = 1$$

(b) If both estimators are equally efficient, then

$$\frac{V(\hat{\mu})}{V(t)} = 1$$

$$\frac{\lambda (n_1 + n_2)^2}{(n_1 + \lambda n_2)(n_1 \lambda + n_2)} = 1$$

$$\lambda (n_1 + n_2)^2 = (n_1 + \lambda n_2)(n_1 \lambda + n_2)$$

$$\lambda n_1^2 + \lambda n_2^2 + 2\lambda n_1 n_2 = n_1^2 \lambda + n_1 n_2 + \lambda^2 n_1 n_2 + \lambda n_2^2$$

$$2\lambda n_1 n_2 = n_1 n_2 + \lambda^2 n_1 n_2$$

$$2\lambda n_1 n_2 = n_1 n_2 (1 + \lambda^2)$$

$$2\lambda = 1 + \lambda^2$$

$$\lambda^2 + 1 - 2\lambda = 0$$

$$(\lambda - 1)^2 = 0 \Rightarrow \lambda = 1$$

2.19 MLE of $\alpha = \mu_1 - \mu_2$. (Two Normal Populations with Different Mean and Variance)

Question 18: Samples of sizes n_1 and n_2 are drawn from two normal populations with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 respectively. What is the MLE of $\alpha = \mu_1 - \mu_2$.

Assuming that the total sample size $n = n_1 + n_2$ is fixed. How should the “n” observations be divided between the two populations in order to minimize the variance of $\hat{\alpha}$.

Solution:

Let $X_{1i}, i = 1, 2, \dots, n_1$ and $X_{2j}, j = 1, 2, \dots, n_2$ be two independent samples from given normal populations respectively. As the samples are independent, the joint likelihood function will be given as

$$L(\mu_1, \mu_2) = \prod_{i=1}^{n_1} \frac{1}{\sigma_1 \sqrt{2\pi}} e^{-\frac{1}{2\sigma_1^2}(x_{1i} - \mu_1)^2} \prod_{j=1}^{n_2} \frac{1}{\sigma_2 \sqrt{2\pi}} e^{-\frac{1}{2\sigma_2^2}(x_{2j} - \mu_2)^2}$$

$$L(\mu_1, \mu_2) = \left(\frac{1}{\sigma_1 \sqrt{2\pi}} \right)^{n_1} \left(\frac{1}{\sigma_2 \sqrt{2\pi}} \right)^{n_2} e^{-\frac{1}{2\sigma_1^2} \sum (x_{1i} - \mu_1)^2} e^{-\frac{1}{2\sigma_2^2} \sum (x_{2j} - \mu_2)^2}$$

$$k = \left(\frac{1}{\sigma_1 \sqrt{2\pi}} \right)^{n_1} \left(\frac{1}{\sigma_2 \sqrt{2\pi}} \right)^{n_2}$$

$$L(\mu_1, \mu_2) = k e^{-\left[\frac{1}{2\sigma_1^2} \sum (x_{1i} - \mu_1)^2 + \frac{1}{2\sigma_2^2} \sum (x_{2j} - \mu_2)^2 \right]}$$

Taking ln both sides

$$L(\mu_1, \mu_2) = \ln k - \left[\frac{1}{2\sigma_1^2} \sum (x_{1i} - \mu_1)^2 + \frac{1}{2\sigma_2^2} \sum (x_{2j} - \mu_2)^2 \right] \quad (1)$$

Diff w.r.t μ_1

$$\frac{\partial}{\partial \mu_1} L(\mu_1, \mu_2) = - \left[\frac{2}{2\sigma_1^2} \sum (x_{1i} - \mu_1)(-1) \right]$$

$$\frac{\partial}{\partial \mu_1} L(\mu_1, \mu_2) = \frac{\sum (x_{1i} - \mu_1)}{\sigma_1^2} \dots \dots \dots (2)$$

Put

$$\begin{aligned} \frac{\partial}{\partial \mu_1} L(\mu_1, \mu_2) &= 0 \\ \frac{\sum (x_{1i} - \mu_1)}{\sigma_1^2} &= 0 \\ \sum x_{1i} - n_1 \mu_1 &= 0 \end{aligned}$$

$$\mu_1 = \frac{\sum x_{1i}}{n_1} = \bar{x}_1$$

Diff w.r.t (1) μ_2

$$\frac{\partial}{\partial \mu_2} L(\mu_1, \mu_2) = - \left[\frac{2}{2\sigma_2^2} \sum (x_{1j} - \mu_2)(-1) \right]$$

$$\frac{\partial}{\partial \mu_2} L(\mu_1, \mu_2) = \frac{\sum (x_{1j} - \mu_2)}{\sigma_2^2} \dots \dots \dots (3)$$

Put

$$\frac{\partial}{\partial \mu_2} L(\mu_1, \mu_2) = 0$$

$$\frac{\sum (x_{1j} - \mu_2)}{\sigma_2^2} = 0$$

$$\sum x_{1j} - n_2 \mu_2 = 0$$

$$\mu_2 = \frac{\sum x_{1j}}{n_2} = \bar{x}_2$$

Diff (2) w.r.t μ_1 and (3) w.r.t μ_2

$$\frac{\partial^2}{\partial \mu_1^2} L(\mu_1, \mu_2) = \frac{-n_1 \mu_1}{\sigma_1^2} < 0, \quad \frac{\partial^2}{\partial \mu_2^2} L(\mu_1, \mu_2) = \frac{-n_2 \mu_2}{\sigma_2^2} < 0$$

Hence, MLE for α

$$\hat{\alpha} = \hat{\mu}_1 - \hat{\mu}_2 = \bar{x}_1 - \bar{x}_2$$

In order to find the that how the total sample size will be divided into two population for minimum variance we have to minimize the variance of $\hat{\alpha}$ for the value of n_1 and n_2

$$V(\hat{\alpha}) = V(\bar{x}_1 - \bar{x}_2) = V(\bar{x}_1) + V(\bar{x}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \dots \dots \dots (A)$$

To find the optimal value of n_1 , take $n_2 = n - n_1$ in (A)

$$V(\hat{\alpha}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n - n_1}$$

Diff w.r.t n_1

$$\frac{\partial}{\partial n_1} V(\hat{\alpha}) = \frac{-\sigma_1^2}{n_1^2} + \frac{\sigma_2^2}{(n - n_1)^2} \dots \dots \dots (B)$$

Put

$$\frac{\partial}{\partial n_1} V(\hat{\alpha}) = 0$$

$$\frac{-\sigma_1^2}{n_1^2} + \frac{\sigma_2^2}{(n - n_1)^2} = 0$$

$$\frac{\sigma_1^2}{n_1^2} = \frac{\sigma_2^2}{(n - n_1)^2}$$

$$(n - n_1)^2 \sigma_1^2 = n_1^2 \sigma_2^2$$

Square root

$$(n - n_1) \sigma_1 = n_1 \sigma_2$$

$$n \sigma_1 - n_1 \sigma_1 = n_1 \sigma_2$$

$$n_1 (\sigma_1 + \sigma_2) = n \sigma_1$$

$$n_1 = \frac{n \sigma_1}{(\sigma_1 + \sigma_2)}$$

Diff w.r.t (B) n_1

$$\frac{\partial^2}{\partial n_1^2} V(\hat{\alpha}) = \frac{2\sigma_1^2}{n_1^3} + \frac{2\sigma_2^2}{(n - n_1)^3} = \frac{2\sigma_1^2}{n_1^3} + \frac{2\sigma_2^2}{n_2^3} > 0, \text{ As } n_1 > 0, n_2 > 0$$

Hence, n_1 will be obtained proportional to the total sample size with proportionality constant

$$\frac{\sigma_1}{(\sigma_1 + \sigma_2)}$$

Which is the ratio of the standard deviation of population-1 to the sum of standard deviation of both populations.

To find the optimal value of n_2 , take $n_1 = n - n_2$ in (A)

$$V(\hat{\alpha}) = \frac{\sigma_2^2}{n_2} + \frac{\sigma_1^2}{n - n_2}$$

Diff w.r.t n_2

$$\frac{\partial}{\partial n_2} V(\hat{\alpha}) = \frac{-\sigma_2^2}{n_2^2} + \frac{\sigma_1^2}{(n - n_2)^2} \dots \dots \dots (C)$$

Put

$$\begin{aligned} \frac{\partial}{\partial n_2} V(\hat{\alpha}) &= 0 \\ \frac{-\sigma_2^2}{n_2^2} + \frac{\sigma_1^2}{(n - n_2)^2} &= 0 \end{aligned}$$

$$\frac{\sigma_2^2}{n_2^2} = \frac{\sigma_1^2}{(n - n_2)^2}$$

$$(n - n_2)^2 \sigma_2^2 = n_2^2 \sigma_1^2$$

Square root

$$(n - n_2) \sigma_2 = n_2 \sigma_1$$

$$n \sigma_2 - n_2 \sigma_2 = n_2 \sigma_1$$

$$n_2 (\sigma_1 + \sigma_2) = n \sigma_2$$

$$n_2 = \frac{n \sigma_2}{(\sigma_1 + \sigma_2)}$$

Diff w.r.t (C) n_2

$$\frac{\partial^2}{\partial n_2^2} V(\hat{\alpha}) = \frac{2\sigma_2^2}{n_2^3} + \frac{2\sigma_1^2}{(n - n_2)^3} = \frac{2\sigma_2^2}{n_2^3} + \frac{2\sigma_1^2}{n_1^3} > 0, \text{ As } n_1 > 0, n_2 > 0$$

Hence, n_2 will be obtained proportional to the total sample size with proportionality constant

$$\frac{\sigma_2}{(\sigma_1 + \sigma_2)}$$

Which is the ratio of the standard deviation of population-2 to the sum of standard deviation of both populations.

2.20 MLE of a,b, and c Sample from Four Normal Populations

Question 19: A sample of size n is drawn from four normal populations all of which have same variance σ^2 and means $a+b+c$, $a+b-c$, $a-b+c$ and $a-b-c$. Find the MLE of a, b , and c

Solution:-

Let $X_{ij}, i = 1, 2, 3, 4$ and $j = 1, 2, \dots, n_2$, where i denotes four normal populations and j denotes random sample of size “ n ”.

The likelihood functions is

$$L(a, b, c) = \prod_{j=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x_{1j}-a-b-c)^2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x_{2j}-a-b+c)^2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x_{3j}-a+b-c)^2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x_{4j}-a+b-c)^2}$$

$$L(a, b, c) = \prod_{j=1}^n \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^4 e^{-\frac{1}{2\sigma^2}(x_{1j}-a-b-c)^2} e^{-\frac{1}{2\sigma^2}(x_{2j}-a-b+c)^2} e^{-\frac{1}{2\sigma^2}(x_{3j}-a+b-c)^2} e^{-\frac{1}{2\sigma^2}(x_{4j}-a+b-c)^2}$$

$$L(a, b, c) = \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^{4n} e^{-\frac{1}{2\sigma^2} [\sum (x_{1j} - a - b - c)^2 + \sum (x_{2j} - a - b + c)^2 + \sum (x_{3j} - a + b + c)^2 + \sum (x_{4j} - a + b - c)^2]}$$

Taking ln both sides

$$\ln L(a, b, c) = \ln \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^{4n} - \frac{1}{2\sigma^2} [\sum (x_{1j} - a - b - c)^2 + \sum (x_{2j} - a - b + c)^2 + \sum (x_{3j} - a + b + c)^2 + \sum (x_{4j} - a + b - c)^2] \dots \dots \dots (1)$$

Diff w.r.t "a"

$$\begin{aligned} \frac{\partial}{\partial a} \ln L(a, b, c) &= -\frac{1}{2\sigma^2} [2\sum (x_{1j} - a - b - c)(-1) + 2\sum (x_{2j} - a - b + c)(-1) \\ &\quad + 2\sum (x_{3j} - a + b + c)(-1) + 2\sum (x_{4j} - a + b - c)(-1)] \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial a} \ln L(a, b, c) &= \frac{2}{2\sigma^2} [\sum x_{1j} + \sum x_{2j} + \sum x_{3j} + \sum x_{4j} - n(a + a + a + a) \\ &\quad - n(b + b - b - b) - n(c - c - c + c)] \end{aligned}$$

$$\frac{\partial}{\partial a} \ln L(a, b, c) = \frac{\sum x_{1j} + \sum x_{2j} + \sum x_{3j} + \sum x_{4j} - 4an}{\sigma^2} \dots \dots \dots (1)$$

Put

$$\begin{aligned} \frac{\partial}{\partial a} \ln L(a, b, c) &= 0 \\ \frac{\sum x_{1j} + \sum x_{2j} + \sum x_{3j} + \sum x_{4j} - 4an}{\sigma^2} &= 0 \\ \sum x_{1j} + \sum x_{2j} + \sum x_{3j} + \sum x_{4j} - 4an &= 0 \\ a &= \frac{\sum x_{1j} + \sum x_{2j} + \sum x_{3j} + \sum x_{4j}}{4n} \end{aligned}$$

Diff (1) w.r.t "a"

$$\frac{\partial^2}{\partial a^2} \ln L(a, b, c) = \frac{-4n}{\sigma^2} < 0$$

Hence, MLE of a is

$$\hat{a} = \frac{\sum x_{1j} + \sum x_{2j} + \sum x_{3j} + \sum x_{4j}}{4n} = \bar{X}$$

Diff (1) w.r.t "b"

$$\begin{aligned} \frac{\partial}{\partial b} \ln L(a, b, c) &= -\frac{1}{2\sigma^2} [2\sum (x_{1j} - a - b - c)(-1) + 2\sum (x_{2j} - a - b + c)(-1) \\ &\quad + 2\sum (x_{3j} - a + b + c)(1) + 2\sum (x_{4j} - a + b - c)(1)] \end{aligned}$$

$$\frac{\partial}{\partial b} \ln L(a, b, c)$$

$$= \frac{2}{2\sigma^2} [\sum x_{1j} + \sum x_{2j} - \sum x_{3j} - \sum x_{4j} - n(a + a - a - a) - n(b + b + b + b) - n(c - c + c - c)]$$

$$\frac{\partial}{\partial b} \ln L(a, b, c) = \frac{\sum x_{1j} + \sum x_{2j} - \sum x_{3j} - \sum x_{4j} - 4bn}{\sigma^2} \dots \dots \dots (2)$$

$$\frac{\partial}{\partial b} \ln L(a, b, c) = 0$$

$$\frac{\sum x_{1j} + \sum x_{2j} - \sum x_{3j} - \sum x_{4j} - 4bn}{\sigma^2} = 0$$

$$\sum x_{1j} + \sum x_{2j} - \sum x_{3j} - \sum x_{4j} - 4bn = 0$$

$$b = \frac{\sum x_{1j} + \sum x_{2j} - \sum x_{3j} - \sum x_{4j}}{4n}$$

Diff (2) w.r.t “b

$$\frac{\partial^2}{\partial b^2} \ln L(a, b, c) = \frac{-4n}{\sigma^2} < 0$$

Hence, MLE of a is

$$\hat{b} = \frac{\sum x_{1j} + \sum x_{2j} - \sum x_{3j} - \sum x_{4j}}{4n} = \frac{\bar{X}_1 + \bar{X}_2 - \bar{X}_3 - \bar{X}_4}{4}$$

Diff (1) w.r.t “c”

$$\frac{\partial}{\partial c} \ln L(a, b, c)$$

$$= -\frac{1}{2\sigma^2} [2\sum(x_{1j} - a - b - c)(-1) + 2\sum(x_{2j} - a - b + c)(1) + 2\sum(x_{3j} - a + b + c)(1) + 2\sum(x_{4j} - a + b - c)(-1)]$$

$$\frac{\partial}{\partial c} \ln L(a, b, c)$$

$$= \frac{2}{2\sigma^2} [\sum x_{1j} - \sum x_{2j} + \sum x_{3j} - \sum x_{4j} - n(a - a + a - a) - n(b - b - b + b) - n(c + c + c + c)]$$

$$\frac{\partial}{\partial c} \ln L(a, b, c) = \frac{\sum x_{1j} - \sum x_{2j} + \sum x_{3j} - \sum x_{4j} - 4cn}{\sigma^2} \dots \dots \dots (3)$$

$$\frac{\partial}{\partial c} \ln L(a, b, c) = 0$$

$$\frac{\sum x_{1j} - \sum x_{2j} + \sum x_{3j} - \sum x_{4j} - 4cn}{\sigma^2} = 0$$

$$\sum x_{1j} - \sum x_{2j} + \sum x_{3j} - \sum x_{4j} - 4cn = 0$$

$$c = \frac{\sum x_{1j} - \sum x_{2j} + \sum x_{3j} - \sum x_{4j}}{4n}$$

Diff (3) w.r.t “c

$$\frac{\partial^2}{\partial c^2} \ln L(a, b, c) = \frac{-4n}{\sigma^2} < 0$$

Hence, MLE of c is

$$\hat{c} = \frac{\sum x_{1j} - \sum x_{2j} + \sum x_{3j} - \sum x_{4j}}{4n} = \frac{\bar{X}_1 - \bar{X}_2 + \bar{X}_3 - \bar{X}_4}{4}$$

2.21 MLE of α and μ From Extreme Value Distribution

Question 20: For a sample size of “n” from the extreme value distribution

$$f(x) = \alpha e^{[-\alpha(x-\mu) - e^{-\alpha(x-\mu)}]}$$

Show that the MLE of α and μ are given

$$\frac{1}{\hat{\alpha}} = \bar{x} - \frac{\sum x_i e^{-\alpha(x_i - \mu)}}{n} \quad \text{and} \quad e^{-\hat{\alpha}\hat{\mu}} = \frac{1}{n} \sum e^{-\hat{\alpha}x_i}$$

Solution:

The likelihood function

$$L(\alpha, \mu) = \prod_{i=1}^n \alpha e^{[-\alpha(x_i - \mu) - e^{-\alpha(x_i - \mu)}]}$$

$$L(\alpha, \mu) = \alpha^n e^{[-\alpha \sum (x_i - \mu) - \sum e^{-\alpha(x_i - \mu)}]}$$

Taking Ln both sides

$$\ln L(\alpha, \mu) = n \ln \alpha - \alpha \sum (x_i - \mu) - \sum e^{-\alpha(x_i - \mu)} \dots \dots \dots (1)$$

Diff w.r.t μ

$$\frac{\partial}{\partial \mu} \ln L(\alpha, \mu) = -\alpha \sum (0 - 1) - \sum e^{-\alpha(x_i - \mu)} (\alpha)$$

$$\frac{\partial}{\partial \mu} \ln L(\alpha, \mu) = n\alpha - \alpha \sum e^{-\alpha(x_i - \mu)} \dots \dots \dots (2)$$

Put

$$\frac{\partial}{\partial \mu} \ln L(\alpha, \mu) = 0$$

$$n\alpha - \alpha \sum e^{-\alpha(x_i - \mu)} = 0$$

$$n\alpha = \alpha \sum e^{-\alpha(x_i - \mu)}$$

$$n = \sum e^{-\alpha(x_i - \mu)} \dots \dots \dots (A)$$

$$n = \sum e^{-\alpha x_i} e^{\alpha \mu}$$

$$e^{\alpha \mu} = \frac{n}{\sum e^{-\alpha x_i}}$$

$$e^{-\alpha \mu} = \frac{1}{n} \sum e^{-\alpha x_i}$$

Diff (2) w.r.t μ

$$\frac{\partial^2}{\partial \mu^2} \ln L(\alpha, \mu) = -\alpha^2 \sum e^{-\alpha(x_i - \mu)} < 0$$

Diff w.r.t (1) α

$$\frac{\partial}{\partial \alpha} \ln L(\alpha, \mu) = \frac{n}{\alpha} - \sum (x_i - \mu) + \sum e^{-\alpha(x_i - \mu)} (x_i - \mu) \dots \dots \dots (4)$$

Put

$$\frac{\partial}{\partial \alpha} \ln L(\alpha, \mu) = 0$$

$$\frac{n}{\alpha} - \sum (x_i - \mu) + \sum e^{-\alpha(x_i - \mu)} (x_i - \mu) = 0$$

$$\frac{n}{\alpha} = \sum (x_i - \mu) - \sum e^{-\alpha(x_i - \mu)} (x_i - \mu)$$

$$\frac{n}{\alpha} = \sum x_i - n\mu - \sum x_i \sum e^{-\alpha(x_i - \mu)} + \mu \sum e^{-\alpha(x_i - \mu)}$$

$$\frac{n}{\alpha} = \sum x_i - \sum x_i \sum e^{-\alpha(x_i - \mu)} + \mu (\sum e^{-\alpha(x_i - \mu)} - n) \dots \dots \dots (B)$$

Using (A)

$$\frac{n}{\alpha} = \sum x_i - \sum x_i \sum e^{-\alpha(x_i - \mu)} + \mu(n - n)$$

$$\frac{n}{\alpha} = \sum x_i - \sum x_i \sum e^{-\alpha(x_i - \mu)}$$

$$\frac{1}{\alpha} = \frac{\sum x_i}{n} - \frac{\sum x_i \sum e^{-\alpha(x_i - \mu)}}{n}$$

$$\frac{1}{\alpha} = \bar{x} - \frac{\sum x_i \sum e^{-\alpha(x_i - \mu)}}{n}$$

Diff w.r.t (4) α

$$\frac{\partial^2}{\partial \alpha^2} \ln L(\alpha, \mu) = \frac{-n}{\alpha^2} - \sum e^{-\alpha(x_i - \mu)} (x_i - \mu)^2$$

$$\frac{\partial^2}{\partial \alpha^2} \ln L(\alpha, \mu) = - \left[\frac{n}{\alpha^2} + \sum e^{-\alpha(x_i - \mu)} (x_i - \mu)^2 \right] < 0$$

Hence MLE of α and μ are.

$$\frac{1}{\hat{\alpha}} = \bar{x} - \frac{\sum x_i \sum e^{-\hat{\alpha}(x_i - \hat{\mu})}}{n} \quad \text{and} \quad e^{-\hat{\alpha}\hat{\mu}} = \frac{1}{n} \sum e^{-\hat{\alpha}x_i}$$

2.22 MLE of Parameters of Gamma Distribution

Question 21: Given the random sample of “n” from Gamma Distribution . Find MLE of two parameters.

$$f(x; \alpha, \beta) = \frac{1}{\Gamma\alpha \beta^\alpha} e^{-\frac{x}{\beta}} x^{\alpha-1}, x > 0$$

Solution:

The likelihood function

$$\begin{aligned} L(\alpha, \beta) &= \prod_{i=1}^n \frac{1}{\Gamma\alpha \beta^\alpha} e^{-\frac{x_i}{\beta}} x_i^{\alpha-1} \\ L(\alpha, \beta) &= \prod_{i=1}^n \frac{1}{(\alpha-1)! \beta^\alpha} e^{-\frac{x_i}{\beta}} x_i^{\alpha-1} \\ L(\alpha, \beta) &= \prod_{i=1}^n \frac{\alpha}{\alpha(\alpha-1)! \beta^\alpha} e^{-\frac{x_i}{\beta}} x_i^{\alpha-1} \\ L(\alpha, \beta) &= \prod_{i=1}^n \frac{\alpha}{\alpha! \beta^\alpha} e^{-\frac{x_i}{\beta}} x_i^{\alpha-1} \\ L(\alpha, \beta) &= \prod_{i=1}^n x_i^{\alpha-1} \frac{\alpha^n}{(\alpha! \beta^\alpha)^n} e^{-\frac{\sum x_i}{\beta}} \end{aligned}$$

Taking ln both sides

$$\ln L(\alpha, \beta) = (\alpha-1) \ln \prod_{i=1}^n x_i + n \ln \alpha - n \ln \alpha! - n \alpha \ln \beta - \frac{\sum x_i}{\beta} \dots \dots \dots (1)$$

Diff (1) w.r.t β

$$\frac{\partial}{\partial \beta} \ln L(\alpha, \beta) = -\frac{n\alpha}{\beta} + \frac{\sum x_i}{\beta^2} \dots \dots \dots (2)$$

Put

$$\frac{\partial}{\partial \beta} \ln L(\alpha, \beta) = 0$$

$$\frac{n\alpha}{\beta} = \frac{\sum x_i}{\beta^2}$$

$$\beta = \frac{\sum x_i}{n\alpha} = \frac{\bar{x}}{\alpha}$$

Diff (2) w.r.t β

$$\frac{\partial^2}{\partial \beta^2} \ln L(\alpha, \beta) = \frac{n\alpha}{\beta^2} - \frac{2\sum x_i}{\beta^3}$$

$$\frac{\partial^2}{\partial \beta^2} \ln L(\alpha, \beta) = \frac{n\alpha\beta - 2\sum x_i}{\beta^3}$$

Using

$$n\alpha\beta = \sum x_i$$

$$\frac{\partial^2}{\partial \beta^2} \ln L(\alpha, \beta) = \frac{\sum x_i - 2\sum x_i}{\beta^3} = \frac{-\sum x_i}{\beta^3} < 0$$

Hence, MLE for β

$$\beta = \frac{\sum x_i}{n\alpha} = \frac{\bar{x}}{\alpha}$$

For MLE of α

Using Starling approximation

$$n! = \sqrt{2\pi} n^{n+\frac{1}{2}} e^{-n}$$

for $\alpha!$ in (1)

$$\ln L(\alpha, \beta) = (\alpha - 1)\sum \ln x_i + n \ln \alpha - n \ln(\sqrt{2\pi} \alpha^{\alpha+\frac{1}{2}} e^{-\alpha}) - n\alpha \ln \beta - \frac{\sum x_i}{\beta}$$

$$\ln L(\alpha, \beta) = (\alpha - 1)\sum \ln x_i + n \ln \alpha - n \ln(\sqrt{2\pi}) - n\left(\alpha + \frac{1}{2}\right) \ln \alpha + n\alpha - n\alpha \ln \beta - \frac{\sum x_i}{\beta}$$

$$\ln L(\alpha, \beta) = (\alpha - 1)\sum \ln x_i + n\left(1 - \alpha - \frac{1}{2}\right) \ln \alpha - n \ln(\sqrt{2\pi}) + n\alpha - n\alpha \ln \beta - \frac{\sum x_i}{\beta}$$

$$\ln L(\alpha, \beta) = (\alpha - 1)\sum \ln x_i + n\left(\frac{1}{2} - \alpha\right) \ln \alpha - n \ln(\sqrt{2\pi}) + n\alpha - n\alpha \ln \beta - \frac{\sum x_i}{\beta}$$

Diff w.r.t α

$$\frac{\partial}{\partial \alpha} \ln L(\alpha, \beta) = \sum \ln x_i + \frac{n\left(\frac{1}{2} - \alpha\right)}{\alpha} - n \ln \alpha + n - n \ln \beta$$

$$\frac{\partial}{\partial \alpha} \ln L(\alpha, \beta) = \sum \ln x_i + \frac{n}{2\alpha} - n - n \ln \alpha + n - n \ln \beta$$

$$\frac{\partial}{\partial \alpha} \ln L(\alpha, \beta) = \sum \ln x_i + \frac{n}{2\alpha} - n \ln \alpha - n \ln \beta \dots \dots \dots (3)$$

Put

$$\frac{\partial}{\partial \alpha} \ln L(\alpha, \beta) = 0$$

$$\sum \ln x_i + \frac{n}{2\alpha} - n \ln \alpha - n \ln \beta = 0$$

Using $\beta = \frac{\bar{x}}{\alpha}$

$$\sum \ln x_i + \frac{n}{2\alpha} - n \ln \alpha - n \ln \left(\frac{\bar{x}}{\alpha}\right) = 0$$

$$\sum \ln x_i + \frac{n}{2\alpha} - n \ln \alpha - n \ln x + n \ln \alpha = 0$$

$$\sum \ln x_i + \frac{n}{2\alpha} - n \ln \bar{x} = 0$$

$$\frac{n}{2\alpha} = n \ln \bar{x} - \sum \ln x_i$$

$$\alpha = \frac{n}{2(n \ln \bar{x} - \sum \ln x_i)}$$

Diff (3) w.r.t α

$$\frac{\partial^2}{\partial \alpha^2} \ln L(\alpha, \beta) = \frac{-n}{2\alpha^2} - \frac{n}{\alpha} = -\left(\frac{n}{2\alpha^2} + \frac{n}{\alpha}\right) < 0$$

Hence, MLE for both parameters of Gamma Distribution are

$$\alpha = \frac{n}{2(n \ln \bar{x} - \sum \ln x_i)} \text{ and } \beta = \frac{\sum x_i}{n\alpha} = \frac{\bar{x}}{\alpha} = \frac{2\bar{x}}{n} (n \ln \bar{x} - \sum \ln x_i)$$

2.23 MLE of Parameters θ and p From Gamma like Distribution. Also Variance-Covariance Matrix

Question 22: Find the MLE of parameters θ and p (p being large) of the distribution with the following p.d.f

$$f(x; \theta, p) = \frac{1}{\Gamma p} \left(\frac{p}{\theta}\right)^p x^{p-1} e^{-\frac{xp}{\theta}}$$

Also find the variance-covariance Matrix

Solution:

$$L(\theta, p) = \prod_{i=1}^n \frac{1}{\Gamma p} \left(\frac{p}{\theta}\right)^p x_i^{p-1} e^{-\frac{x_i p}{\theta}}$$

$$L(\theta, p) = \prod_{i=1}^n x_i^{p-1} \left(\frac{1}{\Gamma p}\right)^n \left(\frac{p}{\theta}\right)^{np} e^{-\frac{p \sum x_i}{\theta}}$$

Taking ln both sides

$$\ln L(\theta, p) = (p-1) \ln \prod_{i=1}^n x_i + \ln \left(\frac{1}{\Gamma p}\right)^n + \ln \left(\frac{p}{\theta}\right)^{np} + \ln e^{-\frac{p \sum x_i}{\theta}}$$

$$\ln L(\theta, p) = (p-1) \sum \ln x_i - n \ln(\Gamma p) + n p \ln p - n p \ln \theta - \frac{p \sum x_i}{\theta} \dots \dots (1)$$

To get MLE of θ Diff wr.t θ

$$\frac{\partial}{\partial \theta} \ln L(\theta, p) = -\frac{np}{\theta} + \frac{p \sum x_i}{\theta^2} \dots \dots \dots (2)$$

Put

$$\begin{aligned}\frac{\partial}{\partial} \ln L(\theta, p) &= 0 \\ -\frac{np}{\theta} + \frac{p \sum x_i}{\theta^2} &= 0\end{aligned}$$

$$\frac{np}{\theta} = \frac{p \sum x_i}{\theta^2}$$

$$\theta = \frac{\sum x_i}{n} = \bar{x}$$

Diff (2) w.r.t to θ

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta, p) = \frac{np}{\theta^2} - \frac{2p \sum x_i}{\theta^3}$$

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta, p) = \frac{np\theta - 2p \sum x_i}{\theta^3}$$

Using $np\theta = p \sum x_i$

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta, p) = \frac{p \sum x_i - 2p \sum x_i}{\theta^3}$$

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta, p) = \frac{-p \sum x_i}{\theta^3} < 0$$

So, MLE of θ

$$\hat{\theta} = \frac{\sum x_i}{n} = \bar{x}$$

For MLE of P

$$\ln L(\theta, p) = (p-1) \sum \ln x_i - n \ln(p-1)! + n \ln p - n \ln \theta - \frac{p \sum x_i}{\theta}$$

$$\ln L(\theta, p) = (p-1) \sum \ln x_i - n \ln \left[\frac{p(p-1)!}{p} \right] + n \ln p - n \ln \theta - \frac{p \sum x_i}{\theta}$$

$$\ln L(\theta, p) = (p-1) \sum \ln x_i - n \ln \left[\frac{p!}{p} \right] + n \ln p - n \ln \theta - \frac{p \sum x_i}{\theta}$$

$$\ln L(\theta, p) = (p-1) \sum \ln x_i - n \ln p! + n \ln p + n \ln p - n \ln \theta - \frac{p \sum x_i}{\theta}$$

Using Starling approximation

$$n! = \sqrt{2\pi} n^{n+\frac{1}{2}} e^{-n}$$

for $p!$ in (1)

$$\ln L(\theta, p) = (p-1) \sum \ln x_i - n \ln \left[\sqrt{2\pi} p^{p+\frac{1}{2}} e^{-p} \right] + n \ln p + n \ln p - n \ln \theta - \frac{p \sum x_i}{\theta}$$

$$\ln L(\theta, p) = (p-1)\sum \ln x_i - n \ln \sqrt{2\pi} - n \left(p + \frac{1}{2} \right) \ln p + np + n \ln p + n p \ln p - n p \ln \theta - \frac{p \sum x_i}{\theta}$$

$$\ln L(\theta, p) = (p-1)\sum \ln x_i - n \ln \sqrt{2\pi} - n \left(p + \frac{1}{2} - p \right) \ln p + n(p + \ln p - p \ln \theta) - \frac{p \sum x_i}{\theta}$$

$$\ln L(\theta, p) = (p-1)\sum \ln x_i - n \ln \sqrt{2\pi} - \frac{n}{2} \ln p + n(p + \ln p - p \ln \theta) - \frac{p \sum x_i}{\theta}$$

Diff w.r.t p

$$\frac{\partial}{\partial p} \ln L(\theta, p) = \sum \ln x_i - \frac{n}{2p} + n \left(1 + \frac{1}{p} - \ln \theta \right) - \frac{\sum x_i}{\theta}$$

$$\frac{\partial}{\partial p} \ln L(\theta, p) = \sum \ln x_i - \frac{n}{2p} + \frac{n}{p} + n - n \ln \theta - \frac{\sum x_i}{\theta}$$

$$\frac{\partial}{\partial p} \ln L(\theta, p) = \sum \ln x_i + \frac{n}{2p} + n - n \ln \theta - \frac{\sum x_i}{\theta} \dots \dots \dots (3)$$

Put

$$\frac{\partial}{\partial p} \ln L(\theta, p) = 0$$

$$\sum \ln x_i + \frac{n}{2p} + n - n \ln \theta - \frac{\sum x_i}{\theta} = 0$$

$$\frac{n}{2p} = n \ln \theta + \frac{\sum x_i}{\theta} - n - \sum \ln x_i$$

Put $\sum x_i = n\theta$

$$\frac{n}{2p} = n \ln \theta + \frac{n\theta}{\theta} - n - \sum \ln x_i$$

$$\frac{n}{2p} = n \ln \theta - \sum \ln x_i$$

$$p = \frac{1}{2(n \ln \theta - \sum \ln x_i)}$$

Diff (3) w.r.t p

$$\frac{\partial^2}{\partial p^2} \ln L(\theta, p) = \frac{-n}{2p^2} < 0$$

Now, for Variance-Covariance Matrix

$$E \left[\frac{\partial^2}{\partial \theta^2} \ln L(\theta, p) \right] = \frac{-p \sum E(x_i)}{\theta^3}$$

$$E(x_i) = \int_0^{\infty} x f(x) dx$$

$$E(x_i) = \int_0^{\infty} x \cdot x^{p-1} \frac{1}{\Gamma p} \left(\frac{p}{\theta}\right)^p e^{-\frac{xp}{\theta}} dx$$

$$E(x_i) = \frac{1}{\Gamma p} \left(\frac{p}{\theta}\right)^p \int_0^{\infty} x^p e^{-\frac{xp}{\theta}} dx$$

Put

$$t = \frac{xp}{\theta}, \quad \frac{t\theta}{p} = x$$

$$\frac{\theta dt}{p} = dx$$

$$E(x_i) = \frac{1}{\Gamma p} \left(\frac{p}{\theta}\right)^p \int_0^{\infty} \left(\frac{t\theta}{p}\right)^p e^{-t} \frac{\theta dt}{p}$$

$$E(x_i) = \frac{1}{\Gamma p} \left(\frac{p}{\theta}\right)^p \left(\frac{\theta}{p}\right)^p \left(\frac{\theta}{p}\right) \int_0^{\infty} t^p e^{-t} dt$$

$$E(x_i) = \frac{1}{\Gamma p} \left(\frac{\theta}{p}\right) \int_0^{\infty} t^{p+1-1} e^{-t} dt$$

$$E(x_i) = \frac{1}{\Gamma p} \left(\frac{\theta}{p}\right) \Gamma(p+1)$$

$$E(x_i) = \frac{\theta p!}{p(p-1)!} = \frac{\theta p!}{p!} = \theta$$

So,

$$E \left[\frac{\partial^2}{\partial \theta^2} \ln L(\theta, p) \right] = \frac{-p \sum \theta}{\theta^3} = -\frac{np}{\theta^2}$$

And

$$E \left[\frac{\partial^2}{\partial p^2} \ln L(\theta, p) \right] = E \left[\frac{-n}{2p^2} \right] = \frac{-n}{2p^2}$$

$$E \left[\frac{\partial^2}{\partial \theta \partial p} \ln L(\theta, p) \right] = E \left[-\frac{\partial}{\partial p} \frac{np}{\theta} + \frac{\partial}{\partial p} \frac{p \sum x_i}{\theta^2} \right] = E \left[-\frac{n}{\theta} + \frac{\sum x_i}{\theta^2} \right] = E \left[-\frac{n}{\theta} + \frac{n\theta}{\theta^2} \right] = 0$$

$$E \left[\frac{\partial^2}{\partial p \partial \theta} \ln L(\theta, p) \right] = E \left[\frac{\partial}{\partial p} \left\{ \sum \ln x_i + \frac{n}{2p} + n - n \ln \theta - \frac{\sum x_i}{\theta} \right\} \right] = E \left[-\frac{n}{\theta} + \frac{\sum x_i}{\theta^2} \right]$$

$$= E \left[-\frac{n}{\theta} + \frac{n\theta}{\theta^2} \right] = 0$$

$$V(\theta, p) = - \begin{bmatrix} E \left[\frac{\partial^2}{\partial \theta^2} \ln L(\theta, p) \right] & E \left[\frac{\partial^2}{\partial \theta \partial p} \ln L(\theta, p) \right] \\ E \left[\frac{\partial^2}{\partial p \partial \theta} \ln L(\theta, p) \right] & E \left[\frac{\partial^2}{\partial p^2} \ln L(\theta, p) \right] \end{bmatrix}^{-1}$$

$$V(\theta, p) = - \begin{bmatrix} -\frac{np}{\theta^2} & 0 \\ 0 & -\frac{n}{2p^2} \end{bmatrix}^{-1}$$

$$V(\theta, p) = \frac{1}{\frac{np}{\theta^2} \cdot \frac{n}{2p^2}} \begin{bmatrix} \frac{n}{2p^2} & 0 \\ 0 & \frac{np}{\theta^2} \end{bmatrix}$$

$$V(\theta, p) = \frac{2p\theta^2}{n^2} \begin{bmatrix} \frac{n}{2p^2} & 0 \\ 0 & \frac{np}{\theta^2} \end{bmatrix}$$

$$V(\theta, p) = \begin{bmatrix} \frac{\theta^2}{np} & 0 \\ 0 & \frac{2p^2}{n} \end{bmatrix}$$

Question 22: If X is $N(\mu, \sigma^2)$ for a random sample of size “n”. Find the MLE of the point A such that $P(X > A) = 0.05$. Also prove that the minimum variance unbiased estimator is

$$\bar{x} + \left[\frac{1.645}{\frac{1}{2^2}} \cdot \frac{\Gamma\left(\frac{n-1}{2}\right) \sqrt{n}}{\Gamma\left(\frac{n}{2}\right)} \right] E(\hat{\sigma})$$

Solution:-

As

$$P(X > A) = 0.05$$

$$\int_A^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} dx = 0.05$$

Let

$$z = \frac{x - \mu}{\sigma}, \Rightarrow x = \mu + z\sigma$$

$$dx = \sigma dz$$

$$\text{when } x = A, \quad z = \frac{A - \mu}{\sigma}, \text{ when } x = \infty, z = \infty$$

$$\int_{\frac{A-\mu}{\sigma}}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-z^2} \sigma dz = 0.05$$

$$\int_{\frac{A-\mu}{\sigma}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2} dz = 0.05$$

$$F(\infty) - F\left(\frac{A - \mu}{\sigma}\right) = 0.05$$

$$1 - F\left(\frac{A - \mu}{\sigma}\right) = 0.05$$

$$F\left(\frac{A - \mu}{\sigma}\right) = 0.95$$

$$\frac{A - \mu}{\sigma} = F^{-1}(0.95) = 1.645$$

$$A = \mu + 1.645\sigma \dots \dots \dots (A)$$

As We have MLE of

$$\hat{\mu} = \bar{x}, \quad \text{and} \quad \hat{\sigma} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

So,

$$A = \bar{x} + 1.645 \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} \dots \dots \dots (1)$$

As we have

$$y = \frac{\sum (x_i - \bar{x})^2}{\sigma^2} = y\sigma^2 \sim \chi_{n-1}^2$$

So,

$$\hat{\sigma} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} = \frac{y^{\frac{1}{2}}\sigma}{\sqrt{n}}$$

Since, $y \sim \chi_{n-1}^2$, its p.d.f will be

$$f(y) = \frac{y^{\frac{n-1}{2}-1} e^{-\frac{y}{2}}}{\Gamma\left(\frac{n-1}{2}\right) 2^{\frac{n-1}{2}}}$$

$$E\left(y^{\frac{1}{2}}\right) = \int_0^{\infty} y^{\frac{1}{2}} \frac{y^{\frac{n-1}{2}-1} e^{-\frac{y}{2}}}{\Gamma\left(\frac{n-1}{2}\right) 2^{\frac{n-1}{2}}} dy$$

$$E\left(y^{\frac{1}{2}}\right) = \frac{1}{\Gamma\left(\frac{n-1}{2}\right) 2^{\frac{n-1}{2}}} \int_0^{\infty} y^{\frac{1}{2}+\frac{n-1}{2}-1} e^{-\frac{y}{2}} dy$$

$$E\left(y^{\frac{1}{2}}\right) = \frac{1}{\Gamma\left(\frac{n-1}{2}\right) 2^{\frac{n-1}{2}}} \int_0^{\infty} y^{\frac{n}{2}-1} e^{-\frac{y}{2}} dy$$

$$E\left(y^{\frac{1}{2}}\right) = \frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n-1}{2}\right) 2^{\frac{n-1}{2}}}$$

$$E(\hat{\sigma}) = \frac{E(y^{\frac{1}{2}})\sigma}{\sqrt{n}} = \frac{\sigma}{\sqrt{n}} \frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n-1}{2}\right) 2^{\frac{n-1}{2}}}$$

Solving for σ

$$\sigma = \frac{\sqrt{n} \Gamma\left(\frac{n-1}{2}\right) 2^{\frac{n-1}{2}}}{\Gamma\left(\frac{n}{2}\right)} E(\hat{\sigma})$$

Putting in A.

$$xA = \bar{x} + 1.645 \frac{\sqrt{n} \Gamma\left(\frac{n-1}{2}\right) 2^{\frac{n-1}{2}}}{\Gamma\left(\frac{n}{2}\right)} E(\hat{\sigma})$$

2.25 General Form of Distribution for which mean (Ungroup) is MLE

Question24 : what is the most general form of the distribution for which the mean is the MLE of θ .

Solution;

$$\hat{\theta} = \frac{\sum x_i}{n}$$

Take $\hat{\theta}$ as θ

$$n\theta = \sum x_i$$

$$n\theta - \sum x_i = 0$$

$$\frac{\partial \ln l}{\partial \theta} = n\theta - \sum x_i$$

Taking single observation

$$\frac{\partial \ln l}{\partial \theta} = (\theta - x)$$

$$\frac{\partial \ln f(x)}{\partial \theta} = \lambda_1(\theta - x)$$

$$\ln f(x) = \int \frac{\partial^2 x}{\partial \theta^2} (\theta - x) d\theta$$

$$\ln f(x) = (\theta - x) \int \frac{\partial^2 x}{\partial \theta^2} d\theta - \int \int \frac{\partial^2 x}{\partial \theta^2} d\theta (1 - 0) d\theta$$

$$\ln f(x) = (\theta - x) \frac{\partial x}{\partial \theta} - \int \frac{\partial x}{\partial \theta} d\theta$$

$$\ln f(x) = (\theta - x) \frac{\partial x}{\partial \theta} - x + B(x)$$

$$f(x) = \exp \left\{ (\theta - x) \frac{\partial x}{\partial \theta} - x + B(x) \right\}$$

Where $B(x)$ is constant of integration.

2.26 General Form of Distribution for which Geometric mean (Ungroup) is MLE

Question25 : what is the most general form of the distribution for which the geometric mean is the MLE of θ .

Solution;

$$\hat{\theta} = (x_1 \cdot x_2 \dots \dots x_n)^{\frac{1}{n}}$$

Take $\hat{\theta}$ as θ

$$\theta = (x_1 \cdot x_2 \dots \dots x_n)^{\frac{1}{n}} = \prod x_i^{\frac{1}{n}}$$

$$\ln \theta = \ln \prod x_i^{\frac{1}{n}} = \frac{1}{n} \sum \ln x_i$$

$$n \ln \theta = \sum \ln x_i$$

$$\sum \ln x_i - n \ln \theta = 0$$

$$\frac{\partial \ln L}{\partial \theta} = \sum \ln x_i - n \ln \theta$$

Taking single observation

$$\frac{\partial \ln f(x)}{\partial \theta} = \lambda_1 (\ln x - \ln \theta)$$

$$\begin{aligned}
 \ln f(x) &= \int \frac{\partial^2 x}{\partial \theta^2} (\ln x - \ln \theta) d\theta \\
 \ln f(x) &= (\ln x - \ln \theta) \int \frac{\partial^2 x}{\partial \theta^2} d\theta - \int \int \frac{\partial^2 x}{\partial \theta^2} d\theta \left(0 - \frac{1}{\theta}\right) d\theta \\
 \ln f(x) &= (\ln x - \ln \theta) \frac{\partial x}{\partial \theta} + \int \left(\frac{1}{\theta}\right) \left(\frac{\partial x}{\partial \theta}\right) d\theta \\
 \ln f(x) &= (\ln x - \ln \theta) \frac{\partial x}{\partial \theta} + \left(\frac{1}{\theta}\right) \int \frac{\partial x}{\partial \theta} d\theta - \int \int \frac{\partial^2 x}{\partial \theta^2} d\theta \left(\frac{-1}{\theta^2}\right) d\theta \\
 \ln f(x) &= (\ln x - \ln \theta) \frac{\partial x}{\partial \theta} + \left(\frac{1}{\theta}\right) x + \int x \left(\frac{1}{\theta^2}\right) d\theta \\
 \ln f(x) &= (\ln x - \ln \theta) \frac{\partial x}{\partial \theta} + \left(\frac{1}{\theta}\right) x + x \left(\frac{\theta^{-1}}{-1}\right) \\
 \ln f(x) &= (\ln x - \ln \theta) \frac{\partial x}{\partial \theta} + \left(\frac{x}{\theta}\right) - \left(\frac{x}{\theta}\right) + B(x) \\
 \ln f(x) &= (\ln x - \ln \theta) \frac{\partial x}{\partial \theta} + B(x) \\
 f(x) &= \exp \left\{ (\ln x - \ln \theta) \frac{\partial x}{\partial \theta} + B(x) \right\}
 \end{aligned}$$

Where $B(x)$ is constant of integration

2.27 General Form of Distribution for which mean (Group) is MLE

Question 26 : What is the most general form of the distribution for which the

$$\text{mean} = \frac{\sum fx}{\sum f}$$

is the MLE of θ .

Solution;

$$\hat{\theta} = \frac{\sum fx}{\sum f}$$

Take $\hat{\theta}$ as θ

$$\theta \sum f = \sum f x_i$$

$$\theta \sum f - \sum f x_i = 0$$

$$\sum f (\theta - x_i) = 0$$

$$\frac{\partial \ln l}{\partial \theta} = \sum f(\theta - x_i)$$

Taking single observation

$$\frac{\partial \ln l}{\partial \theta} = (\theta - x)$$

$$\frac{\partial \ln f(x)}{\partial \theta} = \lambda_1(\theta - x)$$

$$\ln f(x) = \int \frac{\partial^2 x}{\partial \theta^2} (\theta - x) d\theta$$

$$\ln f(x) = (\theta - x) \int \frac{\partial^2 x}{\partial \theta^2} d\theta - \int \int \frac{\partial^2 x}{\partial \theta^2} d\theta (1 - 0) d\theta$$

$$\ln f(x) = (\theta - x) \frac{\partial x}{\partial \theta} - \int \frac{\partial x}{\partial \theta} d\theta$$

$$\ln f(x) = (\theta - x) \frac{\partial x}{\partial \theta} - x + B(x)$$

$$f(x) = \exp \left\{ (\theta - x) \frac{\partial x}{\partial \theta} - x + B(x) \right\}$$

Where $B(x)$ is constant of integration.

2.28 General Form of Distribution for which Harmonic mean (Ungroup) is MLE

Question27 : What is the most general form of the distribution for which the harmonic mean is MLE.

$$\hat{\theta} = \frac{n}{\sum \left(\frac{1}{x_i} \right)}$$

Take $\hat{\theta}$ as θ

$$\theta \sum \left(\frac{1}{x_i} \right) = n$$

$$\theta \sum \left(\frac{1}{x_i} \right) - n = 0$$

$$\frac{\partial \ln l}{\partial \theta} = \theta \sum \left(\frac{1}{x_i} \right) - n$$

Taking single observation

$$\frac{\partial \ln l}{\partial \theta} = \frac{\theta}{x} - 1$$

$$\frac{\partial \ln f(x)}{\partial \theta} = \lambda_1 \left(\frac{\theta}{x} - 1 \right)$$

$$\ln f(x) = \int \frac{\partial^2 x}{\partial \theta^2} \left(\frac{\theta}{x} - 1 \right) d\theta$$

$$\ln f(x) = \left(\frac{\theta}{x} - 1 \right) \int \frac{\partial^2 x}{\partial \theta^2} d\theta - \int \int \frac{\partial^2 x}{\partial \theta^2} d\theta \left(\frac{1}{x} - 0 \right) d\theta$$

$$\ln f(x) = \left(\frac{\theta}{x} - 1 \right) \frac{\partial x}{\partial \theta} - \frac{1}{x} \int \frac{\partial x}{\partial \theta} d\theta$$

$$\ln f(x) = \left(\frac{\theta}{x} - 1 \right) \frac{\partial x}{\partial \theta} - \frac{1}{x} \cdot x + B(x)$$

$$\ln f(x) = \left(\frac{\theta}{x} - 1 \right) \frac{\partial x}{\partial \theta} - 1 + B(x)$$

$$f(x) = \exp \left\{ \left(\frac{\theta}{x} - 1 \right) \frac{\partial x}{\partial \theta} - 1 + B(x) \right\}$$

Where $B(x)$ is constant of integration.

2.29 General Form of Distribution for which Harmonic mean (Group) is MLE

Question 28 : What is the most general form of the distribution for which

$$H.M = \frac{\sum f}{\sum \left(\frac{f}{x} \right)}$$

the harmonic mean is MLE.

$$\hat{\theta} = \frac{\sum f}{\sum \left(\frac{f}{x} \right)}$$

Take $\hat{\theta}$ as θ

$$\theta \sum \left(\frac{f}{x_i} \right) = \sum f$$

$$\theta \sum \left(\frac{f}{x_i} \right) - \sum f = 0$$

$$\frac{\partial \ln l}{\partial \theta} = \theta \sum \left(\frac{f}{x_i} \right) - \sum f$$

Taking single observation

$$\frac{\partial \ln l}{\partial \theta} = \frac{\theta}{x} - 1$$

$$\frac{\partial \ln f(x)}{\partial \theta} = \lambda_1 \left(\frac{\theta}{x} - 1 \right)$$

$$\ln f(x) = \int \frac{\partial^2 x}{\partial \theta^2} \left(\frac{\theta}{x} - 1 \right) d\theta$$

$$\ln f(x) = \left(\frac{\theta}{x} - 1 \right) \int \frac{\partial^2 x}{\partial \theta^2} d\theta - \int \int \frac{\partial^2 x}{\partial \theta^2} d\theta \left(\frac{1}{x} - 0 \right) d\theta$$

$$\ln f(x) = \left(\frac{\theta}{x} - 1 \right) \frac{\partial x}{\partial \theta} - \frac{1}{x} \int \frac{\partial x}{\partial \theta} d\theta$$

$$\ln f(x) = \left(\frac{\theta}{x} - 1 \right) \frac{\partial x}{\partial \theta} - \frac{1}{x} \cdot x + B(x)$$

$$\ln f(x) = \left(\frac{\theta}{x} - 1 \right) \frac{\partial x}{\partial \theta} - 1 + B(x)$$

$$f(x) = \exp \left\{ \left(\frac{\theta}{x} - 1 \right) \frac{\partial x}{\partial \theta} - 1 + B(x) \right\}$$

Where $B(x)$ is constant of integration.

2.30 Appropriate Estimator for Parameter of Cauchy Distribution

Question 29: Obtain the appropriate estimator for parameter θ of the Cauchy distribution

$$f(x) = \frac{1}{\pi \{1 + (x - \theta)^2\}}$$

Solution:

The likelihood function

$$L(\theta) = \prod_{i=1}^n \frac{1}{\pi \{1 + (x_i - \theta)^2\}}$$

$$L(\theta) = \pi^{-n} \prod_{i=1}^n \{1 + (x_i - \theta)^2\}^{-1}$$

Taking ln both sides

$$\ln L(\theta) = -n \ln \pi - \ln \prod_{i=1}^n [1 + (x_i - \theta)^2]$$

$$\ln L(\theta) = -n \ln \pi - \sum \ln [1 + (x_i - \theta)^2]$$

Diff w.r.t θ

$$\frac{\partial}{\partial \theta} \ln L(\theta) = -\sum \frac{1}{[1 + (x_i - \theta)^2]} [0 + 2(x_i - \theta)(0 - 1)]$$

$$\frac{\partial}{\partial \theta} \ln L(\theta) = \sum \frac{2(x_i - \theta)}{[1 + (x_i - \theta)^2]}$$

put

$$\frac{\partial}{\partial \theta} \ln L(\theta) = 0$$

$$\sum \frac{2(x_i - \theta)}{[1 + (x_i - \theta)^2]} = 0$$

$$2 \left[\frac{(x_1 - \theta)}{[1 + (x_1 - \theta)^2]} + \frac{(x_2 - \theta)}{[1 + (x_2 - \theta)^2]} + \dots + \frac{(x_n - \theta)}{[1 + (x_n - \theta)^2]} \right] = 0$$

It is very difficult to solve the above polynomial.

Now, we will use appropriate calculation of MLE for this purpose we use

$$t = t_1 + V(t) \left(\frac{\partial L}{\partial \theta} \right)_{t_1}$$

Where “t” is MLE of θ and “ t_1 ” median of Cauchy distribution which is less efficient than t.

Now we have to find

$$V(t) = \frac{1}{-E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right]} = \frac{1}{-nE \left[\frac{\partial^2 \ln f(x)}{\partial \theta^2} \right]}$$

Now,

$$\ln f(x) = -\ln \pi - \ln [1 + (x - \theta)^2]$$

Diff w.r.t θ

$$\frac{\partial}{\partial \theta} \ln f(x) = -\frac{1}{1 + (x - \theta)^2} [0 + 2(x - \theta)(-1)] = \frac{2(x - \theta)}{1 + (x - \theta)^2}$$

Diff again w.r.t θ

$$\frac{\partial^2}{\partial \theta^2} \ln f(x) = \frac{[1 + (x - \theta)^2](2)(0 - 1) - 2(x - \theta)[0 + 2(x - \theta)(-1)]}{[1 + (x - \theta)^2]^2}$$

$$\frac{\partial^2}{\partial \theta^2} \ln f(x) = \frac{2[-1 - (x - \theta)^2 + 2(x - \theta)^2]}{[1 + (x - \theta)^2]^2} = \frac{2[(x - \theta)^2 - 1]}{[1 + (x - \theta)^2]^2}$$

Now finding expectation

$$E \left[\frac{\partial^2}{\partial \theta^2} \ln f(x) \right] = \int_{-\infty}^{\infty} \frac{2[(x - \theta)^2 - 1]}{[1 + (x - \theta)^2]^2} \frac{1}{\pi [1 + (x - \theta)^2]} dx$$

$$E \left[\frac{\partial^2}{\partial \theta^2} \ln f(x) \right] = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{[(x - \theta)^2 - 1]}{[1 + (x - \theta)^2]^3} dx$$

Let

$$x - \theta = y, \quad dx = dy$$

$$E \left[\frac{\partial^2}{\partial \theta^2} \ln f(x) \right] = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{[y^2 - 1]}{[1 + y^2]^3} dy$$

As it is even function

$$E \left[\frac{\partial^2}{\partial \theta^2} \ln f(x) \right] = \frac{4}{\pi} \int_0^{\infty} \frac{[y^2 - 1]}{[1 + y^2]^3} dy$$

Let

$$y^2 = z, \Rightarrow y = \sqrt{z}, \text{ and } dy = \frac{1}{2\sqrt{z}} dz$$

$$E \left[\frac{\partial^2}{\partial \theta^2} \ln f(x) \right] = \frac{4}{\pi} \int_0^{\infty} \frac{[z - 1]}{[1 + z]^3} \frac{1}{2\sqrt{z}} dz = \frac{2}{\pi} \int_0^{\infty} \frac{[z - 1](z)^{-\frac{1}{2}}}{[1 + z]^3} dz$$

$$E \left[\frac{\partial^2}{\partial \theta^2} \ln f(x) \right] = \frac{2}{\pi} \int_0^{\infty} \frac{z^{\frac{1}{2}} - z^{-\frac{1}{2}}}{[1 + z]^3} dz = \frac{2}{\pi} \left[\int_0^{\infty} \frac{z^{\frac{1}{2}}}{[1 + z]^3} dz - \int_0^{\infty} \frac{z^{-\frac{1}{2}}}{[1 + z]^3} dz \right]$$

Using beta function

$$\beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$$

$$E \left[\frac{\partial^2}{\partial \theta^2} \ln f(x) \right] = \frac{2}{\pi} \left[\int_0^{\infty} \frac{z^{\frac{3}{2}-1}}{[1+z]^{\frac{3}{2}+\frac{3}{2}}} dz - \int_0^{\infty} \frac{z^{\frac{1}{2}-1}}{[1+z]^{\frac{5}{2}+\frac{1}{2}}} dz \right]$$

$$E \left[\frac{\partial^2}{\partial \theta^2} \ln f(x) \right] = \frac{2}{\pi} \left[\beta\left(\frac{3}{2}, \frac{3}{2}\right) - \beta\left(\frac{1}{2}, \frac{5}{2}\right) \right] = \frac{2}{\pi} \left[\frac{\Gamma\left(\frac{3}{2}\right) \Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{3}{2} + \frac{3}{2}\right)} - \frac{\Gamma\left(\frac{1}{2}\right) \Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{1}{2} + \frac{5}{2}\right)} \right]$$

$$E \left[\frac{\partial^2}{\partial \theta^2} \ln f(x) \right] = \frac{2}{\pi} \left[\frac{\Gamma\left(\frac{3}{2}\right) \Gamma\left(\frac{3}{2}\right)}{\Gamma(3)} - \frac{\Gamma\left(\frac{1}{2}\right) \Gamma\left(\frac{5}{2}\right)}{\Gamma(3)} \right] = \frac{2}{\pi \Gamma(3)} \left[\left(\frac{1}{2}\sqrt{\pi}\right) \left(\frac{1}{2}\sqrt{\pi}\right) - (\sqrt{\pi}) \left(\frac{3}{4}\sqrt{\pi}\right) \right]$$

$$E \left[\frac{\partial^2}{\partial \theta^2} \ln f(x) \right] = \frac{2}{\pi \Gamma(3)} \left[\frac{\pi}{4} - \frac{3\pi}{4} \right] = \frac{2}{\pi(2!)} \left[\frac{-2\pi}{4} \right] = -\frac{1}{2}$$

Therefore, we get

$$V(t) = \frac{1}{-nE \left[\frac{\partial^2}{\partial \theta^2} \ln f(x) \right]} = \frac{1}{-n \left(-\frac{1}{2} \right)} = \frac{2}{n}$$

Taking t_1 as the median of the Cauchy distribution and the variance of median of Cauchy distribution is

$$V(t_1) = \frac{\pi^2}{4n}$$

Now relative efficiency of "t" w.r.t " t_1 " is

$$\frac{V(t)}{V(t_1)} = \frac{\frac{2}{n}}{\frac{\pi^2}{4n}} = \frac{8}{\pi^2} < 1, \text{ As } \pi = 3.145$$

Hence, t is less efficient than t_1

Now, As

$$\frac{\partial}{\partial \theta} \ln L(\theta) = \sum \frac{2(x_i - \theta)}{[1 + (x_i - \theta)^2]}$$

$$\left(\frac{\partial}{\partial \theta} \ln L(\theta) \right)_{t_1} = \sum \frac{2(x_i - t_1)}{[1 + (x_i - t_1)^2]}$$

Putting in

$$t = t_1 + V(t) \left(\frac{\partial L}{\partial \theta} \right)_{t_1}$$

$$t = t_1 + \frac{4}{n} \sum \frac{2(x_i - t_1)}{[1 + (x_i - t_1)^2]}$$

2.31 Estimator for Order Statistic From Uniform Distribution.

Question 30: Let $Y_1 < Y_2 < Y_3 \dots \dots \dots < Y_n$ be the order statistic of random sample of size n from the uniform distribution of continuous type over the closed interval $[\theta - p, \theta + p]$. Find the MLE of θ and p . Are these be unbiased.

Solution:

The p.d.f of uniform distribution for the interval $[\theta - p, \theta + p]$. is given as

$$f(y) = \frac{1}{\theta + p - \theta + p} = \frac{1}{2p}, \quad \theta - p \leq Y \leq \theta + p$$

The likelihood function

$$L(\theta, p) = \prod_{i=1}^n \frac{1}{2p} = \frac{1}{(2p)^n}$$

Taking ln both sides

$$L(\theta, p) = \ln 1 - n \ln 2 - n \ln p = -n \ln 2 - n \ln p$$

MLE for θ

The likelihood function is independent of θ , so MLE cannot be obtained.

MLE for p

$$\frac{\partial}{\partial p} L(\theta, p) = -\frac{n}{p}$$

Put

$$\frac{\partial}{\partial p} L(\theta, p) = 0,$$

$$-\frac{n}{p} = 0 \Rightarrow n = 0$$

The estimator vanishes, further more

$$\frac{\partial^2}{\partial p^2} L(\theta, p) = \frac{n}{p^2} = 0, \text{ for } n = 0$$

So, MLE for p does not exist.

As $Y_1 < Y_2 < Y_3 \dots \dots \dots < Y_n$ be order statistic for a random sample of size n so

$$\theta - p \leq Y_1 \quad \text{and} \quad Y_n \leq \theta + p$$

We can take

$$\theta - p = Y_1 \dots \dots (1) \quad \text{and} \quad \theta + p = Y_n \dots \dots (2)$$

Adding (1) and (2) we get

$$\hat{\theta} = \frac{Y_1 + Y_n}{2} \quad \text{and} \quad \hat{p} = \frac{Y_n - Y_1}{2}$$

Now to find the expectations of both estimators we need to find the expectation of Y_1 and Y_n

Now to find the p.d.f of y_1 and y_n

Distribution function of $f(y)$.

$$F(y) = \int_{\theta-p}^y \frac{1}{2p} dy = \frac{1}{2p} y \Big|_{\theta-p}^y = \frac{1}{2p} [y - \theta + p]$$

p.d.f of y_1

$$g(y_1) = n[1 - F(y_1)]^{n-1} f(y_1) = n \left[1 - \frac{1}{2p} \{y_1 - \theta + p\} \right]^{n-1} \left(\frac{1}{2p} \right)$$

$$E(y_1) = \int_{\theta-p}^{\theta+p} y_1 n \left[1 - \frac{1}{2p} \{y_1 - \theta + p\} \right]^{n-1} \left(\frac{1}{2p} \right) dy_1$$

Let

$$z = 1 - \frac{1}{2p} \{y_1 - \theta + p\}, \Rightarrow \frac{1}{2p} \{y_1 - \theta + p\} = 1 - z$$

$$y_1 = 2p(1 - z) + \theta - p$$

$$dy_1 = -2p dz$$

$$\text{when } y_1 = \theta - p \quad \text{then} \quad z = 1 - \frac{1}{2p} [\theta - p - \theta + p] = 1$$

$$\text{when } y_1 = \theta + p \quad \text{then} \quad z = 1 - \frac{1}{2p} [\theta + p - \theta + p] = 1 - \frac{2p}{2p} = 1 - 1 = 0$$

$$E(y_1) = \int_1^0 n[2p(1 - z) + \theta - p] z^{n-1} \left(\frac{1}{2p} \right) (-2p dz)$$

$$E(y_1) = \int_0^1 n[2p(1 - z) + \theta - p] z^{n-1} dz$$

$$E(y_1) = \int_0^1 n(2pz^{n-1} - 2pz^n + \theta z^{n-1} - pz^{n-1}) dz$$

$$E(y_1) = n(2p + \theta - p) \int_0^1 z^{n-1} dz - 2np \int_0^1 z^n dz$$

$$E(y_1) = n(p + \theta) \frac{z^n}{n} \Big|_0^1 - 2pn \frac{z^{n+1}}{n+1} \Big|_0^1 = (p + \theta)[1 - 0] - \frac{2np}{n+1} [1 - 0]$$

$$E(y_1) = p + \theta - \frac{2np}{n+1}$$

p.d.f of y_n

$$g(y_n) = n[F(y_n)]^{n-1} f(y) = n \left[\frac{1}{2p} \{y_n - \theta + p\} \right]^{n-1} \left(\frac{1}{2p} \right)$$

$$E(y_n) = \int_{\theta-p}^{\theta+p} y_n n \left[\frac{1}{2p} \{y_n - \theta + p\} \right]^{n-1} \left(\frac{1}{2p} \right) dy_n$$

Let

$$z = \frac{1}{2p} \{y_n - \theta + p\}$$

$$y_n = 2pz + \theta - p$$

$$dy_n = 2p dz$$

$$\text{when } y_1 = \theta - p \quad \text{then } z = \frac{1}{2p} [\theta - p - \theta + p] = 0$$

$$\text{when } y_1 = \theta + p \quad \text{then } z = \frac{1}{2p} [\theta + p - \theta + p] = \frac{2p}{2p} = 1$$

$$E(y_n) = \int_0^1 n[2pz + \theta - p] z^{n-1} \left(\frac{1}{2p} \right) (2p dz)$$

$$E(y_n) = \int_0^1 n(2pz^n + \theta z^{n-1} - pz^{n-1}) dz$$

$$E(y_n) = 2pn \int_0^1 z^n dz + n(\theta - p) \int_0^1 z^{n-1} dz$$

$$E(y_n) = 2pn \frac{z^{n+1}}{n+1} \Big|_0^1 + n(\theta - p) \frac{z^n}{n} \Big|_0^1$$

$$E(y_n) = \frac{2pn}{n+1} [1 - 0] + (\theta - p)[1 - 0] = \frac{2pn}{n+1} + \theta - p$$

Now, consider

$$\hat{\theta} = \frac{Y_1 + Y_n}{2}$$

$$E(\hat{\theta}) = \frac{E(Y_1) + E(Y_n)}{2} = \frac{p + \theta - \frac{2np}{n+1} + \frac{2pn}{n+1} + \theta - p}{2} = \frac{2\theta}{2} = \theta$$

Hence this estimator is unbiased estimator of θ

Again take

$$\hat{p} = \frac{Y_n - Y_1}{2}$$

$$E(\hat{p}) = \frac{E(Y_n) - E(Y_1)}{2} = \frac{\frac{2pn}{n+1} + \theta - p - p - \theta + \frac{2np}{n+1}}{2} = \frac{2\left(\frac{2np}{n+1}\right)}{2} = \frac{2np}{n+1}$$

This is the biased estimator of p

2.32 ML Estimator of Uniform Distribution of interval (α, β)

Question 31: Obtain the MLE of α and β of the uniform distribution over the interval (α, β) .

Solution:

The likelihood function

$$L(\alpha, \beta) = \prod_{i=1}^n \frac{1}{\beta - \alpha} = \left(\frac{1}{\beta - \alpha}\right)^n$$

Taking ln both sides

$$L(\alpha, \beta) = n \ln 1 - n \ln(\beta - \alpha)$$

Diff w.r.t α

$$\frac{\partial}{\partial \alpha} L(\alpha, \beta) = \frac{n}{(\beta - \alpha)}$$

Put

$$\frac{\partial}{\partial \alpha} L(\alpha, \beta) = 0$$

$$\frac{n}{(\beta - \alpha)} = 0 \Rightarrow n = 0$$

The likelihood function is failed to estimate the estimators for α and β we have to look for some other method

Let $Y_1 < Y_2 \dots \dots \dots < Y_n$ be order statistic, since x lies between α and β so

$$\alpha \leq Y_1 < Y_2 \dots \dots \dots < Y_n \leq \beta$$

$$\hat{\alpha} = Y_1 \text{ and } \hat{\beta} = Y_n$$

3.0 Method of Moments

- It is the oldest method of estimating the parameters of the distribution.
- It was first put forward by Karl Pearson in 1894.
- The method of moments consists of equating the sample moments to the corresponding moments of the distribution, which are functions of unknown parameter.
- We equate as many sample moments as there are unknown parameters.
- Solving the simultaneous equations one can obtain the estimates of the moments if the population in terms of sample variates.

As we have sample moment

$$m_r = E(x^r) = \frac{\sum x_i^r}{n}, r = 1, 2, 3, \dots$$

And the population moment

$$\mu_r = E(X^r) = \frac{\sum X_i^r}{N}, r = 1, 2, 3, \dots$$

Putting r=1

$$m_1 = E(x) = \frac{\sum x_i}{n} = \bar{x} \quad \text{and} \quad \mu_1 = E(X) = \frac{\sum X_i}{N} = \mu$$

Equating m_1 and μ_1

$$m_1 = \mu_1$$

$$\bar{x} = \mu$$

Put r=2

$$m_2 = E(x^2) = \frac{\sum x_i^2}{n} \quad \text{and} \quad \mu_2 = E(X^2) = \frac{\sum X_i^2}{n}$$

$$\text{As } \mu_2 = \mu_1^2 + \sigma^2 \Rightarrow \sigma^2 = \mu_2 - \mu_1^2 \Rightarrow \sigma^2 = \mu_2 - \mu^2$$

Equating m'_2 and μ'_2

$$m'_2 = \mu'_2$$

$$\frac{\sum x_i^2}{n} = \sigma^2 + \mu^2$$

$$\frac{\sum x_i^2}{n} = \sigma^2 + \bar{x}^2$$

$$\frac{\sum x_i^2}{n} - \bar{x}^2 = \sigma^2 \Rightarrow S^2 = \sigma^2$$

3.1 Properties of Moment Estimators

- Under general conditions, the moment estimators will have asymptotically normal distribution for large n.
- The Moment Estimators are consistent estimators of population moments, but may or may not be unbiased, hence they are not efficient in general.
- ME are not necessarily Sufficient.
- The mean of the distribution of estimate will differ from true value of the parameter by the quantity or order $\frac{1}{n}$.
- The variance of the distribution of estimate will be of the type $\frac{c^2}{n}$.
- In general, the deviation estimators obtained by the method of moments are less efficient than the maximum likelihood estimators. In particular cases they are equivalent.
- The ME are identical to MLE if the P.M.F or p.d.f is of the form

$$f(x; \theta) = \exp (b_0 + b_1x + b_2x^2 + \dots)$$

Where b's are independent of x , but may depend upon θ .

The likelihood function $L(x_i, \theta) = \exp (b_0 + b_1\sum x + b_2\sum x^2 + \dots)$

Partial derivation of log likelihood function will yield $\frac{\partial}{\partial \theta_j} \ln L(x_i, \theta) = a_0 + a_1\sum x_i + a_2\sum x_i^2 +$

Thus, both estimation method will yield identical estimates under the condition

“MLE are obtained as the linear combination of the moments”

3.2 Merits and Demerits of ME

- This method of estimation is fairly simple and produce consistent estimators, though these estimators are biased and not efficient in general.
- For known family of probability distribution, the MLE is superior than ME and have higher probability of being close to the population quantities to be estimated.
- For unknown family of probability distribution, the ME may be preferred over MLE.
- MLE are more often unbiased as compare to ME.
- ME are not necessarily Sufficient as these may not take into account the all relevant information in the sample.
- Iteration procedures are fairly simple in ME as compare to MLE which often require the assistance from computer.
- For sample samples it is very frequent that ME are outsidess of the parameter space, it does not make sense to rely on them. This problem is infrequent for large samples.

3.3 importance of Moment Estimators

- Moment Estimators are emerged by considering the sound theoretical background of estimation theory.
- It was considered closer to population parameters as compare o some subjective random guess.
- ME were used as seed value in the Fisher iteration method with the help of computers because these estimators can quickly and easily be calculated.
- ME are used as first approximation to the solution of likelihood equations.

4 Questions Related to Method of Moments

4.1 Moment Estimator for Normal distribution

Question: If $X \sim N(\mu, \sigma^2)$ then obtain the moment estimator of its parameters.

The first population moment

$$\mu'_1 = E(X) = \int_{-\infty}^{\infty} x \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} dx$$

Let

$$z = \frac{x - \mu}{\sigma} \Rightarrow x = \mu + z\sigma \quad \text{and } dx = \sigma dz$$

$$\mu'_1 = \int_{-\infty}^{\infty} (\mu + z\sigma) \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2}} (\sigma dz)$$

$$\mu'_1 = \mu \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz + \sigma \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} z e^{-\frac{z^2}{2}} dz = \mu(1) + \sigma(0) = \mu$$

So,

$$\mu'_1 = \mu$$

Using $m'_1 = \mu'_1$

$$\hat{\mu} = \bar{x}$$

$$\mu'_2 = E(X^2) = \int_{-\infty}^{\infty} x^2 \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} dx$$

Let

$$z = \frac{x - \mu}{\sigma} \Rightarrow x = \mu + z\sigma \quad \text{and } dx = \sigma dz$$

$$\mu'_2 = \int_{-\infty}^{\infty} (\mu + z\sigma)^2 \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2}} (\sigma dz)$$

$$\mu'_2 = \mu^2 \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz + \sigma^2 \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} z^2 e^{-\frac{z^2}{2}} dz + 2\mu\sigma \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} z e^{-\frac{z^2}{2}} dz$$

$$\mu'_2 = \mu^2(1) + \sigma^2(1) + 2\mu\sigma(0) = \mu^2 + \sigma^2$$

Using $m'_2 = \mu'_2$

$$\frac{\sum x_i^2}{n} = \sigma^2 + \mu^2$$

$$\frac{\sum x_i^2}{n} = \sigma^2 + \bar{x}^2$$

$$\frac{\sum x_i^2}{n} - \bar{x}^2 = \sigma^2 \Rightarrow S^2 = \sigma^2$$

$$\hat{\sigma}^2 = S^2$$

4.2 ME for Uniform distribution of interval (0,a)

Question: let x be uniformly distributed on the interval (0,a) show that the moment estimator of “a” is twice the mean.

Solution:

The p.d.f

$$f(x) = \frac{1}{a-0} = \frac{1}{a}, \text{ for } 0 < x < a$$

Population moment

$$\mu'_1 = E(X) = \int_0^a x \frac{1}{a} dx = \frac{1}{a} \cdot \frac{x^2}{2} \Big|_0^a = \frac{1}{2a} (a^2 - 0) = \frac{a}{2}$$

Using $m'_1 = \bar{x} = \mu'_1$

$$\bar{x} = \frac{a}{2}, \text{ Hence we get } \hat{a} = 2\bar{x}$$

4.3 ME for Exponential Distribution

Question: suppose that $T_1, T_2, \dots, T_n$ represent time to failure for a equipment to have exponential (life time) distribution with unknown parameter λ . Further assume that $n=5$ and observed times are 30.4, 7.8, 1.4, 13.1 and 67.5 hours. Obtain Moment Estimator for λ .

The p.d.f

$$f(x) = \lambda e^{-\lambda x}, \text{ for } 0 < x < \infty$$

Population moment

$$\mu_1 = E(X) = \int_0^{\infty} x \lambda e^{-\lambda x} dx$$

Let

$$t = \lambda x \Rightarrow x = \frac{t}{\lambda} \text{ and } dt = \lambda dx$$

$$\mu_1 = E(X) = \int_0^{\infty} \left(\frac{t}{\lambda}\right) \lambda e^{-t} (\lambda dt) = \frac{1}{\lambda} \int_0^{\infty} t^2 e^{-t} dt = \frac{1}{\lambda} \Gamma 2 = \frac{1}{\lambda} (2-1)! = \frac{1}{\lambda}$$

Using $m_1' = \bar{x} = \mu_1$

$$\bar{x} = \frac{1}{\lambda} \Rightarrow \hat{\lambda} = \frac{1}{\bar{x}} = \frac{n}{\sum x}$$

for given sample of n=5

$$\hat{\lambda} = \frac{5}{30.4 + 7.8 + 1.4 + 13.1 + 67.5} = 0.04167$$

4.4 ME of parameters of Binomial Distribution

The probability Mass function of Binomial Distribution

$$f(x) = \binom{n}{x} p^x q^{n-x}, x = 0, 1, \dots, n$$

The population moment

$$\mu_1 = E(X) = \sum x \binom{n}{x} p^x q^{n-x}$$

$$\mu_1 = E(X) = 0 \binom{n}{0} p^0 q^n + 1 \binom{n}{1} p^1 q^{n-1} + 2 \binom{n}{2} p^2 q^{n-2} + \dots \dots \dots n \binom{n}{n} p^n q^0$$

$$\mu_1 = E(X) = np^1 q^{n-1} + \frac{2n(n-1)}{2!} p^2 q^{n-2} + \dots \dots \dots np^n$$

$$\mu_1 = E(X) = np [q^{n-1} + (n-1)p^2 q^{n-2} + \dots \dots \dots p^{n-1}] = np(q+p)^{n-1} = np(1) = np$$

So, Using $m_1' = \bar{x} = \mu_1$

$$\bar{x} = np \dots \dots (A)$$

$$\mu_2 = E(X^2) = E[X(X-1) + X] = E[X(X-1)] + E(X) = E[X(X-1)] + np$$

$$\mu'_2 = 0(0-1)\binom{n}{0}p^0q^n + 1(1-1)\binom{n}{1}p^1q^{n-1} + 2(2-1)\binom{n}{2}p^2q^{n-2} + \dots n(n-1)\binom{n}{n}p^nq^0 + np$$

$$\mu'_2 = \frac{2n(n-1)}{2!}p^2q^{n-2} + \frac{6n(n-1)(n-2)}{3!}p^3q^{n-3} \dots \dots \dots n(n-1)p^n + np$$

$$\mu'_2 = n(n-1)p^2[q^{n-2} + (n-2)p^3q^{n-3} \dots \dots \dots p^{n-2}] + np$$

$$\mu'_2 = n(n-1)p^2[q+p]^{n-2} + np = n(n-1)p^2 + np$$

$$\mu'_2 = n^2p^2 - np^2 + np = np[(1-p) + p]$$

Using (A)

$$\mu'_2 = \bar{x}[(1-p) + \bar{x}] = \bar{x}^2 + \bar{x}(1-p)$$

Using $m'_2 = \mu'_2$

$$\frac{\sum x_i^2}{n} = \bar{x}^2 + \bar{x}(1-p)$$

$$\frac{\sum x_i^2}{n} - \bar{x}^2 = \bar{x}(1-p) \Rightarrow S^2 = \bar{x}(1-p)$$

$$S^2 - \bar{x} = -p\bar{x} \Rightarrow p = -\left(\frac{S^2 - \bar{x}}{\bar{x}}\right) = 1 - \frac{S^2}{\bar{x}}$$

So ME of p is

$$\hat{p} = 1 - \frac{S^2}{\bar{x}}$$

Also we have

$$\bar{x} = np \Rightarrow n = \frac{\bar{x}}{p} = \frac{\bar{x}}{1 - \frac{S^2}{\bar{x}}} = \frac{\bar{x}^2}{\bar{x} - S^2}$$

Alternative Question

If $X \sim b(m, p)$, Obtain ME for p, assuming “m” to be known.

$$m'_1 = mp \dots \dots (1)$$

And

$$\begin{aligned}\dot{m}_2 &= m(m-1)p^2 + mp \\ \dot{m}_2 &= m^2p^2 - (mp)p + mp\end{aligned}$$

Using (1)

$$\begin{aligned}\dot{m}_2 &= \dot{m}_1^2 - \dot{m}_1p + \dot{m}_1 \\ \dot{m}_1p &= \dot{m}_1^2 + \dot{m}_1 - \dot{m}_2 \\ \hat{p} &= \frac{\dot{m}_1^2 + \dot{m}_1 - \dot{m}_2}{\dot{m}_1}\end{aligned}$$

and

$$m = \frac{\dot{m}_1}{p} = \frac{\dot{m}_1}{\frac{\dot{m}_1^2 + \dot{m}_1 - \dot{m}_2}{\dot{m}_1}} = \frac{\dot{m}_1^2}{\dot{m}_1^2 + \dot{m}_1 - \dot{m}_2}$$

4.5 ME of parameters of Negative Binomial Distribution

The probability Mass function of Binomial Distribution

$$f(x) = \binom{x+k-1}{k-1} p^k q^x, x = 0, 1, 2, \dots, \infty, k < n$$

The population moment

$$\begin{aligned}\mu'_1 &= E(X) = \sum x \binom{x+k-1}{k-1} p^k q^x \\ \mu'_1 &= 0 \binom{0+k-1}{k-1} p^k q^0 + 1 \binom{k}{k-1} p^k q^1 + 2 \binom{k+1}{k-1} p^k q^2 + \dots \dots \dots \\ \mu'_1 &= E(X) = kp^k q^1 + \frac{2k(k+1)}{2} p^k q^2 + \dots \dots \dots \\ \mu'_1 &= kp^k q [1 + (k+1)q^1 + \dots \dots \dots] = kp^k q (1-q)^{-(k+1)} \\ \mu'_1 &= kp^k q (1-q)^{-(k+1)} = kp^k q p^{-k-1} = \frac{kq}{p}\end{aligned}$$

Using $m_1' = \bar{x} = \mu'_1$

$$\bar{x} = \frac{kq}{p} \quad \text{--- (A)}$$

$$\begin{aligned}\mu'_2 &= E(X^2) = E[X(X-1) + X] = E[X(X-1)] + E(X) = E[X(X-1)] + \frac{kq}{p} \\ \mu'_2 &= \sum x(x-1) \binom{x+k-1}{k-1} p^k q^x + \frac{kq}{p} \\ \mu'_2 &= 0(0-1) \binom{0+k-1}{k-1} p^k q^0 + 1(1-1) \binom{k}{k-1} p^k q^1 + 2(2-1) \binom{k+1}{k-1} p^k q^2 + \dots + \frac{kq}{p} \\ \mu'_2 &= \frac{2(k+1)(k)}{2!} p^k q^2 + \frac{6(k+2)(k+1)k}{3!} p^k q^3 + \dots + \frac{kq}{p} \\ \mu'_2 &= k(k+1)p^k q^2 [1 + (k+2)p^k q^1 + \dots] + \frac{kq}{p} \\ \mu'_2 &= k(k+1)p^k q^2 [1-q]^{-(k+2)} + \frac{kq}{p} = k(k+1)p^k q^2 p^{-(k+2)} + \frac{kq}{p} \\ \mu'_2 &= k(k+1)p^k q^2 p^{-k} p^{-2} + \frac{kq}{p} = \frac{k(k+1)q^2}{p^2} + \frac{kq}{p} = \frac{k^2 q^2}{p^2} + \frac{kq^2}{p^2} + \frac{kq}{p}\end{aligned}$$

Using (A)

$$\mu'_2 = \bar{x}^2 + \frac{kq^2}{p^2} + \bar{x}$$

Using $m'_2 = \mu'_2$

$$\frac{\sum x_i^2}{n} = \bar{x}^2 + \frac{kq^2}{p^2} + \bar{x}$$

$$\frac{\sum x_i^2}{n} - \bar{x}^2 = \frac{kq^2}{p^2} + \bar{x} \Rightarrow S^2 = \frac{kq^2}{p^2} + \bar{x}$$

$$S^2 = \frac{q}{p} \left(\frac{kq}{p} \right) + \bar{x} \Rightarrow S^2 = \bar{x} \left(\frac{q}{p} + 1 \right)$$

$$S^2 = \bar{x} \left(\frac{q+p}{p} \right) \Rightarrow S^2 = \frac{\bar{x}}{p}$$

So, ME of p is

$$\hat{p} = \frac{\bar{x}}{S^2}$$

putting in (A)

$$\bar{x} = \frac{kq}{\frac{\bar{x}}{S^2}} = \frac{kqS^2}{\bar{x}} \Rightarrow kqS^2 = \bar{x}^2$$

$$k = \frac{\bar{x}^2}{qS^2} = \frac{\bar{x}^2}{S^2(1-p)} = \frac{\bar{x}^2}{S^2 \left(1 - \frac{\bar{x}}{S^2} \right)} = \frac{\bar{x}^2}{S^2 - \bar{x}^2}$$

ME of k

$$\hat{k} = \frac{\bar{x}^2}{S^2 - \bar{x}^2}$$

Alternative Question

The P.M.F of Negative Binomial in the form

$$f(x) = \binom{x + \theta_2 - 1}{\theta_2 - 1} \theta_1^{\theta_2} (1 - \theta_1)^x$$

Here $p = \theta_1$ and $k = \theta_2$

$$\mu'_1 = \frac{\theta_2(1 - \theta_1)}{\theta_1} =$$

$$\mu_2 = \frac{\theta_2^2(1 - \theta_1)^2}{\theta_1^2} + \frac{\theta_2^2(1 - \theta_1)}{\theta_1^2} + \frac{\theta_2(1 - \theta_1)}{\theta_1}$$

$$\hat{\theta}_1 = \frac{\bar{x}}{S^2} \quad \text{and} \quad \hat{\theta}_2 = \frac{\bar{x}^2}{S^2 - \bar{x}^2}$$



4.6 ME of Log Normal Distribution

Question: Log Normal Distribution is given that

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma x} e^{-\frac{1}{2\sigma^2}(\ln x - \mu)^2} \quad 0 < x < \infty$$

Find Moment Estimator and prove that the moment estimators of μ and σ^2 are

$$\ln \left[\frac{(\hat{m}_1)^2}{\hat{m}_2} \right] \text{ and } \ln \left[\frac{\hat{m}_2}{(\hat{m}_1)^2} \right]$$

Solution:

$$\mu'_r = \int_{-\infty}^{\infty} x^r \frac{1}{\sqrt{2\pi}\sigma x} e^{-\frac{1}{2\sigma^2}(\ln x - \mu)^2} dx = \int_{-\infty}^{\infty} x^{r-1} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(\ln x - \mu)^2} dx$$

Let

$$z = \frac{\ln x - \mu}{\sigma} \Rightarrow \ln x = \mu + z\sigma \Rightarrow x = e^{\mu + z\sigma}$$

$$dx = e^{\mu + z\sigma}(\sigma dz)$$

$$\mu'_r = \int_{-\infty}^{\infty} (e^{\mu + z\sigma})^{r-1} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{z^2}{2}} e^{\mu + z\sigma}(\sigma dz) = \int_{-\infty}^{\infty} (e^{\mu + z\sigma})^r \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$\mu'_r = e^{\mu r} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z^2 - 2z\sigma r)} dz = e^{\mu r} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z^2 - 2z\sigma r + r^2\sigma^2 - r^2\sigma^2)} dz$$

$$\mu'_r = e^{\mu r} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z^2 - 2z\sigma r + r^2\sigma^2)} e^{\frac{1}{2}r^2\sigma^2} dz = e^{\mu r + \frac{1}{2}r^2\sigma^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - \sigma r)^2} dz$$

$$\mu'_r = e^{\mu r + \frac{r^2\sigma^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - \sigma r)^2} dz = e^{\mu r + \frac{r^2\sigma^2}{2}} (1)$$

Hence,

$$\mu'_r = e^{\mu r + \frac{r^2\sigma^2}{2}}$$

Put $r=1,2$

$$\mu'_1 = e^{\mu + \frac{6^2}{2}} \quad \text{and} \quad \mu'_2 = e^{2\mu + 2\sigma^2}$$

Using relationship between population and sample moments we get

$$\hat{m}_1 = e^{\mu + \frac{6^2}{2}} \text{ --- (A)}$$

$$\hat{m}_1 = e^{\mu} \cdot e^{\frac{6^2}{2}}$$

$$\frac{\hat{m}_1}{e^{\frac{6^2}{2}}} = e^{\mu} \text{ --- (1)}$$

$$\hat{m}_2 = e^{2\mu + 2\sigma^2} \text{ --- (B)}$$

$$\hat{m}_2 = (e^{\mu})^2 e^{2\sigma^2}$$

Using 1

$$\hat{m}_2 = \left(\frac{\hat{m}_1}{e^{\frac{6^2}{2}}} \right)^2 e^{2\sigma^2} = \frac{(\hat{m}_1)^2 e^{2\sigma^2}}{e^{\sigma^2}} = (\hat{m}_1)^2 e^{\sigma^2}$$

$$e^{\sigma^2} = \frac{\hat{m}_2}{(\hat{m}_1)^2}$$

$$\sigma^2 = \ln \left[\frac{\hat{m}_2}{(\hat{m}_1)^2} \right]$$

Put in (1)

$$e^{\mu} = \frac{\hat{m}_1}{e^{\frac{1}{2} \ln \left[\frac{\hat{m}_2}{(\hat{m}_1)^2} \right]}} = e^{\mu} = \frac{\hat{m}_1}{e^{\ln \left[\frac{\hat{m}_2}{(\hat{m}_1)^2} \right]^{\frac{1}{2}}}} = \frac{\hat{m}_1}{\left[\frac{\hat{m}_2}{(\hat{m}_1)^2} \right]^{\frac{1}{2}} \ln e} = \frac{\hat{m}_1}{\left[\frac{\hat{m}_2}{(\hat{m}_1)^2} \right]^{\frac{1}{2}}}$$

$$e^{\mu} = \frac{\hat{m}_1 \hat{m}_1}{\sqrt{\hat{m}_2}} = \frac{(\hat{m}_1)^2}{\sqrt{\hat{m}_2}}$$

$$\mu = \ln \left[\frac{(\hat{m}_1)^2}{\sqrt{\hat{m}_2}} \right]$$

Hence it is proved

$$\hat{\mu} = \ln \left[\frac{(\hat{m}_1)^2}{\sqrt{\hat{m}_2}} \right] \quad \text{and} \quad \hat{\sigma}^2 = \ln \left[\frac{\hat{m}_2}{(\hat{m}_1)^2} \right]$$

Alternative Method for Solving

$$\hat{m}_1 = e^{\mu + \frac{\sigma^2}{2}}$$

$$\ln(\hat{m}_1) = \mu + \frac{\sigma^2}{2} \dots \dots \dots (1)$$

$$\hat{m}_2 = e^{2\mu + 2\sigma^2}$$

$$\ln(\hat{m}_2) = 2\mu + 2\sigma^2 \dots \dots \dots (2)$$

Eq(2)-2Eq(1)

$$\ln(\hat{m}_2) - 2\ln(\hat{m}_1) = 2\mu + 2\sigma^2 - 2\mu - \sigma^2$$

$$\ln \left[\frac{\hat{m}_2}{(\hat{m}_1)^2} \right] = \sigma^2$$

$$\ln(\hat{m}_2) = 2\mu + 2 \ln \left[\frac{\hat{m}_2}{(\hat{m}_1)^2} \right]$$

$$\ln(\hat{m}_2) - 2 \ln \left[\frac{\hat{m}_2}{(\hat{m}_1)^2} \right] = 2\mu$$

$$\ln(\hat{m}_2) - 2 \ln(\hat{m}_2) + 2 \ln(\hat{m}_1)^2 = 2\mu$$

$$2 \ln(\hat{m}_1)^2 - \ln(\hat{m}_2) = 2\mu$$

$$2 \left[\ln(\hat{m}_1)^2 - \frac{1}{2} \ln(\hat{m}_2) \right] = 2\mu$$

$$\ln \left[\frac{(\hat{m}_1)^2}{\sqrt{\hat{m}_2}} \right] = \mu$$

4.7 ME of Gamma Distribution

Question: suppose the time “X” between order of a given part at a large warehouse is a Gamma random variable with parameter “r” and “λ”. Obtain the moment estimator for “r” and “λ”. Ten observed values for “X” be 15.5,4.5,6.8,34.5,4.7,20.9,8.2,14.9,17.7.

Solution:

The p.d.f of Gamma Distribution is

$$f(x) = \frac{1}{\Gamma r \lambda^r} e^{-\frac{x}{\lambda}} x^{r-1}, \quad 0 < x < \infty$$

The population Moment

$$\mu'_R = \int_0^\infty x^R \frac{1}{\Gamma r \lambda^r} e^{-\frac{x}{\lambda}} x^{r-1} dx = \int_0^\infty x^{R+r-1} \frac{1}{\Gamma r \lambda^r} e^{-\frac{x}{\lambda}} dx$$

Let

$$t = \frac{x}{\lambda} \Rightarrow x = t\lambda \Rightarrow dx = \lambda dt$$

$$\mu'_R = \frac{1}{\Gamma r \lambda^r} \int_0^\infty (t\lambda)^{R+r-1} e^{-t} \lambda dt = \frac{\lambda^R}{\Gamma r} \int_0^\infty t^{R+r-1} e^{-t} dt$$

$$\mu'_R = \frac{\lambda^R}{\Gamma r} \Gamma(R+r)$$

Put R=1

$$\mu'_1 = \frac{\lambda}{\Gamma r} \Gamma(r+1) = \frac{\lambda r!}{(r-1)!} = \frac{\lambda r(r-1)!}{(r-1)!} = \lambda r$$

Put R=2

$$\mu'_2 = \frac{\lambda^2}{\Gamma r} \Gamma(r+2) = \frac{\lambda^2 (r+1)!}{(r-1)!} = \frac{\lambda^2 r(r+1)(r-1)!}{(r-1)!} = \lambda^2 r(r+1)$$

Using Relationship between Sample and Population Moments

$$\mu'_1 = m'_1 \Rightarrow \lambda r = \bar{x} \dots \dots \dots (1)$$

$$\mu'_2 = m'_2 \Rightarrow \lambda^2 r(r+1) = \frac{\sum x_i^2}{n} \Rightarrow \lambda^2 r(r+1) = S^2 + \bar{x}^2 \dots \dots (2)$$

In Equation (2)

$$\lambda^2 r^2 + (\lambda r)\lambda = S^2 + \bar{x}^2$$

From (1)

$$\bar{x}^2 + \bar{x}\lambda = S^2 + \bar{x}^2 \Rightarrow \bar{x}\lambda = S^2$$

Hence Moment Estimator for r is

$$\hat{\lambda} = \frac{S^2}{\bar{x}}$$

Put in (1)

$$\left(\frac{S^2}{\bar{x}}\right)r = \bar{x}$$

Hence Moment Estimator for r

$$r = \frac{\bar{x}^2}{S^2}$$

Now for Observed sample of size 10

S.N	1	2	3	4	5	Total
x	15.5	4	5	6.8	34.5	
x^2	240	16	25	46.2	1190	
S.N	6	7	8	9	10	$\sum x = 132$
x	4.7	20.9	8.2	14.9	17.7	
x^2	22.1	437	67.2	222	313	$\sum x^2 = 2579$

$$\bar{x} = \frac{132.2}{10} = 13.22, S^2 = \frac{2579.2}{10} - (13.22)^2 = 257.92 - 174.76 = 83.16$$

Estimated Moment Estimators

$$\hat{\lambda} = \frac{83.16}{13.22} = 6.29 \text{ and } r = \frac{(13.22)^2}{83.16} = 2.10$$

4.8 ME of Beta Distribution

Question: Let $X_1, X_2, \dots, X_n$ be random sample from Beta Distribution with p.d.f

$$f(x) = \frac{1}{\beta(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1$$

Prove that Moment Estimators for α and β are

$$\frac{\bar{X}[\bar{X}(1 - \bar{X})^2 + S^2\bar{X} - S^2]}{S^2(1 - \bar{X})} \quad \text{and} \quad \frac{\bar{X}(1 - \bar{X})^2 + S^2\bar{X} - S^2}{S^2(1 - \bar{X})}$$

Where S^2 is the common variance.

Solution:

The r^{th} population Moment

$$\mu'_r = \int_0^1 x^r \frac{1}{\beta(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} dx = \frac{1}{\beta(\alpha, \beta)} \int_0^1 x^{r+\alpha-1} (1-x)^{\beta-1} dx$$

$$\mu'_r = \frac{1}{\beta(\alpha, \beta)} \beta(r + \alpha, \beta)$$

Put $r=1$

$$\mu'_1 = \frac{1}{\beta(\alpha, \beta)} \beta(\alpha + 1, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma\alpha\Gamma\beta} X \frac{\Gamma(\alpha + 1)\Gamma\beta}{\Gamma(\alpha + \beta + 1)} = \frac{(\alpha + \beta - 1)!}{(\alpha - 1)!(\beta - 1)!} X \frac{\alpha!(\beta - 1)!}{(\alpha + \beta)!}$$

$$\mu'_1 = \frac{(\alpha + \beta - 1)!}{(\alpha - 1)!} X \frac{\alpha(\alpha - 1)!}{(\alpha + \beta)(\alpha + \beta - 1)!} = \frac{\alpha}{\alpha + \beta}$$

Put $r=2$

$$\mu'_2 = \frac{1}{\beta(\alpha, \beta)} \beta(\alpha + 2, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma\alpha\Gamma\beta} X \frac{\Gamma(\alpha + 2)\Gamma\beta}{\Gamma(\alpha + \beta + 2)} = \frac{(\alpha + \beta - 1)!}{(\alpha - 1)!(\beta - 1)!} X \frac{(\alpha + 1)!(\beta - 1)!}{(\alpha + \beta + 1)!}$$

$$\mu'_2 = \frac{1}{\beta(\alpha, \beta)} \beta(\alpha + 2, \beta) = \frac{(\alpha + \beta - 1)!}{(\alpha - 1)!} X \frac{(\alpha + 1)\alpha(\alpha - 1)!}{(\alpha + \beta + 1)(\alpha + \beta)(\alpha + \beta - 1)!}$$

$$= \frac{(\alpha + 1)\alpha}{(\alpha + \beta + 1)(\alpha + \beta)}$$

$$\mu'_1 = E(X) \Rightarrow \frac{\alpha}{\alpha + \beta} = \bar{X} \text{ --- (1)}$$

$$\mu'_2 = E(X^2) = \frac{(\alpha + 1)\alpha}{(\alpha + \beta + 1)(\alpha + \beta)} \text{ --- (2)}$$

Taking

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{(\alpha + 1)\alpha}{(\alpha + \beta + 1)(\alpha + \beta)} - \left(\frac{\alpha}{\alpha + \beta}\right)^2$$

$$Var(X) = \left(\frac{\alpha}{\alpha + \beta}\right) \left[\frac{(\alpha + 1)}{(\alpha + \beta + 1)} - \left(\frac{\alpha}{\alpha + \beta}\right) \right] = \left(\frac{\alpha}{\alpha + \beta}\right) \left[\frac{(\alpha + 1)(\alpha + \beta) - \alpha(\alpha + \beta + 1)}{(\alpha + \beta + 1)(\alpha + \beta)} \right]$$

$$Var(X) = \left(\frac{\alpha}{\alpha + \beta}\right) \left[\frac{\alpha^2 + \alpha\beta + \alpha + \beta - \alpha^2 - \alpha\beta - \alpha}{(\alpha + \beta + 1)(\alpha + \beta)} \right] = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

$$\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = S^2$$

From 1

$$\alpha = \bar{X}(\alpha + \beta) \Rightarrow \alpha = \frac{\beta\bar{X}}{1 - \bar{X}} \text{ --- (3)}$$

Putting value of α

$$\frac{\beta \left(\frac{\beta\bar{X}}{1 - \bar{X}} \right)}{\left(\frac{\beta\bar{X}}{1 - \bar{X}} + \beta \right)^2 \left(\frac{\beta\bar{X}}{1 - \bar{X}} + \beta + 1 \right)} = S^2$$

$$\frac{\left(\frac{\beta^2\bar{X}}{1 - \bar{X}} \right)}{\left(\frac{\beta\bar{X} + \beta - \beta\bar{X}}{1 - \bar{X}} \right)^2 \left(\frac{\beta\bar{X} + \beta + 1 - \bar{X} - \beta\bar{X}}{1 - \bar{X}} \right)} = S^2$$

$$\frac{\left(\frac{\beta^2\bar{X}}{1 - \bar{X}} \right)}{\left(\frac{\beta}{1 - \bar{X}} \right)^2 \left(\frac{\beta + 1 - \bar{X}}{1 - \bar{X}} \right)} = S^2$$

$$\left(\frac{\beta^2\bar{X}}{1 - \bar{X}} \right) \frac{(1 - \bar{X})^3}{\beta^2(\beta + 1 - \bar{X})} = S^2$$

$$\frac{\bar{X}(1 - \bar{X})^2}{(\beta + 1 - \bar{X})} = S^2 \Rightarrow \beta + 1 - \bar{X} = \frac{\bar{X}(1 - \bar{X})^2}{S^2}$$

$$\beta = \frac{\bar{X}(1 - \bar{X})^2}{S^2} + \bar{X} - 1 = \frac{\bar{X}(1 - \bar{X})^2 + S^2\bar{X} - S^2}{S^2}$$

Hence, Moment Estimator for β

$$\hat{\beta} = \frac{\bar{X}(1 - \bar{X})^2 + S^2\bar{X} - S^2}{S^2}$$

Putting in (3)

$$\hat{\alpha} = \frac{\bar{X}[\bar{X}(1 - \bar{X})^2 + S^2\bar{X} - S^2]}{(1 - \bar{X})S^2}$$

4.9 ME for Uniform over interval $(-\lambda, \lambda)$

Assume X is a uniform random variable over the interval $(-\lambda, \lambda)$, where λ is any unknown. Obtain the method of moments estimator for λ . If the observed values in a sample of size 4 are -0.003, 2.590, 2.319 and -0.268.

Solution:

The p.d.f of uniform distribution is

$$f(x) = \frac{1}{\lambda - (-\lambda)} = \frac{1}{2\lambda}, \quad -\lambda < x < \lambda$$

The r^{th} Population Moment

$$\mu_r = \int_{-\lambda}^{\lambda} x^r \frac{1}{2\lambda} dx = \frac{1}{2\lambda} \frac{x^{r+1}}{r+1} \Big|_{-\lambda}^{\lambda} = \frac{1}{2\lambda(r+1)} [\lambda^{r+1} - (-\lambda)^{r+1}]$$

Put $r=1,2$

$$\begin{aligned} \mu_1 &= \frac{1}{2\lambda(1+1)} [\lambda^2 - (-\lambda)^2] = 0 \quad \text{and} \quad \mu_2 = \frac{1}{2\lambda(2+1)} [\lambda^3 - (-\lambda)^3] \\ &= \frac{2\lambda^3}{2\lambda(3)} = \frac{\lambda^2}{3} \end{aligned}$$

Using Relationship between population and sample moments

$$\mu_1 = m_1 \Rightarrow m_1 = 0$$

And

$$\mu'_2 = m'_2 \Rightarrow \frac{\lambda^2}{3} = \frac{\sum x^2}{n} \Rightarrow \lambda = \sqrt{\frac{3\sum x^2}{n}}$$

Hence, The Moment Estimator for λ

$$\hat{\lambda} = \sqrt{\frac{3\sum x^2}{n}}$$

For observed sample of size 4

$$\begin{aligned}\hat{\lambda} &= \sqrt{\frac{3}{4} [(-0.003)^2 + (2.590)^2 + (2.319)^2 + (-0.268)^2]} = \sqrt{\frac{3}{4} (12.1346)} \\ &= 3.0168\end{aligned}$$

4.10 ME for Uniform over interval (a, b)

Question: Find the Moment Estimator for “a” and “b” from the p.d.f

$$f(x; a, b) = \frac{1}{b-a}, \quad a < x < b$$

Solution:

The r^{th} Population Moment

$$\mu'_r = \int_a^b x^r \frac{1}{b-a} dx = \frac{1}{(b-a)} \frac{x^{r+1}}{r+1} \Big|_a^b = \frac{1}{(b-a)(r+1)} [b^{r+1} - a^{r+1}]$$

Put $r=1$

$$\mu'_1 = \frac{1}{(b-a)(1+1)} [b^{1+1} - a^{1+1}] = \frac{b^2 - a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{b+a}{2}$$

Put $r=2$

$$\begin{aligned}\mu'_2 &= \frac{1}{(b-a)(2+1)} [b^{2+1} - a^{2+1}] = \frac{b^3 - a^3}{3(b-a)} = \frac{(b-a)(b^2 + ab + a^2)}{3(b-a)} \\ &= \frac{b^2 + ab + a^2}{3}\end{aligned}$$

Now, Equating Population and sample moments

$$\mu'_1 = m'_1 \Rightarrow \frac{a+b}{2} = m'_1 \Rightarrow a+b = 2m'_1 \dots \dots \dots (1)$$

$$\mu'_2 = m'_2 \Rightarrow \frac{a^2 + b^2 + ab}{3} = m'_2 \Rightarrow a^2 + b^2 + ab = 3m'_2 \dots \dots \dots (2)$$

From Eq(1)

$$a = 2m'_1 - b \dots \dots \dots (3)$$

Put in Eq(2)

$$\begin{aligned}(2m'_1 - b)^2 + b^2 + b(2m'_1 - b) &= 3m'_2 \\ 4(m'_1)^2 + b^2 - 4m'_1b + b^2 + 2m'_1b - b^2 - 3m'_2 &= 0 \\ b^2 - (2m'_1)b + (4m'^2_1 - 3m'_2) &= 0\end{aligned}$$

Using Quadratic formula

$$\begin{aligned}b &= \frac{-(-2m'_1) \pm \sqrt{(-2m'_1)^2 - 4(1)(4m'^2_1 - 3m'_2)}}{2} \\ &= \frac{2m'_1 \pm \sqrt{4m'^2_1 - 4(1)(4m'^2_1 - 3m'_2)}}{2} \\ b &= \frac{2 \left[m'_1 \pm \sqrt{m'^2_1 - (4m'^2_1 - 3m'_2)} \right]}{2} = m'_1 \pm \sqrt{m'^2_1 - 4m'^2_1 + 3m'_2}\end{aligned}$$

Hence ME for “b”

$$\hat{b} = \hat{m}_1 \pm \sqrt{3(m_2 - \hat{m}_1^2)}$$

Putting in (3)

$$a = 2\hat{m}_1 - \left[\hat{m}_1 \pm \sqrt{3(m_2 - \hat{m}_1^2)} \right]$$

Hence, ME for “a”

$$\hat{a} = \hat{m}_1 \mp \sqrt{3(m_2 - \hat{m}_1^2)}$$



4.11 ME for θ

Question: Let $Y_1, Y_2, \dots, Y_n$ be a random sample from the density

$$f(y; \theta) = (\theta + 1)y^\theta, \quad 0 < y < 1$$

Prove that the Moment Estimator of θ is

$$\frac{1 - 2\bar{y}}{\bar{y} - 1}$$

Solution:

The Population Moment

$$\mu'_1 = \int_0^1 y (\theta + 1) y^\theta dy = (\theta + 1) \int_0^1 y^{\theta+1} dy = \frac{(\theta + 1)}{(\theta + 2)} y^{\theta+1} \Big|_0^1$$

$$\mu'_1 = \frac{(\theta + 1)}{(\theta + 2)} [(1)^{\theta+1} - (0)^{\theta+1}] = \frac{(\theta + 1)}{(\theta + 2)}$$

Using

$$\mu'_1 = m'_1 \Rightarrow \frac{(\theta + 1)}{(\theta + 2)} = \bar{y}$$

$$(\theta + 1) = (\theta + 2)\bar{y} \Rightarrow \theta + 1 = \theta\bar{y} + 2\bar{y} \Rightarrow 1 - 2\bar{y} = \theta(\bar{y} - 1)$$

Hence, the ME for θ

$$\hat{\theta} = \frac{1 - 2\bar{y}}{\bar{y} - 1}$$

4.12 ME for α and β from Pearson type-II Distribution.

Question: Estimate α and β from the Pearson Type-II distribution having the p.d.f

$$f(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma\alpha} x^{\alpha-1} e^{-x\beta}, \quad 0 < x < \infty$$

Solution

The r^{th} Population Moment

$$\mu'_r = \int_0^\infty x^r \frac{\beta^\alpha}{\Gamma\alpha} x^{\alpha-1} e^{-x\beta} dx = \frac{\beta^\alpha}{\Gamma\alpha} \int_0^\infty x^{r+\alpha-1} e^{-x\beta} dx$$

Let

$$t = \beta x \Rightarrow x = \frac{t}{\beta} \quad \text{and} \quad dx = \frac{dt}{\beta}$$

$$\mu'_r = \frac{\beta^\alpha}{\Gamma\alpha} \int_0^\infty \left(\frac{t}{\beta}\right)^{r+\alpha-1} e^{-t} \left(\frac{dt}{\beta}\right) = \frac{\beta^\alpha}{\Gamma\alpha \beta^{r+\alpha-1+1}} \int_0^\infty t^{r+\alpha-1} e^{-t} dt$$

$$\mu'_r = \frac{1}{\Gamma\alpha \beta^r} \int_0^\infty t^{r+\alpha-1} e^{-t} dt = \frac{1}{\Gamma\alpha \beta^r} \Gamma(r+\alpha)$$

Putting $r=1$

$$\mu'_1 = \frac{1}{\Gamma\alpha \beta^1} \Gamma(\alpha+1) = \frac{\alpha!}{(\alpha-1)!\beta} = \frac{\alpha}{\beta}$$

Putting $r=2$

$$\mu'_2 = \frac{1}{\Gamma\alpha \beta^2} \Gamma(\alpha+2) = \frac{(\alpha+1)!}{(\alpha-1)!\beta^2} = \frac{\alpha(\alpha+1)(\alpha-1)!}{(\alpha-1)!\beta^2} = \frac{\alpha(\alpha+1)}{\beta^2}$$

Using $\mu'_1 = m'_1$

$$\frac{\alpha}{\beta} = m'_1 \quad \text{--- (1)}$$

Using $\mu'_2 = m'_2$

$$m'_2 = \frac{\alpha(\alpha+1)}{\beta^2} = \frac{\alpha^2}{\beta^2} + \left(\frac{\alpha}{\beta}\right) \frac{1}{\beta}$$

Using (1)

$$m'_2 = (m'_1)^2 - \frac{m'_1}{\beta} \Rightarrow \frac{m'_1}{\beta} = (m'_1)^2 - m'_2$$

Hence, Moment Estimator for β

$$\hat{\beta} = \frac{m'_1}{(m'_1)^2 - m'_2}$$

Put in (1)

$$\frac{\alpha}{\frac{m'_1}{(m'_1)^2 - m'_2}} = m'_1$$

$$\hat{\alpha} = \frac{(m'_1)^2}{(m'_1)^2 - m'_2}$$

4.13 ME of Beta Type Distribution

Question: Let $X_1, X_2, \dots, X_n$ be a random sample from beta distribution of the form

$$f(x; \theta) = \theta(1 + \theta)(1 - x)x^{\theta-1}, 0 < x < 1$$

Estimate θ by Method of Moments

Solution:

The population Moment

$$\begin{aligned}\mu'_1 &= \int_0^1 x \theta(1 + \theta)(1 - x)x^{\theta-1} dx = \theta(1 + \theta) \int_0^1 x^{\theta+1-1} (1 - x)^{2-1} dx \\ \mu'_1 &= \theta(1 + \theta)\beta(\theta + 1, 2) = \theta(1 + \theta) \frac{\Gamma(\theta + 1) \Gamma 2}{\Gamma(\theta + 3)} = \theta(1 + \theta) \frac{\theta! \cdot 1!}{(\theta + 2)!} \\ &= \theta(1 + \theta) \frac{\theta(\theta - 1)!}{(\theta + 2)(\theta + 1)\theta(\theta - 1)!} \\ \mu'_1 &= \frac{\theta}{\theta + 2}\end{aligned}$$

Using $\mu'_1 = m'_1 = \bar{x}$

$$\bar{x} = \frac{\theta}{\theta + 2} \Rightarrow \bar{x}(\theta + 2) = \theta \Rightarrow 2\bar{x} = (1 - \bar{x})\theta$$

Hence Moment Estimator for θ

$$\hat{\theta} = \frac{2\bar{x}}{1 - \bar{x}}$$

4.14 Comparison of ME and MLE for θ

Question: Let $Y_1, Y_2, \dots, Y_n$ be a random sample from the density

$$f(y; \theta) = (\theta + 1)y^\theta, \quad 0 < y < 1$$

Prove that the Moment Estimator of θ is

$$\frac{1 - 2\bar{y}}{\bar{y} - 1}$$

Also find the MLE for θ and compare it with ME.

Solution:

(a) Moment Estimator

The Population Moment

$$\mu'_1 = \int_0^1 y(\theta + 1)y^\theta dy = (\theta + 1) \int_0^1 y^{\theta+1} dy = \frac{(\theta + 1)}{(\theta + 2)} y^{\theta+1} \Big|_0^1$$

$$\mu'_1 = \frac{(\theta + 1)}{(\theta + 2)} [(1)^{\theta+1} - (0)^{\theta+1}] = \frac{(\theta + 1)}{(\theta + 2)}$$

Using

$$\mu'_1 = m'_1 \Rightarrow \frac{(\theta + 1)}{(\theta + 2)} = \bar{y}$$

$$(\theta + 1) = (\theta + 2)\bar{y} \Rightarrow \theta + 1 = \theta\bar{y} + 2\bar{y} \Rightarrow 1 - 2\bar{y} = \theta(\bar{y} - 1)$$

Hence, the ME for θ

$$\hat{\theta} = \frac{1 - 2\bar{y}}{\bar{y} - 1}$$

(b) MLE

The likelihood function

$$L(\theta) = \prod_{i=1}^n (\theta + 1)y_i^\theta = (\theta + 1)^n \prod (y_i)^\theta$$

Taking Ln Both sides

$$\ln L(\theta) = n \ln(\theta + 1) + \theta \ln(\prod y_i) = n \ln(\theta + 1) + \theta \sum \ln y_i$$

Diff w.r.t θ

$$\frac{\partial}{\partial \theta} \ln L(\theta) = \frac{n}{\theta + 1} + \sum \ln y_i \quad \text{--- (1)}$$

Put

$$\frac{\partial}{\partial \theta} \ln L(\theta) = 0$$

$$\frac{n}{\theta + 1} + \sum \ln y_i = 0 \Rightarrow n = -\sum \ln y_i (\theta + 1) \Rightarrow \frac{-n}{\sum \ln y_i} = \theta + 1$$

$$\theta = \frac{-n}{\sum \ln y_i} - 1 = \frac{-n - \sum \ln y_i}{\sum \ln y_i}$$

Diff (1) w.r.t θ

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = \frac{-n}{(\theta + 1)^2} < 0$$

So, MLE of θ is

$$\hat{\theta} = \frac{-n - \sum \ln y_i}{\sum \ln y_i}$$

Now comparing ME and MLE of θ is

- (i) Both Estimators are Different in their functional form.
- (ii) The moment Estimator is Simpler than the Maximum likelihood Estimator.

4.15 ME and MLE for Parameter β of Laplace Distribution

Question: Consider the Laplace distribution centre at origin with shape parameter β having p.d.f

$$f(x; \beta) = \frac{1}{2\beta} e^{-\frac{|x|}{\beta}}, -\infty < x < \infty, \beta > 0$$

Find the moments estimator as well as the maximum likelihood estimator of β .

Solution

(a) Moment Estimator

The population moment

$$\mu'_1 = \int_{-\infty}^{\infty} x \frac{1}{2\beta} e^{-\frac{|x|}{\beta}} dx = \frac{2}{2\beta} \int_0^{\infty} x e^{-\frac{x}{\beta}} dx$$

Let

$$t = \frac{x}{\beta} \Rightarrow t\beta = x \Rightarrow \beta dt = dx$$

$$\mu'_1 = \frac{1}{\beta} \int_0^{\infty} t\beta e^{-t} (\beta dt) = \beta \int_0^{\infty} t^{2-1} e^{-t} dt = \beta \Gamma 2 = \beta(2-1)! = \beta$$

Using $\mu'_1 = m'_1 = \bar{x}$

$$\beta = \bar{x}$$

Which is the Moment Estimator of β .

(b) Maximum Likelihood Estimator

The likelihood function

$$L(\beta) = \prod_{i=1}^n \frac{1}{2\beta} e^{-\frac{|x_i|}{\beta}} = \left(\frac{1}{2\beta}\right)^n e^{-\frac{\sum |x_i|}{\beta}}$$

Taking ln both sides

$$\ln (L(\beta)) = n \ln 1 - n \ln \beta - n \ln 2 - \frac{\sum |x_i|}{\beta}$$

Diff w.r.t β

$$\frac{\partial}{\partial \beta} \ln (L(\beta)) = -\frac{n}{\beta} + \frac{\sum |x_i|}{\beta^2} \dots \dots \dots (1)$$

Put

$$\begin{aligned} \frac{\partial}{\partial \beta} \ln (L(\beta)) &= 0 \\ -\frac{n}{\beta} + \frac{\sum |x_i|}{\beta^2} &= 0 \Rightarrow n\beta = \sum |x_i| \\ \beta &= \frac{\sum |x_i|}{n} \end{aligned}$$

Diff (1) w.r.t β

$$\frac{\partial^2}{\partial \beta^2} \ln (L(\beta)) = \frac{n}{\beta^2} - \frac{2\sum |x_i|}{\beta^3} = \frac{n\beta - 2\sum |x_i|}{\beta^3}$$

Using $n\beta = \sum |x_i|$

$$\frac{\partial^2}{\partial \beta^2} \ln (L(\beta)) = \frac{\sum |x_i| - 2\sum |x_i|}{\beta^3} = \frac{-\sum |x_i|}{\beta^3} < 0$$

Hence, MLE for β

$$\hat{\beta} = \frac{\sum |x_i|}{n}$$

4.16 ME for Double Poisson Distribution.

Question: For the double Poisson distribution

$$f(x) = \frac{1}{2} \frac{e^{-\lambda_1} \lambda_1^x}{x!} + \frac{1}{2} \frac{e^{-\lambda_2} \lambda_2^x}{x!}$$

Show that the estimates of λ_1 and λ_2 by method of moments are

$$\hat{m}_1 + \sqrt{\hat{m}_2 - \hat{m}_1 - \hat{m}_1^2}$$

Solution:

The first Population moment

$$\begin{aligned}\mu'_1 &= \sum_{i=0}^{\infty} xP(x) = \sum_{i=0}^{\infty} x \left[\frac{1}{2} \frac{e^{-\lambda_1} \lambda_1^x}{x!} + \frac{1}{2} \frac{e^{-\lambda_2} \lambda_2^x}{x!} \right] \\ &= \frac{e^{-\lambda_1}}{2} \sum_{i=0}^{\infty} x \frac{\lambda_1^x}{x(x-1)!} + \frac{e^{-\lambda_2}}{2} \sum_{i=0}^{\infty} x \frac{\lambda_2^x}{x(x-1)!}\end{aligned}$$

$$\mu'_1 = \frac{e^{-\lambda_1} \lambda_1}{2} \sum_{i=1}^{\infty} \frac{\lambda_1^{x-1}}{(x-1)!} + \frac{e^{-\lambda_2} \lambda_2}{2} \sum_{i=1}^{\infty} \frac{\lambda_2^{x-1}}{(x-1)!} = \frac{e^{-\lambda_1} \lambda_1}{2} e^{\lambda_1} + \frac{e^{-\lambda_2} \lambda_2}{2} e^{\lambda_2}$$

$$\mu_1 = \frac{1}{2}(\lambda_1 + \lambda_2)$$

$$E[X(X-1)] = \sum_{i=0}^{\infty} x(x-1)P(x) = \sum_{i=0}^{\infty} x(x-1) \left[\frac{1}{2} \frac{e^{-\lambda_1} \lambda_1^x}{x!} + \frac{1}{2} \frac{e^{-\lambda_2} \lambda_2^x}{x!} \right]$$

$$\begin{aligned}E[X(X-1)] &= \frac{e^{-\lambda_1}}{2} \sum_{i=0}^{\infty} x(x-1) \frac{\lambda_1^x}{x(x-1)(x-2)!} \\ &\quad + \frac{e^{-\lambda_2}}{2} \sum_{i=0}^{\infty} x(x-1) \frac{\lambda_2^x}{x(x-1)(x-2)!}\end{aligned}$$

$$E[X(X-1)] = \frac{e^{-\lambda_1} \lambda_1^2}{2} \sum_{i=2}^{\infty} \frac{\lambda_1^{x-2}}{(x-2)!} + \frac{e^{-\lambda_2} \lambda_2^2}{2} \sum_{i=2}^{\infty} \frac{\lambda_2^{x-2}}{(x-2)!} = \frac{e^{-\lambda_1} \lambda_1^2}{2} e^{\lambda_1} + \frac{e^{-\lambda_2} \lambda_2^2}{2} e^{\lambda_2}$$

$$E[X(X-1)] = \frac{1}{2}[\lambda_1^2 + \lambda_2^2]$$

$$\mu_2 = E[X(X-1)] + E(X) = \frac{1}{2}(\lambda_1^2 + \lambda_2^2) + \frac{1}{2}(\lambda_1 + \lambda_2)$$

Equating Population and sample moments

$$m'_1 = \frac{1}{2}(\lambda_1 + \lambda_2) \Rightarrow \lambda_1 + \lambda_2 = 2m'_1 \text{ --- (1)}$$

$$\hat{m}_2 = \frac{1}{2}(\lambda_1^2 + \lambda_2^2) + \frac{1}{2}(\lambda_1 + \lambda_2)$$

From (1)

$$\lambda_2 = 2\hat{m}_1 - \lambda_1 \dots \dots \dots (2)$$

$$\hat{m}_2 = \frac{\lambda_1^2}{2} + \frac{1}{2}(2\hat{m}_1 - \lambda_1)^2 + \frac{1}{2}(2\hat{m}_1)$$

$$2\hat{m}_2 = \lambda_1^2 + 4\hat{m}_1^2 + \lambda_1^2 - 4\hat{m}_1\lambda_1 + 2\hat{m}_1$$

$$0 = 2\lambda_1^2 - 4\hat{m}_1\lambda_1 + 4\hat{m}_1^2 - 2\hat{m}_2 + 2\hat{m}_1$$

$$2\lambda_1^2 - (4\hat{m}_1)\lambda_1 + 2(2\hat{m}_1^2 + \hat{m}_1 - \hat{m}_2) = 0$$

$$\lambda_1^2 - (2\hat{m}_1)\lambda_1 + (2\hat{m}_1^2 + \hat{m}_1 - \hat{m}_2) = 0$$

$$\lambda = \frac{-(-2\hat{m}_1) \pm \sqrt{(-2\hat{m}_1)^2 - 4(1)(2\hat{m}_1^2 + \hat{m}_1 - \hat{m}_2)}}{2(1)} = \frac{2\hat{m}_1 \pm \sqrt{4\hat{m}_1^2 - 8\hat{m}_1^2 - 4\hat{m}_1 + 4\hat{m}_2}}{2}$$

$$\lambda_1 = \frac{2\hat{m}_1 \pm \sqrt{4(\hat{m}_2 - \hat{m}_1 - \hat{m}_1^2)}}{2} = \frac{2\left[\hat{m}_1 \pm \sqrt{(\hat{m}_2 - \hat{m}_1 - \hat{m}_1^2)}\right]}{2} = \hat{m}_1 \pm \sqrt{(\hat{m}_2 - \hat{m}_1 - \hat{m}_1^2)}$$

Put in (2)

$$\lambda_2 = 2\hat{m}_1 - \left[\hat{m}_1 \pm \sqrt{(\hat{m}_2 - \hat{m}_1 - \hat{m}_1^2)}\right] = \hat{m}_1 \mp \sqrt{(\hat{m}_2 - \hat{m}_1 - \hat{m}_1^2)}$$

5 Bayesian Estimation

5.1 Introduction to Bayesian Estimation

In planning of any investigation or experiment some important information is available on the statistical properties of the variable to be studied. For instance, the type of distribution, the range of the parameters involved, etc. Heretofore, the methods of estimation did not make use of any such information. But Sir Thomas Bayes propounded the theory which makes use of the prior information to get the estimates of population parameters. This procedure is known as Bayesian Estimation.

The Bayesian theory has always been a matter of controversy. Bayesians believe that it is generally very effective, whereas, non-Bayesians consider it irrelevant

5.2 Comparison Between Bayesian and Non-Bayesian(Classical) Estimation

- In Classical inferences the parameters are fixed values, whereas, in Bayesian the true value of a parameter is taken as random variable to which we assign probability distribution known as Prior distribution.
- In classical, we do not have any prior information, whereas, in Bayesian analysis we have prior information before taking the sample.
- In classical inference we have to rely on large sample approximation by assuming asymptotic normality. In contrast to it in Bayesian inferences can be computed exactly even in highly complex situations.
- The Bayesian approach enables direct probability statements to be made about the parameters of interest. But in classical methods we make indirect inferences by considering the sample data.
- Classical $100(1 - \alpha)\%$ confidence interval does not provide estimate about the true value of the parameter with probability $100(1 - \alpha)$, but the Bayesian approach allows to construct the confidence interval which has probabilistic interpretation.

- Classical methods regard probability as a limiting, long term approach and only applies to the events which are repeatable. But in Bayesian approach the probability is regarded as measure of personal belief about the value of the unknown parameter.
- In classical the base of inferences is likelihood function and in Bayesian the base of inferences is posterior distribution.
- In classical analysis the estimate of parameter is Maximum Likelihood Estimate, but in Bayesian analysis the estimate of parameter is the mean, median or mode of the posterior distribution.

5.3 Prior Distribution

- Prior distribution represents additional (External Information) that is available to the investigator about the parameter.
- It can be viewed as representing the current description of uncertainty about the model parameters prior to the data being observed.
- It is used to quantify our knowledge about the parameters before we gather data.
- Let $X_1, X_2, \dots, X_n$ be a random sample from the density $f(x|\theta)$, where θ be a value of a random variable Θ (Parameter Space). $g(\theta)$ is known and contain the unknown parameter, then $g(\theta)$ is said to be prior distribution.

5.4 Posterior Distribution

- Posterior distribution is the conditional probability that is assigned after the relevant evidence is taken into account.
- The posterior distribution of θ is a weighted compromise between the prior information and the sample data.
- It expresses what is known about a set of parameters based upon on both sample data and prior information.
- Let $X_1, X_2, \dots, X_n$ denotes the random sample of size “n” having likelihood function $p(x|\theta)$ and let the prior distribution is $p(\theta)$ then the posterior probability distribution is defined as

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)} = \frac{p(x|\theta)p(\theta)}{\int p(x|\theta)p(\theta)} \propto p(\theta)L(\theta|x)$$

It can be written in the form

$$\text{Posterior Distribution} \propto \text{Likelihood Function} \times \text{Prior Probability}$$

5.5 Bayes Estimator

“The mean or median of the posterior distribution of the unknown parameter θ is called Bayes estimator”.

Or

An estimator or decision rule that minimizes the posterior expected value of a loss function or it maximizes the posterior expectation of the utility function is known as Bayes Estimator”.

If θ be any parameter and $g(\theta)$ is prior probability distribution, also $L(x|\theta)$ be the likelihood function then the Bayes estimator is denoted by $E(\theta)$ is calculated as

$$\hat{\theta} = E(\theta) = \int \theta \frac{L(x|\theta)g(\theta)}{\int L(x|\theta)g(\theta)} d\theta$$

5.6 Properties of Bayes Estimator

- Bayes estimator is always a function of minimal sufficient statistic.
- For large n , Bayes estimators always tend to maximum likelihood estimators, whatever may be the prior density $g(\theta)$.
- In many cases Bayes estimators are asymptotically consistent.

5.7 Construction of Posterior Distribution

A typical Bayesian analysis can be outlined in the following steps.

- 1- Formulate a probability model for data.
- 2- Decide on the prior distribution, which quantifies the uncertainty on the values of the unknown model parameters before the data is observed.
- 3- Observed the data and construct the likelihood function based upon the data and the probability model formulated in step 1.
- 4- The likelihood function is then combined with the prior distribution to determine the posterior distribution which quantifies the uncertainty in the value of the unknown model parameter after the data are observed.

- 5- summarize the important features of posterior distribution or calculate quantities of interest based upon the posterior distribution. These quantities constitute statistical output, such as point estimate and interval estimate.



6. Questions Related to Bayesian Estimation

6.1 Bayesian Estimator from Exponential Distribution

Question: Suppose that the time to failure “T” of an electronic tube is an exponential random variable with parameter λ . We also assume the prior for λ as $g(y) = 160e^{-160\lambda}$. Obtain the Bayes estimator for λ

Solution:-

The prior distribution is

$$g(\lambda) = 160e^{-160\lambda}$$

The sampling Distribution is (Exponential Distribution)

$$f(t) = \lambda e^{-\lambda t}, t > 0$$

The likelihood function

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda t_i} = \lambda^n e^{-\lambda \sum t_i}$$

Now,

$$L(\lambda)g(\lambda) = (\lambda^n e^{-\lambda \sum t_i})(160e^{-160\lambda}) = 160\lambda^n e^{-\lambda(\sum t_i + 160)}$$

Integrate both sides

$$\int_0^{\infty} L(\lambda)g(\lambda)d\lambda = \int_0^{\infty} 160\lambda^n e^{-\lambda(\sum t_i + 160)} d\lambda$$

Put

$$z = \lambda(\sum t_i + 160), \Rightarrow \lambda = \frac{z}{\sum t_i + 160} \Rightarrow d\lambda = \frac{dz}{\sum t_i + 160}$$

$$\int_0^{\infty} L(\lambda)g(\lambda)d\lambda = 160 \int_0^{\infty} \left(\frac{z}{\sum t_i + 160}\right)^n e^{-z} \left(\frac{dz}{\sum t_i + 160}\right) = \frac{160}{(\sum t_i + 160)^{n+1}} \int_0^{\infty} z^{n+1-1} e^{-z} dz$$

$$\int_0^{\infty} L(\lambda)g(\lambda)d\lambda = \frac{160}{(\sum t_i + 160)^{n+1}} \Gamma(n+1)$$

Now, the Posterior Distribution

$$P(\lambda|t) = \frac{L(\lambda)g(\lambda)}{\int_0^\infty L(\lambda)g(\lambda)d\lambda} = \frac{160\lambda^n e^{-\lambda(\sum t + 160)}}{\frac{160}{(\sum t + 160)^{n+1}} \Gamma(n+1)} = \frac{\lambda^n (\sum t + 160)^{n+1} e^{-\lambda(\sum t + 160)}}{\Gamma(n+1)}$$

Now, The Bayes Estimator

$$E(\lambda) = \int_0^\infty \lambda \frac{\lambda^n (\sum t + 160)^{n+1} e^{-\lambda(\sum t + 160)}}{\Gamma(n+1)} d\lambda = \frac{(\sum t + 160)^{n+1}}{\Gamma(n+1)} \int_0^\infty \lambda^{n+1} e^{-\lambda(\sum t + 160)} d\lambda$$

Let

$$z = \lambda(\sum t + 160), \Rightarrow \lambda = \frac{z}{\sum t + 160} \Rightarrow d\lambda = \frac{dz}{\sum t + 160}$$

$$\begin{aligned} E(\lambda) &= \frac{(\sum t + 160)^{n+1}}{\Gamma(n+1)} \int_0^\infty \left(\frac{z}{\sum t + 160}\right)^{n+1} e^{-z} \left(\frac{dz}{\sum t + 160}\right) \\ &= \frac{(\sum t + 160)^{n+1}}{\Gamma(n+1)(\sum t + 160)^{n+2}} \int_0^\infty z^{n+2-1} e^{-z} dz \end{aligned}$$

$$E(\lambda) = \frac{\Gamma(n+2)}{\Gamma(n+1)(\sum t + 160)} = \frac{(n+1)n!}{(\sum t + 160)n!} = \frac{(n+1)}{(\sum t + 160)}$$

Hence the Bayes Estimator for λ is

$$\hat{\lambda} = \frac{(n+1)}{(\sum t + 160)}$$

6.2 Bayesian Estimator from Exponential Distribution

Question: Suppose the length of nylon string which can be extended by a machine without a break occurring is an exponential random variable X with parameter λ . Also assume an exponential prior density with parameter $\frac{1}{1000}$. Given a sample of 1 observation (and we find $x=1000$), compute the posterior distribution for λ and also Bayes estimator.

Solution

The prior distribution is (Since Prior Distribution only depends upon parameter)

$$g(\lambda) = \frac{1}{1000} e^{-\frac{\lambda}{1000}}$$

The sampling Distribution is (Exponential Distribution)

$$f(x) = \lambda e^{-\lambda x}, x > 0$$

The likelihood function (Sample of one observation)

$$L(\lambda) = \lambda e^{-\lambda x}$$

Now,

$$L(\lambda)g(\lambda) = (\lambda e^{-\lambda x}) \left(\frac{1}{1000} e^{-\frac{\lambda}{1000}} \right) = \frac{\lambda}{1000} e^{-\lambda(x + \frac{1}{1000})}$$

Integrate both sides

$$\int_0^{\infty} L(\lambda)g(\lambda)d\lambda = \int_0^{\infty} \frac{\lambda}{1000} e^{-\lambda(x + \frac{1}{1000})} d\lambda$$

Put

$$z = \lambda \left(x + \frac{1}{1000} \right), \Rightarrow \lambda = \frac{z}{\left(x + \frac{1}{1000} \right)} \Rightarrow d\lambda = \frac{dz}{\left(x + \frac{1}{1000} \right)}$$

$$\int_0^{\infty} L(\lambda)g(\lambda)d\lambda = \frac{1}{1000} \int_0^{\infty} \frac{z}{\left(x + \frac{1}{1000} \right)} e^{-z} \frac{dz}{\left(x + \frac{1}{1000} \right)} = \frac{1}{1000 \left(x + \frac{1}{1000} \right)^2} \int_0^{\infty} z^{2-1} e^{-z} dz$$

$$\int_0^{\infty} L(\lambda)g(\lambda)d\lambda = \frac{1}{1000 \left(x + \frac{1}{1000} \right)^2} \Gamma 2 = \frac{1}{1000 \left(x + \frac{1}{1000} \right)^2}$$

Now, the Posterior Distribution

$$P(\lambda|x) = \frac{L(\lambda)g(\lambda)}{\int_0^{\infty} L(\lambda)g(\lambda)d\lambda} = \frac{\frac{\lambda}{1000} e^{-\lambda(x + \frac{1}{1000})}}{\frac{1}{1000 \left(x + \frac{1}{1000} \right)^2}} = \left(x + \frac{1}{1000} \right)^2 \lambda e^{-\lambda(x + \frac{1}{1000})}$$

Now, The Bayes Estimator

$$E(\lambda) = \int_0^{\infty} \lambda \left(x + \frac{1}{1000} \right)^2 \lambda e^{-\lambda(x + \frac{1}{1000})} d\lambda = \left(x + \frac{1}{1000} \right)^2 \int_0^{\infty} \lambda^2 e^{-\lambda(x + \frac{1}{1000})} d\lambda$$

Let

$$z = \lambda \left(x + \frac{1}{1000} \right), \Rightarrow \lambda = \frac{z}{\left(x + \frac{1}{1000} \right)} \Rightarrow d\lambda = \frac{dz}{\left(x + \frac{1}{1000} \right)}$$

$$E(\lambda) = \left(x + \frac{1}{1000} \right)^2 \int_0^\infty \frac{z^2}{\left(x + \frac{1}{1000} \right)^2} e^{-z} \frac{dz}{\left(x + \frac{1}{1000} \right)} = \frac{1}{\left(x + \frac{1}{1000} \right)} \int_0^\infty z^{3-1} e^{-z} dz$$

$$E(\lambda) = \frac{\Gamma 3}{\left(x + \frac{1}{1000} \right)} = \frac{(3-1)!}{\left(x + \frac{1}{1000} \right)} = \frac{2}{\left(x + \frac{1}{1000} \right)}$$

Hence Bayes Estimator for λ for sample of one observation is

$$\hat{\lambda} = \frac{2}{\left(x + \frac{1}{1000} \right)}$$

6.3 Bayesian Estimator from Exponential Distribution

Question: The random variable “X” follows the exponential distribution with parameter λ . we assume the prior density for unknown λ as

$$f(\lambda) = \frac{1}{m!} \left(\frac{m+1}{\lambda_0} \right)^{m+1} \lambda^m e^{-\frac{(m+1)\lambda}{\lambda_0}}$$

Given the random sample of size “n”. Estimate the Bayes estimator.

Solution:

The prior distribution is

$$g(\lambda) = \frac{1}{m!} \left(\frac{m+1}{\lambda_0} \right)^{m+1} \lambda^m e^{-\frac{(m+1)\lambda}{\lambda_0}}$$

The sampling Distribution is (Exponential Distribution)

$$f(x) = \lambda e^{-\lambda x}, t > 0$$

The likelihood function

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum x_i}$$

Now,

$$\begin{aligned} L(\lambda)g(\lambda) &= (\lambda^n e^{-\lambda \sum x_i}) \frac{1}{m!} \left(\frac{m+1}{\lambda_0} \right)^{m+1} \lambda^m e^{-\frac{(m+1)\lambda}{\lambda_0}} \\ &= \frac{1}{m!} \left(\frac{m+1}{\lambda_0} \right)^{m+1} \lambda^{m+n} e^{-\lambda \left[\frac{(m+1)}{\lambda_0} + \sum x_i \right]} \end{aligned}$$

$$L(\lambda)g(\lambda) = K \lambda^{m+n} e^{-\lambda \left[\frac{(m+1)}{\lambda_0} + \sum x_i \right]}$$

$$K = \frac{1}{m!} \left(\frac{m+1}{\lambda_0} \right)^{m+1}$$

Integrate both sides

$$\int_0^{\infty} L(\lambda)g(\lambda)d\lambda = \int_0^{\infty} K \lambda^{m+n} e^{-\lambda \left[\frac{(m+1)}{\lambda_0} + \sum x_i \right]} d\lambda$$

Let

$$z = \lambda \left[\frac{(m+1)}{\lambda_0} + \sum x \right] \Rightarrow \lambda = \frac{z}{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]} \Rightarrow d\lambda = \frac{dz}{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]}$$

$$\int_0^\infty L(\lambda)g(\lambda)d\lambda = K \int_0^\infty \left(\frac{z}{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]} \right)^{m+n} e^{-z} \frac{dz}{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]}$$

$$\int_0^\infty L(\lambda)g(\lambda)d\lambda$$

$$= \frac{K}{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]^{m+n+1}} \int_0^\infty z^{m+n+1-1} e^{-z} dz = \frac{K \Gamma(m+n+1)}{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]^{m+n+1}}$$

$$\begin{aligned} P(\lambda|x) &= \frac{L(\lambda)g(\lambda)}{\int_0^\infty L(\lambda)g(\lambda)d\lambda} = \frac{K \lambda^{m+n} e^{-\lambda \left[\frac{(m+1)}{\lambda_0} + \sum x \right]}}{\frac{K \Gamma(m+n+1)}{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]^{m+n+1}}} \\ &= \frac{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]^{m+n+1}}{\Gamma(m+n+1)} \lambda^{m+n} e^{-\lambda \left[\frac{(m+1)}{\lambda_0} + \sum x \right]} \end{aligned}$$

Bayes Estimator

$$E(\lambda) = \frac{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]^{m+n+1}}{\Gamma(m+n+1)} \int_0^\infty \lambda^{m+n+1} e^{-\lambda \left[\frac{(m+1)}{\lambda_0} + \sum x \right]} d\lambda$$

Let

$$z = \lambda \left[\frac{(m+1)}{\lambda_0} + \sum x \right] \Rightarrow \lambda = \frac{z}{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]} \Rightarrow d\lambda = \frac{dz}{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]}$$

$$E(\lambda) = \frac{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]^{m+n+1}}{\Gamma(m+n+1)} \int_0^\infty \left(\frac{z}{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]} \right)^{m+n+1} e^{-z} \frac{dz}{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]}$$

$$E(\lambda) = \frac{1}{\Gamma(m+n+1) \left[\frac{(m+1)}{\lambda_0} + \sum x \right]} \int_0^{\infty} z^{m+n+2-1} e^{-z} dz$$

$$= \frac{\Gamma(m+n+2)}{\Gamma(m+n+1) \left[\frac{(m+1)}{\lambda_0} + \sum x \right]}$$

$$E(\lambda) = \frac{(m+n+1)(m+n)!}{(m+n)! \left[\frac{(m+1)}{\lambda_0} + \sum x \right]} = \frac{(m+n+1)}{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]}$$

Hence, Bayes Estimator for λ

$$\hat{\lambda} = \frac{(m+n+1)}{\left[\frac{(m+1)}{\lambda_0} + \sum x \right]}$$

6.3 Bayesian Estimator from Exponential Distribution

Question: The random variable “X” follows the Poisson distribution with parameter λ . we assume the prior density for unknown λ as

$$f(\lambda) = \frac{1}{m!} \left(\frac{m+1}{\lambda_0} \right)^{m+1} \lambda^m e^{-\frac{(m+1)\lambda}{\lambda_0}}$$

Given the random sample of size “n”. Estimate the Bayes estimator.

Solution:

The prior distribution is

$$g(\lambda) = \frac{1}{m!} \left(\frac{m+1}{\lambda_0} \right)^{m+1} \lambda^m e^{-\frac{(m+1)\lambda}{\lambda_0}}$$

The sampling Distribution is (Poisson Distribution)

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}, x > 0$$

The likelihood function

$$L(\lambda) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \frac{\lambda^{\sum x_i} e^{-n\lambda}}{\prod x_i!}$$

Now,

$$\begin{aligned} L(\lambda)g(\lambda) &= \left(\frac{\lambda^{\sum x_i} e^{-n\lambda}}{\prod x_i!} \right) \frac{1}{m!} \left(\frac{m+1}{\lambda_0} \right)^{m+1} \lambda^m e^{-\frac{(m+1)\lambda}{\lambda_0}} \\ &= \left[\frac{\left(\frac{m+1}{\lambda_0} \right)^{m+1}}{\prod x_i m!} \right] \lambda^{m+\sum x_i} e^{-\lambda \left[\frac{(m+1)}{\lambda_0} + n \right]} \end{aligned}$$

$$L(\lambda)g(\lambda) = K \lambda^{m+\sum x_i} e^{-\lambda \left[\frac{(m+1)}{\lambda_0} + n \right]}$$

$$K = \left[\frac{\left(\frac{m+1}{\lambda_0} \right)^{m+1}}{\prod x_i m!} \right]$$

Integrate both sides

$$\int_0^{\infty} L(\lambda)g(\lambda)d\lambda = \int_0^{\infty} K \lambda^{m+\Sigma x} e^{-\lambda\left[\frac{(m+1)}{\lambda_0}+n\right]}d\lambda$$

Let

$$z = \lambda \left[\frac{(m+1)}{\lambda_0} + n \right] \Rightarrow \lambda = \frac{z}{\left[\frac{(m+1)}{\lambda_0} + n \right]} \Rightarrow d\lambda = \frac{dz}{\left[\frac{(m+1)}{\lambda_0} + n \right]}$$

$$\int_0^{\infty} L(\lambda)g(\lambda)d\lambda = K \int_0^{\infty} \left(\frac{z}{\left[\frac{(m+1)}{\lambda_0} + n \right]} \right)^{m+\Sigma x} e^{-z} \frac{dz}{\left[\frac{(m+1)}{\lambda_0} + n \right]}$$

$$\begin{aligned} \int_0^{\infty} L(\lambda)g(\lambda)d\lambda &= \frac{K}{\left[\frac{(m+1)}{\lambda_0} + n \right]^{m+\Sigma x+1}} \int_0^{\infty} z^{m+\Sigma x+1-1} e^{-z} dz \\ &= \frac{K \Gamma(m + \Sigma x + 1)}{\left[\frac{(m+1)}{\lambda_0} + n \right]^{m+\Sigma x+1}} \end{aligned}$$

Posterior Distribution

$$\begin{aligned} P(\lambda|x) &= \frac{L(\lambda)g(\lambda)}{\int_0^{\infty} L(\lambda)g(\lambda)d\lambda} = \frac{K \lambda^{m+\Sigma x} e^{-\lambda\left[\frac{(m+1)}{\lambda_0}+n\right]}}{\frac{K \Gamma(m + \Sigma x + 1)}{\left[\frac{(m+1)}{\lambda_0} + n \right]^{m+\Sigma x+1}}} \\ &= \frac{\left[\frac{(m+1)}{\lambda_0} + n \right]^{m+\Sigma x+1}}{\Gamma(m + \Sigma x + 1)} \lambda^{m+\Sigma x} e^{-\lambda\left[\frac{(m+1)}{\lambda_0}+n\right]} \end{aligned}$$

Bayes Estimator

$$E(\lambda) = \frac{\left[\frac{(m+1)}{\lambda_0} + n\right]^{m+\sum x+1}}{\Gamma(m+\sum x+1)} \int_0^\infty \lambda^{m+\sum x+1} e^{-\lambda\left[\frac{(m+1)}{\lambda_0} + n\right]} d\lambda$$

Let

$$z = \lambda \left[\frac{(m+1)}{\lambda_0} + n\right] \Rightarrow \lambda = \frac{z}{\left[\frac{(m+1)}{\lambda_0} + n\right]} \Rightarrow d\lambda = \frac{dz}{\left[\frac{(m+1)}{\lambda_0} + n\right]}$$

$$E(\lambda) = \frac{\left[\frac{(m+1)}{\lambda_0} + n\right]^{m+\sum x+1}}{\Gamma(m+\sum x+1)} \int_0^\infty \left(\frac{z}{\left[\frac{(m+1)}{\lambda_0} + n\right]}\right)^{m+\sum x+1} e^{-z} \frac{dz}{\left[\frac{(m+1)}{\lambda_0} + n\right]}$$

$$\begin{aligned} E(\lambda) &= \frac{1}{\Gamma(m+\sum x+1) \left[\frac{(m+1)}{\lambda_0} + n\right]} \int_0^\infty z^{m+\sum x+2-1} e^{-z} dz \\ &= \frac{\Gamma(m+\sum x+2)}{\Gamma(m+\sum x+1) \left[\frac{(m+1)}{\lambda_0} + n\right]} \end{aligned}$$

$$E(\lambda) = \frac{(m+\sum x+1)(m+\sum x)!}{(m+\sum x)! \left[\frac{(m+1)}{\lambda_0} + n\right]} = \frac{(m+\sum x+1)}{\left[\frac{(m+1)}{\lambda_0} + n\right]}$$

Hence, Bayes Estimator for λ

$$\hat{\lambda} = \frac{(m+\sum x+1)}{\left[\frac{(m+1)}{\lambda_0} + n\right]} = \frac{\lambda_0(m+\sum x+1)}{n\lambda_0 + m + 1}$$

6.5 Bayesian Estimator for Bernoulli Distribution

Question: Let $X_1, X_2, \dots, X_n$ be random sample from Bernoulli distribution. Assume, that the prior distribution of θ is given by

$$g(\theta) = I_{(0,1)}(\theta)$$

Find the posterior Bayes Estimator θ and $t(\theta) = \theta(\theta - 1)$.

Solution:-

The prior distribution is

$$g(\theta) = I_{(0,1)}(\theta)$$

The sampling Distribution is

$$f(\theta) = \theta^x (1 - \theta)^{1-x}, x = 0, 1$$

The likelihood function

$$L(\theta) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^{\sum x_i} (1 - \theta)^{n - \sum x_i}$$

Now,

$$L(\theta)g(\theta) = \theta^{\sum x} (1 - \theta)^{n - \sum x}$$

Integrate both sides

$$\begin{aligned} \int_0^1 L(\theta)g(\theta)d\theta &= \int_0^1 \theta^{\sum x} (1 - \theta)^{n - \sum x} d\theta = \int_0^1 \theta^{\sum x + 1 - 1} (1 - \theta)^{n - \sum x + 1 - 1} d\theta \\ &= \beta(\sum x + 1, n - \sum x + 1) \end{aligned}$$

Posterior Distribution

$$P(\theta|x) = \frac{L(\theta)g(\theta)}{\int_0^1 L(\theta)g(\theta)d\theta} = \frac{\theta^{\sum x} (1 - \theta)^{n - \sum x}}{\beta(\sum x + 1, n - \sum x + 1)}$$

Bayes Estimator for θ

$$E(\theta) = \frac{1}{\beta(\sum x + 1, n - \sum x + 1)} \int_0^1 \theta^{\sum x + 1} (1 - \theta)^{n - \sum x} d\theta$$

$$E(\theta) = \frac{1}{\beta(\sum x + 1, n - \sum x + 1)} \int_0^1 \theta^{\sum x + 2 - 1} (1 - \theta)^{n - \sum x + 1 - 1} d\theta = \frac{\beta(\sum x + 2, n - \sum x + 1)}{\beta(\sum x + 1, n - \sum x + 1)}$$

$$E(\theta) = \frac{\Gamma(\sum x + 2) \Gamma(n - \sum x + 1)}{\Gamma(n - \sum x + 1 + \sum x + 2)} X \frac{\Gamma(n - \sum x + 1 + \sum x + 1)}{\Gamma(\sum x + 1) \Gamma(n - \sum x + 1)}$$

$$E(\theta) = \frac{\Gamma(\sum x + 2)}{\Gamma(n + 3)} X \frac{\Gamma(n + 2)}{\Gamma(\sum x + 1)}$$

$$E(\theta) = \frac{(\sum x + 1)(\sum x)!}{(n + 2)(n + 1)!} X \frac{(n + 1)!}{(\sum x)!} = \frac{(\sum x + 1)}{(n + 2)}$$

-0Hence, Bayes Estimator for θ

$$\hat{\theta} = \frac{(\sum x + 1)}{(n + 2)}$$

Now, Posterior Bayes Estimator for $t(\theta) = \theta(\theta - 1)$

We have to define risk function as

$$R(t) = E[L(t; \theta)] = E[\theta(1 - \theta)]$$

$$R(t) = \frac{1}{\beta(\sum x + 1, n - \sum x + 1)} \int_0^1 \theta(1 - \theta) \theta^{\sum x} (1 - \theta)^{n - \sum x} d\theta$$

$$R(t) = \frac{1}{\beta(\sum x + 1, n - \sum x + 1)} \int_0^1 \theta^{\sum x + 2 - 1} (1 - \theta)^{n - \sum x + 2 - 1} d\theta = \frac{\beta(\sum x + 2, n - \sum x + 2)}{\beta(\sum x + 1, n - \sum x + 1)}$$

$$R(t) = \frac{\Gamma(\sum x + 2) \Gamma(n - \sum x + 2)}{\Gamma(n - \sum x + 2 + \sum x + 2)} X \frac{\Gamma(n - \sum x + 1 + \sum x + 1)}{\Gamma(\sum x + 1) \Gamma(n - \sum x + 1)}$$

$$R(t) = \frac{\Gamma(\sum x + 2) \Gamma(n - \sum x + 2)}{\Gamma(n + 4)} X \frac{\Gamma(n + 2)}{\Gamma(\sum x + 1) \Gamma(n - \sum x + 1)}$$

$$R(t) = \frac{(\sum x + 1)(\sum x)! (n - \sum x + 1)(n - \sum x)!}{(n + 3)(n + 2)(n + 1)!} X \frac{(n + 1)!}{(\sum x)! (n - \sum x)!}$$

$$R(t) = \frac{(\sum x + 1) (n - \sum x + 1)}{(n + 3)(n + 2)}$$

Which is the posterior Bayes Estimator for $t(\theta)$.



6.6 Bayes Estimator for Bernoulli distribution with Loss function

Question: Let “x” be a random sample from $f(x|\theta) = \theta^x(1 - \theta)^{1-x}$, $x = 0, 1$ Where θ is distributed a prior as uniform over (0,1). Find the Bayes estimator of θ w.r.t the Loss function

$$L(\theta; t) = \frac{(\theta - t)^2}{\theta(1 - \theta)}$$

Solution:

The prior distribution is

$$g(\theta) = I_{(0,1)}(\theta)$$

The sampling Distribution is

$$f(\theta) = \theta^x(1 - \theta)^{1-x}, x = 0, 1$$

The likelihood function

$$L(\theta) = \prod_{i=1}^n \theta^{x_i}(1 - \theta)^{1-x_i} = \theta^{\sum x_i}(1 - \theta)^{n - \sum x_i}$$

Now,

$$L(\theta)g(\theta) = \theta^{\sum x} (1 - \theta)^{n - \sum x}$$

Integrate both sides

$$\begin{aligned} \int_0^1 L(\theta)g(\theta)d\theta &= \int_0^1 \theta^{\sum x} (1 - \theta)^{n - \sum x} d\theta = \int_0^1 \theta^{\sum x + 1 - 1} (1 - \theta)^{n - \sum x + 1 - 1} d\theta \\ &= \beta(\sum x + 1, n - \sum x + 1) \end{aligned}$$

Posterior Distribution

$$P(\theta|x) = \frac{L(\theta)g(\theta)}{\int_0^1 L(\theta)g(\theta)d\theta} = \frac{\theta^{\sum x} (1 - \theta)^{n - \sum x}}{\beta(\sum x + 1, n - \sum x + 1)}$$

Now, to find Bayes Estimator for θ w.r.t to given Loss function we have to define the risk function as

$$R(t) = \int L(t; \theta)P(\theta|x)d\theta$$

$$R(t) = \int_0^1 \frac{(\theta - t)^2}{\theta(1 - \theta)} \frac{\theta^{\sum x} (1 - \theta)^{n - \sum x}}{\beta(\sum x + 1, n - \sum x + 1)} d\theta$$

We have to minimize the Risk function w.r.t “t”

$$\frac{\partial}{\partial t} R(t) = \frac{1}{\beta(\sum x + 1, n - \sum x + 1)} \int_0^1 \frac{2(\theta - t)(-1)}{\theta(1 - \theta)} \theta^{\sum x} (1 - \theta)^{n - \sum x} d\theta$$

$$\frac{\partial}{\partial t} R(t) = \frac{2}{\beta(\sum x + 1, n - \sum x + 1)} \left[t \int_0^1 \frac{1}{\theta(1 - \theta)} \theta^{\sum x} (1 - \theta)^{n - \sum x} d\theta - \int_0^1 \frac{\theta}{\theta(1 - \theta)} \theta^{\sum x} (1 - \theta)^{n - \sum x} d\theta \right]$$

$$\frac{\partial}{\partial t} R(t) = \frac{2}{\beta(\sum x + 1, n - \sum x + 1)} \left[t \int_0^1 \theta^{\sum x - 1} (1 - \theta)^{n - \sum x - 1} d\theta - \int_0^1 \theta^{\sum x + 1 - 1} (1 - \theta)^{n - \sum x - 1} d\theta \right]$$

$$\frac{\partial}{\partial t} R(t) = \frac{2}{\beta(\sum x + 1, n - \sum x + 1)} [t\beta(\sum x, n - \sum x) - \beta(\sum x + 1, n - \sum x)]$$

$$\frac{\partial}{\partial t} R(t) = \frac{2}{\beta(\sum x + 1, n - \sum x + 1)} \left[t \frac{\Gamma(\sum x) \Gamma(n - \sum x)}{\Gamma(n - \sum x + \sum x)} - \frac{\Gamma(\sum x + 1) \Gamma(n - \sum x)}{\Gamma(n - \sum x + \sum x + 1)} \right]$$

$$\frac{\partial}{\partial t} R(t) = \frac{2}{\beta(\sum x + 1, n - \sum x + 1)} \left[t \frac{\Gamma(\sum x) \Gamma(n - \sum x)}{\Gamma(n)} - \frac{\Gamma(\sum x + 1) \Gamma(n - \sum x)}{\Gamma(n + 1)} \right]$$

$$\frac{\partial}{\partial t} R(t) = \frac{2}{\beta(\sum x + 1, n - \sum x + 1)} \left[t \frac{(\sum x - 1)! (n - \sum x - 1)!}{(n - 1)!} - \frac{(\sum x)(\sum x - 1)! (n - \sum x - 1)!}{(n)(n - 1)!} \right]$$

$$\frac{\partial}{\partial t} R(t) = \frac{2(\sum x - 1)! (n - \sum x - 1)!}{\beta(\sum x + 1, n - \sum x + 1)(n - 1)!} \left[t - \frac{\sum x}{n} \right] \dots \dots \dots (1)$$

Put

$$\frac{\partial}{\partial t} R(t) = 0$$

$$\frac{2(\sum x - 1)! (n - \sum x - 1)!}{\beta(\sum x + 1, n - \sum x + 1)(n - 1)!} \left[t - \frac{\sum x}{n} \right] = 0$$

$$t - \frac{\sum x}{n} = 0 \Rightarrow t = \bar{x}$$

Diff (1) w.r.t “t”

$$\frac{\partial}{\partial t} R(t) = \frac{2(\sum x - 1)! (n - \sum x - 1)!}{\beta(\sum x + 1, n - \sum x + 1)(n - 1)!} [1] > 0$$

Hence, the Risk function is minimized so the Bayes estimator w.r.t to given loss function is

$$t = \bar{x}$$

6.7 Bayes Estimator for Uniform distribution with Loss function

Question: Based upon random sample of size n from the density

$$f(x) = \frac{1}{\theta}, 0 < x < \theta \text{ or } f(x|\theta) = \frac{1}{\theta} I_{(0,\theta)}(x)$$

With prior distribution as $g(\theta) = 1; 0 < \theta < 1$. Obtain Bayes estimator of θ w.r.t the Loss function

$$L(\theta, t) = \frac{(t - \theta)^2}{\theta^2}$$

Solution:-

The prior distribution is

$$g(\theta) = 1$$

The sampling Distribution is

$$f(\theta) = \frac{1}{\theta}, 0 < x < \theta$$

The likelihood function

$$L(\theta) = \prod_{i=1}^n \frac{1}{\theta} = \left(\frac{1}{\theta}\right)^n$$

Now,

$$L(\theta)g(\theta) = \left(\frac{1}{\theta}\right)^n = \frac{1}{\theta^n}$$

Since, the range of X involves θ so, we have to merge the range for x and range for θ

$$0 < x < \theta ; 0 < \theta < 1$$

$$0 < y_1 < y_2 \dots < y_n < \theta ; 0 < \theta < 1$$

$$y_n < \theta < 1$$

Integrate both sides

$$\int_{y_n}^1 L(\theta)g(\theta)d\theta = \int_{y_n}^1 \left(\frac{1}{\theta}\right)^n d\theta = \frac{1}{1-n} \theta^{1-n} \Big|_{y_n}^1 = \frac{1}{1-n} [1 - y_n^{1-n}]$$

Posterior Distribution

$$\begin{aligned}
 P(\theta|x) &= \frac{L(\theta)g(\theta)}{\int_{y_n}^1 L(\theta)g(\theta)d\theta} = \frac{\frac{1}{\theta^n}}{\frac{1}{1-n}[1-y_n^{1-n}]} = \frac{1-n}{\theta^n[1-y_n^{1-n}]} = \frac{-(n-1)}{-\theta^n y_n^{1-n} \left[1 - \frac{1}{y_n^{1-n}}\right]} \\
 &= \frac{y_n^{n-1}(n-1)}{\theta^n(1-y_n^{n-1})}
 \end{aligned}$$

Now, to find Bayes Estimator for θ w.r.t to given Loss function we have to define the risk function as

$$R(t) = \int L(t; \theta)P(\theta|x)d\theta$$

$$R(t) = \int_{y_n}^1 \frac{(t-\theta)^2}{\theta^2} \frac{y_n^{n-1}(n-1)}{\theta^n(1-y_n^{n-1})} d\theta = \frac{y_n^{n-1}(n-1)}{1-y_n^{n-1}} \int_{y_n}^1 \frac{(t-\theta)^2}{\theta^{n+2}} d\theta$$

We have to minimize the Risk function w.r.t “t”

Diff w.r.t “t”

$$\frac{\partial}{\partial t} R(t) = \frac{y_n^{n-1}(n-1)}{1-y_n^{n-1}} \int_{y_n}^1 \frac{2(t-\theta)}{\theta^{n+2}} d\theta = \frac{2y_n^{n-1}(n-1)}{1-y_n^{n-1}} \left[t \int_{y_n}^1 \theta^{-n-2} d\theta - \int_{y_n}^1 \theta^{-n-1} d\theta \right]$$

$$\frac{\partial}{\partial t} R(t) = \frac{2y_n^{n-1}(n-1)}{1-y_n^{n-1}} \left[t \frac{1}{-n-1} \theta^{-n-1} \Big|_{y_n}^1 - \frac{1}{-n} \theta^{-n} \Big|_{y_n}^1 \right]$$

$$\frac{\partial}{\partial t} R(t) = \frac{2y_n^{n-1}(n-1)}{1-y_n^{n-1}} \left[\frac{t}{-(n+1)} \{1 - y_n^{-(n+1)}\} + \frac{1}{n} \{1 - y_n^{-n}\} \right]$$

$$\frac{\partial}{\partial t} R(t) = \frac{2y_n^{n-1}(n-1)}{1-y_n^{n-1}} \left[\frac{-t}{(n+1)} \left\{ \frac{y_n^{(n+1)} - 1}{y_n^{(n+1)}} \right\} + \frac{1}{n} \left\{ \frac{y_n^n - 1}{y_n^n} \right\} \right]$$

$$\frac{\partial}{\partial t} R(t) = \frac{2y_n^{n-1}(n-1)}{1-y_n^{n-1}} \left[\frac{t}{(n+1)} \left\{ \frac{1 - y_n^{(n+1)}}{y_n^{(n+1)}} \right\} - \frac{1}{n} \left\{ \frac{1 - y_n^n}{y_n^n} \right\} \right] \dots \dots (1)$$

$$\frac{\partial}{\partial t} R(t) = 0$$

$$\frac{2y_n^{n-1}(n-1)}{1-y_n^{n-1}} \left[\frac{t}{(n+1)} \left\{ \frac{1-y_n^{(n+1)}}{y_n^{(n+1)}} \right\} - \frac{1}{n} \left\{ \frac{1-y_n^n}{y_n^n} \right\} \right] = 0$$

$$\frac{t}{(n+1)} \left\{ \frac{1-y_n^{(n+1)}}{y_n^{(n+1)}} \right\} - \frac{1}{n} \left\{ \frac{1-y_n^n}{y_n^n} \right\} = 0$$

$$\frac{t}{(n+1)} \left\{ \frac{1-y_n^{(n+1)}}{y_n^{(n+1)}} \right\} = \frac{1}{n} \left\{ \frac{1-y_n^n}{y_n^n} \right\}$$

$$t = \frac{n+1}{n} \left\{ \frac{1-y_n^n}{y_n^n} \right\} \left\{ \frac{y_n^{(n+1)}}{1-y_n^{(n+1)}} \right\}$$

$$t = \left(\frac{n+1}{n} \right) \left\{ \frac{y_n(1-y_n^n)}{1-y_n^{(n+1)}} \right\}$$

Diff (1) w.r.t “t”

$$\frac{\partial^2}{\partial t^2} R(t) = \frac{2y_n^{n-1}(n-1)}{1-y_n^{n-1}} \left[\frac{1}{(n+1)} \frac{1-y_n^{(n+1)}}{y_n^{(n+1)}} \right] = \frac{2(n-1)(1-y_n^{(n+1)})}{(n+1)y_n^2(1-y_n^{n-1})} > 0$$

6.8 Bayes Estimator for Uniform distribution with Loss function

Question: Based upon random sample of size n from the density

$$f(x|\theta) = e^{-(x-\theta)} I_{(\theta, \infty)}(\theta)$$

With prior distribution as $g(\theta) = e^{-\theta} I_{(0, \infty)}(\theta)$; . Obtain Bayes estimator of θ

Solution:-

The prior distribution is

$$g(\theta) = e^{-\theta}, \quad 0 < \theta < \infty$$

The sampling Distribution is

$$f(\theta) = e^{-(x-\theta)}, \quad \theta < x < \infty$$

The likelihood function

$$L(\theta) = \prod_{i=1}^n e^{-(x-\theta)} = e^{-\sum x + n\theta}$$

Now,

$$L(\theta)g(\theta) = e^{-\sum x + n\theta} e^{-\theta} = e^{-\sum x + \theta(n-1)}$$

Since, the range of X involves θ so, we have to merge the range for x and range for θ

$$\theta < x < \infty ; 0 < \theta < \infty$$

$$\theta < y_1 < y_2 \dots < y_n < \infty ; 0 < \theta < \infty$$

$$0 < \theta < y_1 < y_2 \dots < y_n < \infty$$

$$0 < \theta < y_1$$

Integrate both sides

$$\begin{aligned} \int_0^{y_1} L(\theta)g(\theta)d\theta &= \int_0^{y_1} e^{-\sum x + \theta(n-1)} d\theta = e^{-\sum x} \int_0^{y_1} e^{\theta(n-1)} d\theta = e^{-\sum x} \frac{1}{n-1} e^{\theta(n-1)} \Big|_0^{y_1} \\ &= \frac{e^{-\sum x}}{n-1} [e^{y_1(n-1)} - 1] \end{aligned}$$

Posterior Distribution

$$P(\theta|x) = \frac{L(\theta)g(\theta)}{\int_0^{y_1} L(\theta)g(\theta)d\theta} = \frac{e^{-\sum x + \theta(n-1)}}{\frac{e^{-\sum x}}{n-1} [e^{y_1(n-1)} - 1]} = \frac{(n-1)e^{\theta(n-1)}}{e^{y_1(n-1)} - 1}$$

The posterior Bayes Estimator

$$E(\theta) = \int_0^{y_1} \theta \frac{(n-1)e^{\theta(n-1)}}{e^{y_1(n-1)} - 1} d\theta = \frac{n-1}{e^{y_1(n-1)} - 1} \int_0^{y_1} \theta e^{\theta(n-1)} d\theta$$

Integrate by Parts

$$E(\theta) = \frac{n-1}{e^{y_1(n-1)} - 1} \left[\frac{\theta e^{\theta(n-1)}}{n-1} - \frac{e^{\theta(n-1)}}{(n-1)^2} \right] \Big|_0^{y_1}$$

$$E(\theta) = \frac{n-1}{e^{y_1(n-1)} - 1} \left[\left\{ \frac{y_1 e^{y_1(n-1)}}{n-1} - \frac{e^{y_1(n-1)}}{(n-1)^2} \right\} - \left\{ 0 - \frac{1}{(n-1)^2} \right\} \right]$$

$$E(\theta) = \frac{n-1}{(n-1)e^{y_1(n-1)} - 1} \left[y_1 e^{y_1(n-1)} - \frac{e^{y_1(n-1)}}{n-1} + \frac{1}{n-1} \right]$$

$$E(\theta) = \frac{1}{e^{y_1(n-1)} - 1} \left[y_1 e^{y_1(n-1)} - \frac{1}{n-1} (e^{y_1(n-1)} - 1) \right]$$

$$E(\theta) = \frac{y_1 e^{y_1(n-1)}}{e^{y_1(n-1)} - 1} - \frac{1}{n-1}$$

Which is the required Posterior Bayes Estimator of θ

7 Method of Least Square

- The idea of least square estimation has emerged from the method of maximum likelihood estimation.
- Consider the log likelihood function of normal distribution

$$\ln L(\mu) = -\frac{n}{2} \ln(\sigma^2) - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x - \mu)^2$$

This would be maximized if

$$\sum_{i=1}^n (x - \mu)^2$$

Is minimized

- The method of least square estimation is mostly used in estimating the parameters of linear functions.
- Method of least square estimation can be extended to multiple linear models and curvilinear models.
- The principal of LSE state that “the estimator which minimize the sum of square deviation between the true value and the estimated value will be chosen”.

7.1 Properties of Least Estimates

- Least Square Estimators are unbiased in case of linear models.
- The least square method is a device to obtain Best Linear Unbiased Estimators (BLUEs) under linear set up.
- In linear setup because LSE are BLUE so these estimators will be efficient.
- LSE are best for any sample size and even best for small sample size.
- In simple linear regression the least square estimators are Uniformly Minimum Variance Unbiased Estimators (UMVUEs)

7.2 Uses of Least Square Estimators

- To find the estimates of simple linear and multiple linear regression models.

- LSE provides BLUE and have certainly minimum variance when the response variable has linear functional relationship with the predictors.
- In case of small samples there is no comparison of LSE.
- LSE perform better than MLE in case of small sample.

7.3 Best Linear Unbiased Estimator (BLUE)

An estimator among the class of all possible linear unbiased estimators which has the smallest variance possible variance than any other estimator, such estimator is called minimum variance linear unbiased estimator. It is generally, called Best Linear Unbiased Estimator (BLUE).

7.4 Minimum Variance Quadratic Unbiased Estimator (MVQUE)

If an unbiased quadratic estimator of the variance σ^2 exists and it possesses the minimum variance out of the class of available unbiased estimators. It is known as Minimum Variance Quadratic Estimator (MVQUE).

8 Questions Related to Least Square Estimation

8.1 Question 1

Let there are $2n$ observations $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$ and $z_1, z_2, \dots, z_n$, having same unknown variance σ^2 . The mean values of the observations are

$$E(x_i) = \theta_1 + 2\theta_2 + 3\theta_3, E(y_i) = 2\theta_1 + 3\theta_2 + \theta_3 \text{ and } E(z_i) = 3\theta_1 + \theta_2 + 2\theta_3$$

Where θ_1, θ_2 and θ_3 are unknown parameters.

Apply method of least square to derive the estimates of contrast (i) $\theta_1 - \theta_2$ (ii) $\theta_2 - \theta_3$ and (iii) $\theta_3 - \theta_1$ by using $\theta_1 + \theta_2 + \theta_3 = 0$ (iv) Obtain the unbiased estimate of σ^2 .

Also prove that $V(\hat{\theta}_1) = \frac{14\sigma^2}{9n}$ (v) Also compute variance of each contrast.

Solution:

Let S be sum of squares

$$S = \sum_{i=1}^n (x_i - \theta_1 - 2\theta_2 - 3\theta_3)^2 + \sum_{i=1}^n (y_i - 2\theta_1 - 3\theta_2 - \theta_3)^2 + \sum_{i=1}^n (z_i - 3\theta_1 - \theta_2 - 2\theta_3)^2$$

Diff w.r.t θ_1

$$\frac{\partial S}{\partial \theta_1} = - \sum_{i=1}^n (x_i - \theta_1 - 2\theta_2 - 3\theta_3) - 2 \sum_{i=1}^n (y_i - 2\theta_1 - 3\theta_2 - \theta_3) - 3 \sum_{i=1}^n (z_i - 3\theta_1 - \theta_2 - 2\theta_3)$$

$$\text{Put } \frac{\partial S}{\partial \theta_1} = 0$$

$$- \sum_{i=1}^n (x_i - \theta_1 - 2\theta_2 - 3\theta_3) - 2 \sum_{i=1}^n (y_i - 2\theta_1 - 3\theta_2 - \theta_3) - 3 \sum_{i=1}^n (z_i - 3\theta_1 - \theta_2 - 2\theta_3) = 0$$

$$-\sum x_i + n\theta_1 + 2n\theta_2 + 3n\theta_3 - 2\sum y_i + 4n\theta_1 + 6n\theta_2 + 2n\theta_3 - 3\sum z_i + 9n\theta_1 + 3n\theta_2 + 6n\theta_3 = 0$$

$$14n\theta_1 + 11n\theta_2 + 11n\theta_3 - \sum x_i - 2\sum y_i - 3\sum z_i = 0$$

$$14n\theta_1 + 11n\theta_2 + 11n\theta_3 - n\bar{x} - 2n\bar{y} - 3n\bar{z} = 0$$

$$\bar{x} + 2\bar{y} + 3\bar{z} = 14\theta_1 + 11\theta_2 + 11\theta_3$$

$$\bar{x} + 2\bar{y} + 3\bar{z} = 11(\theta_1 + \theta_2 + \theta_3) + 3\theta_1$$

Using $\theta_1 + \theta_2 + \theta_3 = 0$

$$\bar{x} + 2\bar{y} + 3\bar{z} = 3\theta_1$$

$$\hat{\theta}_1 = \frac{\bar{x} + 2\bar{y} + 3\bar{z}}{3}$$

Diff w.r.t θ_2

$$\frac{\partial S}{\partial \theta_2} = -2 \sum_{i=1}^n (x_i - \theta_1 - 2\theta_2 - 3\theta_3) - 3 \sum_{i=1}^n (y_i - 2\theta_1 - 3\theta_2 - \theta_3) - \sum_{i=1}^n (z_i - 3\theta_1 - \theta_2 - 2\theta_3)$$

$$\text{Put } \frac{\partial S}{\partial \theta_2} = 0$$

$$-2 \sum_{i=1}^n (x_i - \theta_1 - 2\theta_2 - 3\theta_3) - 3 \sum_{i=1}^n (y_i - 2\theta_1 - 3\theta_2 - \theta_3) - \sum_{i=1}^n (z_i - 3\theta_1 - \theta_2 - 2\theta_3) = 0$$

$$-2\sum x_i + 2n\theta_1 + 4n\theta_2 + 6n\theta_3 - 3\sum y_i + 6n\theta_1 + 9n\theta_2 + 3n\theta_3 - \sum z_i + 3n\theta_1 + n\theta_2 + 2n\theta_3 = 0$$

$$11n\theta_1 + 14n\theta_2 + 11n\theta_3 - 2\sum x_i - 3\sum y_i - \sum z_i = 0$$

$$11n\theta_1 + 14n\theta_2 + 11n\theta_3 - 2n\bar{x} - 3n\bar{y} - n\bar{z} = 0$$

$$2\bar{x} + 3\bar{y} + \bar{z} = 11\theta_1 + 14\theta_2 + 11\theta_3$$

$$2\bar{x} + 3\bar{y} + \bar{z} = 11(\theta_1 + \theta_2 + \theta_3) + 3\theta_2$$

Using $\theta_1 + \theta_2 + \theta_3 = 0$

$$2\bar{x} + 3\bar{y} + \bar{z} = 3\theta_2$$

$$\hat{\theta}_2 = \frac{2\bar{x} + 3\bar{y} + \bar{z}}{3}$$

Diff w.r.t θ_3

$$\frac{\partial S}{\partial \theta_3} = -3 \sum_{i=1}^n (x_i - \theta_1 - 2\theta_2 - 3\theta_3) - \sum_{i=1}^n (y_i - 2\theta_1 - 3\theta_2 - \theta_3) - 2 \sum_{i=1}^n (z_i - 3\theta_1 - \theta_2 - 2\theta_3)$$

Put $\frac{\partial S}{\partial \theta_2} = 0$

$$-3 \sum_{i=1}^n (x_i - \theta_1 - 2\theta_2 - 3\theta_3) - \sum_{i=1}^n (y_i - 2\theta_1 - 3\theta_2 - \theta_3) - 2 \sum_{i=1}^n (z_i - 3\theta_1 - \theta_2 - 2\theta_3) = 0$$

$$-3\sum x_i + 3n\theta_1 + 6n\theta_2 + 9n\theta_3 - \sum y_i + 2n\theta_1 + 3n\theta_2 + n\theta_3 - 2\sum z_i + 6n\theta_1 + 2n\theta_2 + 4n\theta_3 = 0$$

$$11n\theta_1 + 11n\theta_2 + 14n\theta_3 - 3\sum x_i - \sum y_i - 2\sum z_i = 0$$

$$11n\theta_1 + 11n\theta_2 + 14n\theta_3 - 3n\bar{x} - n\bar{y} - 2n\bar{z} = 0$$

$$3\bar{x} + \bar{y} + 2\bar{z} = 11\theta_1 + 11\theta_2 + 14\theta_3$$

$$3\bar{x} + \bar{y} + 2\bar{z} = 11(\theta_1 + \theta_2 + \theta_3) + 3\theta_3$$

Using $\theta_1 + \theta_2 + \theta_3 = 0$

$$3\bar{x} + \bar{y} + 2\bar{z} = 3\theta_3$$

$$\hat{\theta}_3 = \frac{3\bar{x} + \bar{y} + 2\bar{z}}{3}$$

Now as given

$$\hat{\theta}_1 + \hat{\theta}_2 + \hat{\theta}_3 = 0$$

$$\frac{\bar{x} + 2\bar{y} + 3\bar{z}}{3} + \frac{2\bar{x} + 3\bar{y} + \bar{z}}{3} + \frac{3\bar{x} + \bar{y} + 2\bar{z}}{3} = 0$$

$$6\bar{x} + 6\bar{y} + 6\bar{z} = 0 \quad \Rightarrow \quad \bar{x} + \bar{y} + \bar{z} = 0 \dots \dots (1)$$

(i) Contract $\hat{\theta}_1 - \hat{\theta}_2$

$$\hat{\theta}_1 - \hat{\theta}_2 = \frac{\bar{x} + 2\bar{y} + 3\bar{z}}{3} - \frac{2\bar{x} + 3\bar{y} + \bar{z}}{3} = \frac{-\bar{x} - \bar{y} + 2\bar{z}}{3} = \frac{-(\bar{x} + \bar{y} + \bar{z}) + 3\bar{z}}{3}$$

Using (1)

$$\hat{\theta}_1 - \hat{\theta}_2 = \frac{0 + 3\bar{z}}{3} = \bar{z}$$

(ii) Contract $\hat{\theta}_2 - \hat{\theta}_3$

$$\hat{\theta}_2 - \hat{\theta}_3 = \frac{2\bar{x} + 3\bar{y} + \bar{z}}{3} - \frac{3\bar{x} + \bar{y} + 2\bar{z}}{3} = \frac{-\bar{x} + 2\bar{y} - 2\bar{z}}{3} = \frac{-(\bar{x} + \bar{y} + \bar{z}) + 3\bar{y}}{3}$$

Using (1)

$$\hat{\theta}_2 - \hat{\theta}_3 = \frac{0 + 3\bar{y}}{3} = \bar{y}$$

iii) Contrast $\hat{\theta}_3 - \hat{\theta}_1$

$$\hat{\theta}_3 - \hat{\theta}_1 = \frac{3\bar{x} + \bar{y} + 2\bar{z}}{3} - \frac{\bar{x} + 2\bar{y} + 3\bar{z}}{3} = \frac{2\bar{x} - 2\bar{y} - \bar{z}}{3} = \frac{-(\bar{x} + \bar{y} + \bar{z}) + 3\bar{x}}{3}$$

Using (1)

$$\hat{\theta}_3 - \hat{\theta}_1 = \frac{0 + 3\bar{x}}{3} = \bar{x}$$

(iv) Variance of $\hat{\theta}_1$

$$V(\hat{\theta}_1) = V\left[\frac{\bar{x} + 2\bar{y} + 3\bar{z}}{3}\right] = \frac{1}{9}[V(\bar{x}) + 4V(\bar{y}) + 9V(\bar{z})] = \frac{1}{9}\left[\frac{\sigma^2}{n} + \frac{4\sigma^2}{n} + \frac{9\sigma^2}{n}\right] = \frac{14\sigma^2}{9n}$$

Variance of $\hat{\theta}_2$

$$V(\hat{\theta}_2) = V\left[\frac{2\bar{x} + 3\bar{y} + \bar{z}}{3}\right] = \frac{1}{9}[4V(\bar{x}) + 9V(\bar{y}) + V(\bar{z})] = \frac{1}{9}\left[\frac{4\sigma^2}{n} + \frac{9\sigma^2}{n} + \frac{\sigma^2}{n}\right] = \frac{14\sigma^2}{9n}$$

Variance of $\hat{\theta}_3$

$$V(\hat{\theta}_3) = V\left[\frac{3\bar{x} + \bar{y} + 2\bar{z}}{3}\right] = \frac{1}{9}[9V(\bar{x}) + V(\bar{y}) + 4V(\bar{z})] = \frac{1}{9}\left[\frac{9\sigma^2}{n} + \frac{\sigma^2}{n} + \frac{4\sigma^2}{n}\right] = \frac{14\sigma^2}{9n}$$

(v) Variance of each Contrast

$$V(\hat{\theta}_1 - \hat{\theta}_2) = V(\hat{\theta}_1) + V(\hat{\theta}_2) = \frac{14\sigma^2}{9n} + \frac{14\sigma^2}{9n} = \frac{28\sigma^2}{9n}$$

$$V(\hat{\theta}_2 - \hat{\theta}_3) = V(\hat{\theta}_2) + V(\hat{\theta}_3) = \frac{14\sigma^2}{9n} + \frac{14\sigma^2}{9n} = \frac{28\sigma^2}{9n}$$

$$V(\hat{\theta}_3 - \hat{\theta}_1) = V(\hat{\theta}_3) + V(\hat{\theta}_1) = \frac{14\sigma^2}{9n} + \frac{14\sigma^2}{9n} = \frac{28\sigma^2}{9n}$$

If Condition $\bar{x} + \bar{y} + \bar{z} = 0$ is used than

$$V(\hat{\theta}_1 - \hat{\theta}_2) = V(\bar{z}) = \frac{\sigma^2}{n}, V(\hat{\theta}_2 - \hat{\theta}_3) = V(\bar{y}) = \frac{\sigma^2}{n}, V(\hat{\theta}_3 - \hat{\theta}_1) = V(\bar{x}) = \frac{\sigma^2}{n}$$

(vi) Unbiased Estimator of σ^2

Let M be the minimum sum of square of sample space

$$M = \sum_{i=1}^n (x_i - \bar{x})^2 + \sum_{i=1}^n (y_i - \bar{y})^2 + \sum_{i=1}^n (z_i - \bar{z})^2$$

The unbiased estimator for σ^2 will be

$$\hat{\sigma}^2 = \frac{M}{3n - 3}$$

8.2 Question 2

Given the rectangular distribution

$$f(x; \mu, \sigma) = \frac{1}{\sigma}, \quad \mu - \frac{1}{2}\sigma \leq x \leq \mu + \frac{1}{2}\sigma$$

For

$$\alpha_r = \left(\frac{r}{n+1}\right) - \frac{1}{2}, \quad V_{rs} = \frac{r(n-s+1)}{(n+1)^2(n+2)}, \quad r \leq s, \quad r = 1, 2, \dots, n, \quad s = 1, 2, \dots, n$$

Find the order statistic for $\hat{\mu}, \hat{\sigma}$ and $V(\hat{\mu}), V(\hat{\sigma})$

Solution:-

Since, $r = 1, 2, \dots, n, \quad s = 1, 2, \dots, n$ all possible combination of r and s will give n by n matrix for V_{rs}

$$V = \begin{bmatrix} V_{11} & V_{12} & V_{13} & \dots & V_{1n} \\ V_{21} & V_{22} & V_{23} & \dots & V_{2n} \\ V_{31} & V_{32} & V_{33} & \dots & V_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ V_{n1} & V_{n2} & V_{n3} & \dots & V_{nn} \end{bmatrix}$$

$$V = \begin{bmatrix} n & n-1 & n-2 & \dots & 1 \\ n-1 & n & 2(n-1) & \dots & 2 \\ n-2 & n-1 & 3(n-1) & \dots & 3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 2 & 3 & \dots & n \end{bmatrix}$$

Now by induction method we determine inverse of V_{rs}

For $n=2$

$$V_2 = \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix} = \frac{1}{(2+1)^2(2+2)} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Now to find V_2^{-1}

$$V_2^{-1} = (2+1)^2(2+2) \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}^{-1} = \frac{(2+1)^2(2+2)}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = (2+1)(2+2) \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

For $n=3$

$$V_3 = \begin{bmatrix} V_{11} & V_{12} & V_{13} \\ V_{21} & V_{22} & V_{23} \\ V_{31} & V_{32} & V_{33} \end{bmatrix} = \frac{1}{(3+1)^2(3+2)} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Now to find the inverse of V_3

$$|V_3^*| = \begin{vmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{vmatrix} = 3(12 - 4) - 2(6 - 2) + 1(4 - 4) = 24 - 8 + 0 = 16$$

$$(V_3^*)^{-1} = \frac{1}{16} \begin{bmatrix} \begin{vmatrix} 4 & 2 \\ 2 & 3 \end{vmatrix} & -\begin{vmatrix} 2 & 2 \\ 1 & 3 \end{vmatrix} & \begin{vmatrix} 2 & 4 \\ 1 & 2 \end{vmatrix} \\ -\begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} & \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} & -\begin{vmatrix} 3 & 2 \\ 1 & 2 \end{vmatrix} \\ \begin{vmatrix} 2 & 1 \\ 4 & 2 \end{vmatrix} & -\begin{vmatrix} 3 & 1 \\ 2 & 2 \end{vmatrix} & \begin{vmatrix} 3 & 2 \\ 2 & 4 \end{vmatrix} \end{bmatrix}^t$$

$$(V_3^*)^{-1} = \frac{1}{16} \begin{bmatrix} 8 & -4 & 0 \\ -4 & 8 & -4 \\ 0 & -4 & 8 \end{bmatrix}^t = \frac{1}{16} \begin{bmatrix} 8 & -4 & 0 \\ -4 & 8 & -4 \\ 0 & -4 & 8 \end{bmatrix} = \frac{4}{16} \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

$$V_3^{-1} = (3+1)^2(3+2)(V_3^*)^{-1} = \frac{(3+1)^2(3+2)}{4} \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

$$= (3+1)(3+2) \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Now, Generalizing

$$V^{-1} = (n+1)(n+2) \begin{bmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & 0 & \dots & 0 \\ 0 & -1 & 2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 2 \end{bmatrix}$$

$$\hat{\mu} = \frac{\hat{1}V^{-1}Y}{\hat{1}V^{-1}\hat{1}}$$

$$\hat{1}V^{-1} = [1 \ 1 \ \dots \ 1](n+1)(n+2) \begin{bmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & 0 & \dots & 0 \\ 0 & -1 & 2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 2 \end{bmatrix}$$

$$= (n+1)(n+2)[1 \ 0 \ 0 \ \dots \ 1]$$

$$\hat{1}V^{-1}y = (n+1)(n+2)[1 \ 0 \ 0 \ \dots \ 1] \begin{bmatrix} y_1 \\ y_2 \\ \cdot \\ y_n \end{bmatrix} = (n+1)(n+2)(y_1 + y_n)$$

$$\hat{1}V^{-1}1 = (n+1)(n+2)[1 \ 0 \ 0 \ \dots \ 1] \begin{bmatrix} 1 \\ 1 \\ \cdot \\ 1 \end{bmatrix} = 2(n+1)(n+2)$$

$$\hat{\mu} = \frac{(n+1)(n+2)(y_1 + y_n)}{2(n+1)(n+2)} = \frac{y_1 + y_n}{2}$$

And

$$V(\hat{\mu}) = \frac{\sigma^2}{\hat{1}V^{-1}\hat{1}} = \frac{\sigma^2}{2(n+1)(n+2)}$$

Also given

$$\alpha_r = \frac{r}{n+1} - \frac{1}{2}$$

$$\alpha = \begin{bmatrix} \frac{1}{n+1} - \frac{1}{2} \\ \frac{2}{n+1} - \frac{1}{2} \\ \cdot \\ \frac{n}{n+1} - \frac{1}{2} \end{bmatrix} = \frac{1}{2(n+1)} \begin{bmatrix} -(n-1) \\ -(n-3) \\ \cdot \\ n-3 \\ n-1 \end{bmatrix}$$

$$\acute{\alpha} = \frac{1}{2(n+1)} [-(n-1) \quad -(n-3) \quad \dots \quad n-3 \quad n-1]$$

$$\hat{\sigma} = \frac{\acute{\alpha}V^{-1}Y}{\acute{\alpha}V^{-1}\alpha}$$

$$\acute{\alpha}V^{-1} = \frac{1}{2(n+1)} [-(n-1) \quad -(n-3) \quad \dots \quad n-3 \quad n-1] (n+1)(n$$

$$+ 2) \begin{bmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & 0 & \dots & 0 \\ 0 & -1 & 2 & \dots & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & \dots & 2 \end{bmatrix}$$

$$\acute{\alpha}V^{-1} = \frac{n+2}{2} [-2n+2+n-3 \quad n-1-2n+6+n-5 \quad \dots \quad -n+3+2n-2]$$

$$\acute{\alpha}V^{-1} = \frac{n+2}{2} [-(n+1) \quad 0 \quad 0 \quad \dots \dots \quad n+1]$$

$$\acute{\alpha}V^{-1}y = \frac{n+2}{2} [-(n+1) \quad 0 \quad 0 \quad \dots \dots \quad n+1] \begin{bmatrix} y_1 \\ y_2 \\ \cdot \\ \cdot \\ y_n \end{bmatrix} = \frac{n+2}{2} [-y_1(n+1) + (n+1)y_n]$$

$$\acute{\alpha}V^{-1}y = \frac{(n+1)(n+2)}{2} (y_n - y_1)$$

And

$$\acute{\alpha}V^{-1}\alpha = \frac{n+2}{2} [-(n+1) \quad 0 \quad 0 \quad \dots \dots \quad n+1] \frac{1}{2(n+1)} \begin{bmatrix} -(n-1) \\ -(n-3) \\ \cdot \\ \cdot \\ \cdot \\ n-3 \\ n-1 \end{bmatrix}$$

$$\acute{\alpha}V^{-1}\alpha = \frac{n+2}{4(n+1)} [(n+1)(n-1) + (n+1)(n-1)] = \frac{2(n+1)(n-1)(n+2)}{4(n+1)} = \frac{(n-1)(n+2)}{2}$$

So ,

$$\hat{\sigma} = \frac{\frac{(n+1)(n+2)}{2} (y_n - y_1)}{\frac{(n-1)(n+2)}{2}} = \frac{(n+1)(y_n - y_1)}{(n-1)}$$

And

$$V(\hat{\sigma}) = \frac{\sigma^2}{\acute{\alpha}V^{-1}\alpha} = \frac{\sigma^2}{\frac{(n-1)(n+2)}{2}} = \frac{2\sigma^2}{(n+1)(n+2)}$$

8.3 Question 3

Let μ and σ be location and shape parameters such that $z = \frac{y - \mu}{\sigma}$, where (y) is r^{th} order statistic. If $E(z) = \alpha$ and $V(z) = v$. find the ordered least square estimator of μ and σ . Also find their variance and covariances.

Solution:-

As Given

$$z_r = \frac{y_r - \mu}{\sigma} \Rightarrow y_r = \mu + \sigma z_r$$

In matrix form

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix} + \sigma \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}$$

$$\mathbf{y} = \boldsymbol{\mu} + \sigma \mathbf{z}$$

Taking Expectation

$$E(\mathbf{y}) = E(\boldsymbol{\mu}) + \sigma E(\mathbf{z})$$

As given $E(z) = \alpha$

$$E(\mathbf{y}) = \boldsymbol{\mu} + \sigma \alpha$$

$$E(\mathbf{y}) = [\mathbf{1} \quad \alpha] \begin{bmatrix} \boldsymbol{\mu} \\ \sigma \end{bmatrix} = \mathbf{X}\boldsymbol{\beta}$$

in GLS

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{Y}$$

$$\begin{bmatrix} \hat{\mu} \\ \hat{\sigma} \end{bmatrix} = \{[\mathbf{1} \quad \alpha]' \mathbf{V}^{-1} [\mathbf{1} \quad \alpha]\}^{-1} [\mathbf{1} \quad \alpha]' \mathbf{V}^{-1} \mathbf{Y}$$

Let

$$A = \begin{bmatrix} 1 & \alpha \end{bmatrix} V^{-1} \begin{bmatrix} 1 \\ \alpha \end{bmatrix}, \quad B = \begin{bmatrix} 1 & \alpha \end{bmatrix} V^{-1} Y$$

$$\begin{bmatrix} \hat{\mu} \\ \hat{\sigma} \end{bmatrix} = A^{-1} B$$

$$A = V^{-1} \begin{bmatrix} 1 \\ \alpha \end{bmatrix} \begin{bmatrix} 1 & \alpha \end{bmatrix} = \begin{bmatrix} 1' V^{-1} 1 & 1' V^{-1} \alpha \\ \alpha' V^{-1} 1 & \alpha' V^{-1} \alpha \end{bmatrix}$$

$$A^{-1} = \frac{1}{(1' V^{-1} 1)(\alpha' V^{-1} \alpha) - (1' V^{-1} \alpha)(\alpha' V^{-1} 1)} \begin{bmatrix} \alpha' V^{-1} \alpha & -1' V^{-1} \alpha \\ -\alpha' V^{-1} 1 & 1' V^{-1} 1 \end{bmatrix} = \frac{1}{\Delta} \begin{bmatrix} \alpha' V^{-1} \alpha & -1' V^{-1} \alpha \\ -\alpha' V^{-1} 1 & 1' V^{-1} 1 \end{bmatrix}$$

Where

$$(1' V^{-1} 1)(\alpha' V^{-1} \alpha) - (1' V^{-1} \alpha)(\alpha' V^{-1} 1) = \Delta$$

$$B = \begin{bmatrix} 1 & \alpha \end{bmatrix} V^{-1} Y = V^{-1} \begin{bmatrix} 1 \\ \alpha \end{bmatrix} Y = \begin{bmatrix} V^{-1} 1' Y \\ V^{-1} \alpha' Y \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mu} \\ \hat{\sigma} \end{bmatrix} = \frac{1}{\Delta} \begin{bmatrix} \alpha' V^{-1} \alpha & -1' V^{-1} \alpha \\ -\alpha' V^{-1} 1 & 1' V^{-1} 1 \end{bmatrix} \begin{bmatrix} V^{-1} 1' Y \\ V^{-1} \alpha' Y \end{bmatrix} = \begin{bmatrix} \frac{(\alpha' V^{-1} \alpha 1' - 1' V^{-1} \alpha \alpha') V^{-1} Y}{\Delta} \\ \frac{(-\alpha' V^{-1} 1 1' + 1' V^{-1} 1 \alpha') V^{-1} Y}{\Delta} \end{bmatrix}$$

Hence

$$\hat{\mu} = \frac{(\alpha' V^{-1} \alpha 1' - 1' V^{-1} \alpha \alpha') V^{-1} Y}{\Delta} \quad \text{and} \quad \hat{\sigma} = \frac{(-\alpha' V^{-1} 1 1' + 1' V^{-1} 1 \alpha') V^{-1} Y}{\Delta}$$

Again consider

$$y = \mu + \sigma z$$

Taking Variance both sides

$$V(\mathbf{y}) = \sigma^2 V(\mathbf{z}) = \sigma^2 V$$

In GLS

$$V(\hat{\beta}) = (X'V^{-1}X)^{-1} \sigma^2$$

$$V \begin{bmatrix} \hat{\mu} \\ \hat{\sigma} \end{bmatrix} = A^{-1} \sigma^2 = \frac{1}{\Delta} \begin{bmatrix} \alpha' V^{-1} \alpha & -\mathbf{1}' V^{-1} \alpha \\ -\alpha' V^{-1} \mathbf{1} & \mathbf{1}' V^{-1} \mathbf{1} \end{bmatrix} \sigma^2$$

So,

$$V(\hat{\mu}) = \frac{\alpha' V^{-1} \alpha \sigma^2}{\Delta} \quad , \quad V(\hat{\sigma}) = \frac{\mathbf{1}' V^{-1} \mathbf{1} \sigma^2}{\Delta} \quad \text{and} \quad V(\hat{\mu}, \hat{\sigma}) = \frac{-\mathbf{1}' V^{-1} \alpha \sigma^2}{\Delta}$$



9 Minimum Chi-Square Method

- The minimum chi-square makes use of Pearson's chi-square statistic.
- This method can be used in case of discrete distribution or for the group data of continuous distribution.
- It is the method of estimating unobserved quantities based upon the observed data.
- Chi-square test involves choosing the number and the width of classes.

Let $f_1, f_2, \dots, f_k$ be the observed frequencies in K groups or classes and unknown probabilities that f_i elements belong to the i^{th} group or class be $p_i (i = 1, 2, \dots, k)$. p_i 's are the functions of the unknown parameters $\theta_1, \theta_2, \dots, \theta_m$. Thus, $p_i = p_i(\theta')$, where $\theta' = (\theta_1, \theta_2, \dots, \theta_m)$. Suppose the total sample size is n . Therefore, $\sum f_i = n$. The expected frequencies are $np_1(\theta'), np_2(\theta') \dots \dots np_k(\theta')$. We know that Pearsonian Chi-Square statistic

$$\chi^2 = \sum_{i=1}^k \frac{[f_i - np_i(\theta')]^2}{np_i(\theta')} = \sum_{i=1}^k \frac{f_i^2}{np_i(\theta')} - n$$

Under the method of minimum Chi-square one has to choose $(\theta_1, \theta_2, \dots, \theta_m)$, which minimize χ^2 . This will be minimize when $np_i(\theta')$ is as close as possible to f_i .

To obtain the estimates of θ_i 's, partially differentiate χ^2 w.r.t $\theta_i (i = 1, 2, \dots, m)$ successively and equate to zero.

Also check that the second derivatives are non-negative. i.e.

$$\frac{\partial \chi^2}{\partial \theta_i} = 0 \text{ for } i = 1, 2, \dots, m \quad \text{and} \quad \frac{\partial^2 \chi^2}{\partial \theta_i^2} > 0$$

$\frac{\partial \chi^2}{\partial \theta_i} = 0$, provides m simultaneous equations in m unknowns parameters. Solving these m - equations one gets the estimated values of $(\theta_1, \theta_2, \dots, \theta_m)$

9.1 Properties of Minimum Chi-Square Estimators

- The Minimum Chi-Square estimators are consistent.
- The Minimum Chi-Square estimators are asymptotically normal.
- The Minimum Chi-Square estimators are efficient.

- The Minimum Chi-Square estimators are not necessarily unbiased.

9.2 Comparison of MCSE and MLE.

- The procedure of finding MCSE and MLE is similar.
- MCSE and MLE have the similar properties.
- It is trivial to prove that for large n the estimates obtained by the method of MCS are almost equal to estimates obtained through MLE.
- In 1955 J.Berkson showed that in particular case the mean square error of estimators under MCSE is smaller as compare to mean square error of MLE.
- MLE is Considered superior to MCSE.
- MCSE are termed as having 2nd order efficiency.

9.3 Use of MCSE

- Minimum Chi-square estimators are rarely used in practice.
- It is used only when it is difficult to solve the simultaneous equations obtained under maximum likelihood estimation method.

9.4 Modified Minimum Chi-Square Estimator

- Expected frequencies $np_i(\theta')$ in the denominator of χ^2 -statistic cause certain problems. Hence, a modification has been suggested which makes the process of differentiation easier.

Let $f_1, f_2, \dots, f_k$ be the observed frequencies in K groups or classes and unknown probabilities that f_i element belong to the i^{th} group or class be $p_i (i = 1, 2, \dots, k)$ p's are the functions of the unknown parameters $\theta_1, \theta_2, \dots, \theta_m$. Thus, $p_i = p_i(\theta')$, where $\theta' = (\theta_1, \theta_2, \dots, \theta_m)$. Suppose the total sample size is n. Therefore, $\sum f_i = n$. The expected frequencies are $np_1(\theta'), np_2(\theta') \dots \dots np_k(\theta')$. We know that Pearsonian Chi-Square statistic

$$\chi^2 = \sum_{i=1}^k \frac{[np_i(\theta') - f_i]^2}{f_i} = \sum_{i=1}^k \frac{n^2 p_i^2(\theta')}{f_i} - n$$

Under the method of minimum Chi-square one has to choose $(\theta_1, \theta_2, \dots, \theta_m)$, which minimize χ^2 . This will be minimize when $np_i(\theta')$ is as close as possible to f_i .

To obtain the estimates of θ_i 's, partially differentiate χ^2 w.r.t $\theta_i (i = 1, 2, \dots, m)$ successively and equate to zero.

Also check that the second derivatives are non-negative. i.e.

$$\frac{\partial \chi^2}{\partial \theta_i} = 0 \text{ for } i = 1, 2, \dots, m \quad \text{and} \quad \frac{\partial^2 \chi^2}{\partial \theta_i^2} > 0$$

$\frac{\partial \chi^2}{\partial \theta_i} = 0$, provides m simultaneous equations in m unknowns parameters. Solving these m - equations one gets the estimated values of $(\theta_1, \theta_2, \dots, \theta_m)$

10 Questions Related to Minimum Chi-Square Method

10.1 MCSE for Binomial Distribution

$$f(x; \theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}, x = 0, 1, 2, \dots, n$$

$$\pi_j = \binom{n}{j} \theta^j (1 - \theta)^{n-j}, j = 1, 2, \dots, n$$

Diff w.r.t to θ

$$\frac{\partial \pi_j}{\partial \theta} = \binom{n}{j} [j\theta^{j-1}(1 - \theta)^{n-j} - \theta^j(n - j)(1 - \theta)^{n-j-1}]$$

$$\frac{\partial \pi_j}{\partial \theta} = \binom{n}{j} \theta^j (1 - \theta)^{n-j} \left[\frac{j}{\theta} - \frac{n - j}{(1 - \theta)} \right] = \binom{n}{j} \theta^j (1 - \theta)^{n-j} \left[\frac{j - j\theta - n\theta + j\theta}{\theta(1 - \theta)} \right]$$

$$\frac{\partial \pi_j}{\partial \theta} = \binom{n}{j} \theta^j (1 - \theta)^{n-j} \left[\frac{j - n\theta}{\theta(1 - \theta)} \right] = \pi_j \left[\frac{j - n\theta}{\theta(1 - \theta)} \right]$$

Now, Consider

$$\chi^2 = \sum n_i \sum \frac{p_i^2}{\pi_j} - 1$$

Diff w.r.t θ

$$\frac{\partial \chi^2}{\partial \theta} = -\sum n_i \sum \left(\frac{p_i}{\pi_j} \right)^2 \frac{\partial \pi_j}{\partial \theta} = -\sum n_i \sum \left(\frac{p_i}{\pi_j} \right)^2 \pi_j \left[\frac{j - n\theta}{\theta(1 - \theta)} \right]$$

For minimization drop the factor n put equal to zero

$$\frac{\partial \chi^2}{\partial \theta} = -\sum \left(\frac{p_i}{\pi_j} \right)^2 \pi_j \left[\frac{j - n\theta}{\theta(1 - \theta)} \right] = 0$$

$$\frac{j - n\theta}{\theta(1 - \theta)} = 0 \quad \Rightarrow \quad \theta = \frac{j}{n}$$

The above equation holds for

$$\theta = \frac{j}{n}$$



10.2 MCSE for Bernoulli Distribution

Question: Based upon the random sample of size n from

$$f(x; \theta) = \theta^x(1 - \theta)^{1-x}, x = 0, 1$$

Obtain the MCSE for θ

Solution

Let

$$p_i = \frac{n_j}{n} \quad \text{and} \quad \pi_j = \theta^j(1 - \theta)^{1-j} \quad j = 0, 1$$

	0	1	Σ
n_j	n_0	n_1	N
p_j	p_0	p_1	1
π_j	$\pi_0 = (1 - \theta)$	$\pi_1 = \theta$	1

$$\frac{\partial \pi_0}{\partial \theta} = -1, \frac{\partial \pi_1}{\partial \theta} = 1$$

The modified chi-square

$$\chi^2 = n \left[\sum \left(\frac{\pi_j^2}{p_j} \right) - 1 \right] = n \left[\frac{\pi_0^2}{p_0} + \frac{\pi_1^2}{p_1} - 1 \right]$$

Diff w.r.t θ

$$\frac{\partial \chi^2}{\partial \theta} = n \left[2 \frac{\pi_0}{p_0} \frac{\partial \pi_0}{\partial \theta} + 2 \frac{\pi_1}{p_1} \frac{\partial \pi_1}{\partial \theta} \right] = n \left[2 \frac{(1 - \theta)}{\frac{n_0}{n}} (-1) + 2 \frac{\theta}{\frac{n_1}{n}} (1) \right]$$

$$\frac{\partial \chi^2}{\partial \theta} = 2n^2 \left[\frac{(\theta - 1)}{n_0} + \frac{\theta}{n_1} \right]$$

$$\frac{(\theta - 1)}{n_0} + \frac{\theta}{n_1} = 0 \Rightarrow \frac{(\theta - 1)}{\theta} = -\frac{n_0}{n_1} \Rightarrow 1 - \frac{1}{\theta} = -\frac{n_0}{n_1}$$

$$1 + \frac{n_0}{n_1} = \frac{1}{\theta} \Rightarrow \frac{n_0 + n_1}{n_1} = \frac{1}{\theta} \Rightarrow \frac{n}{n_1} = \frac{1}{\theta}$$

Hence,

$$\theta = \frac{n_1}{n} = p_1$$

Is the Requires MCSE of θ .

Also

$$\frac{\partial^2 \chi^2}{\partial \theta^2} = 2n^2 \left[\frac{1}{n_0} + \frac{1}{n_1} \right] = 2n^2 \left[\frac{n_0 + n_1}{n_0 n_1} \right] = \frac{2n^3}{n_0 n_1}$$



11.1 Confidence Interval Estimation

11.1.1 Confidence Interval

- In statistic, confidence interval is a type of interval estimate of population parameter and is used to indicate the reliability of an estimate.
- It consists of range of values that acts as good estimates of the unknown population parameter.
- Between the interval the true value of the parameter is expected to fall.
- The level of confidence $(1 - \alpha)$ is the probability which indicates that how frequently the observed interval will contain the true value of the population parameter.
- Mathematically,

Let $X_1, X_2, \dots, X_n$ be a collection of random variables with the joint distribution $f(\theta)$. Where $\theta \in \Omega \subseteq R$ is unknown. Let $L(x)$ and $U(x)$ be two statistics., the interval $[L(x), U(x)]$ is $(1 - \alpha)$ interval estimator of θ if

$$P(\theta \in [L(x), U(x)]) \geq 1 - \alpha \text{ for all } \theta \in \Omega$$

With equality for at least one value of θ .

- Furthermore, if we observe a realization of the data $x_1, x_2, \dots, x_n$, we say that $[L(x), U(x)]$ is a $100(1 - \alpha)\%$ confidence interval for θ ., where $L(x), U(x)$ are lower and upper values of confidence interval.

11.1.2 Confidence Region

- Sometimes an investigator aspires to extend the idea of interval estimation to more than parameters simultaneously. For instant one may wish to estimate the parameters μ and σ^2 of normal distribution simultaneously by some confidence region in a plan.
- A Confidence region is a multi-dimensional generalization of a confidence interval.
- It is set of points in an n-dimensional space, often representing an ellipsoid around a point which is an estimated solution of a problem, although other shapes can also occur.

- Mathematically,

Let $X = (X_1, X_2, \dots, X_n)$ be a collection of random variables with the joint distribution $f(\theta)$. Where $\theta \in \Omega \subseteq R^d$ is unknown. Let $G(X) \subseteq R^d$ be a random set such that

$$P(\theta \in G(X)) \geq 1 - \alpha \text{ for all } \theta \in \Omega$$

With equality for at least one value of θ . The set $G(X)$ is called a $(1 - \alpha)$ -set estimator for θ .

- Furthermore, if we observe a realization of the data $x_1, x_2, \dots, x_n$, we say that $G(X)$, is a $100(1 - \alpha)\%$ confidence region for θ .
- The number $r = 1 - \alpha$ is called the **Coverage Probability**.
- Example:** Suppose θ_1 and θ_2 are two parameters for which the confidence region has to be worked out. Let (t_1, t_2) and (m_1, m_2) be the statistics which are to constitute a simultaneous interval leading to the confidence region of confidence coefficient $(1 - \alpha)$ in such a way that

$$P(t_1 \leq \theta_1 \leq t_2 \text{ and } m_1 \leq \theta_2 \leq m_2) = 1 - \alpha$$

11.1.3 shortest Set of Confidence Interval

- Shortest set of confidence interval is obtained by the direct analysis of coverage probability function.
- A simple sufficient and necessary condition for the existence of the smallest, two-sided confidence interval is provided and the smallest interval is derived if it exists.
- These proposed shortest confidence intervals are uniformly most accurate and have the uniform minimum expected length as well.
- Suppose we have two statistics T_1 and T_2 which are functions of sample variates $X_1, X_2, \dots, X_n$ from a density $f(x; \theta)$ such that the area to the left of T_1 is α_1 and to the right of T_2 is α_2 , both $\alpha_1, \alpha_2 > 0$ in such a way that $\alpha_1 + \alpha_2 = \alpha$. There are innumerable number of values of α_1 and α_2 which satisfy the equation $\alpha_1 + \alpha_2 = \alpha$, leading to innumerable number of confidence intervals. Now, problem arises which one to choose out of countless number of confidence intervals
- One widely accepted criterion is the width of the confidence interval. an interval with the smaller width is better than other having same confidence co-efficient

$$T_2 - T_1 \leq \hat{T}_2 - \hat{T}_1 \text{ for all } \theta \in \Omega$$

Hence, the confidence interval $(T_2 - T_1)$ which is uniformly shorter than any other confidence interval $(\hat{T}_2 - \hat{T}_1)$ is known as shortest confidence interval and is the most preferred one.

- In most of the problem it is not possible to construct confidence interval of shortest width for a given confidence co-efficient $1 - \alpha$. Hence, equal tail confidence intervals are generally used

11.1.4 One Sided Confidence Interval

Let $X_1, X_2, \dots, X_n$ be a random sample from the density $f(x; \theta)$ and

$T_1 = t_1(X_1, X_2, \dots, X_n)$ be any statistic such that

$$P(t_1 \leq \theta) = 1 - \alpha \text{ for all } \theta$$

Then t_1 is known as the **lower confidence bound** for the parameter θ with confidence co-efficient $(1 - \alpha)$. The interval (t_1, θ) is termed as the **lower one-sided confidence interval** for θ with confidence co-efficient $(1 - \alpha)$.

Similarly, any other statistic, $T_2 = t_2(X_1, X_2, \dots, X_n)$ be any statistic such that;-

$$P(t_2 \geq \theta) = 1 - \alpha \text{ for all } \theta$$

Then t_2 is known as the **upper confidence bound** for the parameter θ with confidence co-efficient $(1 - \alpha)$. The interval $(-\infty, t_2)$ is termed as the **upper one-sided confidence interval** for θ with confidence co-efficient $(1 - \alpha)$.

11.1.5 Large Sample Confidence Interval

Let $X_1, X_2, \dots, X_n$ be a random sample from the distribution $f(x; \theta)$ and let the likelihood function is the joint p.d.f

$$L(\theta) = L = \prod_{i=1}^n f(x_i, \theta)$$

The large sample confidence interval can be computer by using the following formula

$$\Psi = \frac{\left(\frac{\partial \ln L}{\partial \theta}\right)}{\left[E\left(\frac{\partial \ln L}{\partial \theta}\right)^2\right]^{\frac{1}{2}}}$$

Or

$$\Psi = \frac{\left(\frac{\partial \ln L}{\partial \theta}\right)}{\left[-E\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)\right]^{\frac{1}{2}}}$$

This will follow standard normal distribution with zero mean and unit variance.

11.1.6: Example: Large Sample Confidence Interval For Normal Distribution

The pdf of normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

The likelihood function

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

$$L(\mu) = \frac{1}{\sigma^n (2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2\sigma^2} \sum (x-\mu)^2}$$

Taking ln both sides

$$\ln L(\mu) = \ln \left[\frac{1}{\sigma^n (2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x-\mu)^2} \right]$$

$$\ln L(\mu) = \ln 1 - \frac{n}{2} \ln(\sigma^2) - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x-\mu)^2 \ln e$$

$$\ln L(\mu) = -\frac{n}{2} \ln(\sigma^2) - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x - \mu)^2$$

Differentiate partially w.r.t μ

$$\frac{\partial}{\partial \mu} \ln L(\mu) = -\frac{1}{\sigma^2} \sum_{i=1}^n (x - \mu) (-1)$$

$$\frac{\partial}{\partial \mu} \ln L(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (x - \mu) = \frac{1}{\sigma^2} (\sum x - n\mu) = \frac{n}{\sigma^2} (\bar{x} - \mu) \dots \dots \dots (1)$$

Differentiate, (1) w.r.t μ

$$\frac{\partial^2}{\partial \mu^2} \ln L(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (0 - 1) = -\frac{n}{\sigma^2}$$

$$-E \left[\frac{\partial^2}{\partial \mu^2} \ln L(\mu) \right] = \frac{n}{\sigma^2}$$

Using

$$\Psi = \frac{\left(\frac{\partial \ln L}{\partial \theta} \right)}{\left[-E \left(\frac{\partial^2 \ln L}{\partial \theta^2} \right) \right]^{\frac{1}{2}}} = \frac{\frac{n}{\sigma^2} (\bar{x} - \mu)}{\left(\frac{n}{\sigma^2} \right)^{\frac{1}{2}}} = \frac{n}{\sigma^2} (\bar{x} - \mu) \times \frac{\sigma}{\sqrt{n}} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$$

$$P \left(-Z < \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} < Z \right) = 1 - \alpha$$

$$P \left(-Z \frac{\sigma}{\sqrt{n}} - \bar{x} < \mu < Z \frac{\sigma}{\sqrt{n}} - \bar{x} \right) = 1 - \alpha$$

$$P \left(-\left\{ \bar{x} + Z \frac{\sigma}{\sqrt{n}} \right\} < \mu < -\left\{ \bar{x} - Z \frac{\sigma}{\sqrt{n}} \right\} \right) = 1 - \alpha$$

$$P \left(\bar{x} - Z \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + Z \frac{\sigma}{\sqrt{n}} \right) = 1 - \alpha$$

11. 2 Pivotal Quantity

- A pivotal quantity or pivot is a function of observation and unobservable parameter whose probability distribution does not depend upon the unknown parameter
- A pivot quantity need not to be a statistic.
- Its function and its values can depend upon the parameters of the model but its distribution must not.
- If pivot is a statistic, then it is called ancillary statistic.
- Let $X_1, X_2, \dots, X_n$ be a random sample from the distribution depend upon the parameter θ . Let $g(x; \theta)$ be a random variable whose distribution is the same for all θ , then g is called pivot quantity.

11.2.1 Uses of Pivot Quantity

- Pivotal quantities are commonly used for normalization to allow data comparison of different kind. It is relatively easy to construct pivots for location and scale parameters. For the location parameters we form differences to make it unitless and cancelation of location. For scale parameters we form ratio, so the scale cancel.
- Pivot quantities are fundamentals are fundamentals to the construction of test statistic, as they allow the statistic to not depend on parameter. For example, student's t distribution is for a normal distribution with unknown variance and mean.
- They also provide one method of constructing the confidence intervals.
- The use of pivotal quantities improves the performance of the bootstrap.
- In the form of ancillary statistic, pivot quantities are used to construct frequentist prediction interval

11.2.2 Examples of Pivotal Quantities

- One of the simple pivotal quantities is standard normal variate or z-score. Given the normal distribution with mean μ and variance σ^2 and observation X , the z-score is

$$Z = \frac{x - \mu}{\sigma} \sim N(0,1)$$

- For sample mean from a random sample of size n which has normal distribution $N\left(\mu, \frac{\sigma^2}{n}\right)$ the z-score is

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$$

- Given n independently, identically distributed (I.I.D) observations $X = (X_1, X_2, \dots, X_n)$ from normal distribution with unknown mean and variance σ^2 , a pivotal quantity can be obtained from the function

$$g(x, X) = \frac{\sqrt{n}(x - \bar{x})}{s}$$

Where,

$$\bar{x} = \frac{\sum X_i}{n} \quad \text{and} \quad s^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$$

Are unbiased estimators of μ and σ^2 .

The function $g(x, X)$ is the student's t-statistic for a new variable x . To be drawn from the same population as the already set of values X .

Using $x = \mu$, the function $g(\mu, X)$ becomes a pivotal quantity, which is also distributed by student's t-distribution with $n-1$ degree of freedom. As required even though μ appears as argument to the function g , the distribution of $g(\mu, X)$ does not depend upon the parameters μ and σ^2 of normal distribution.

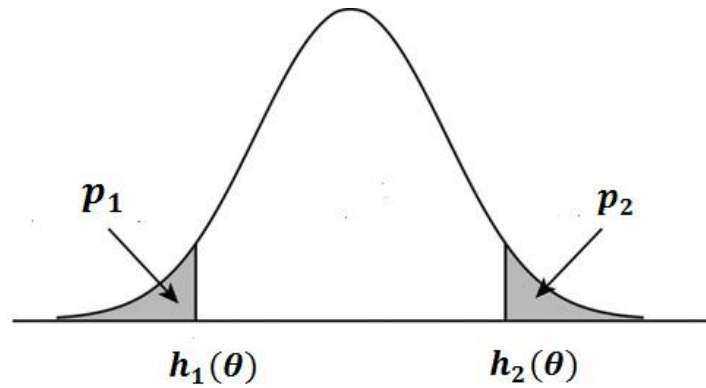
11.3 Statistical Method for Construction of Confidence Interval

Let $X_1, X_2, \dots, X_n$ be a random sample from the density $f(x; \theta)$. Further, we assume that the parameter θ_0 is real and parametric space Ω is some interval. We seek a confidence interval for θ_0 itself.

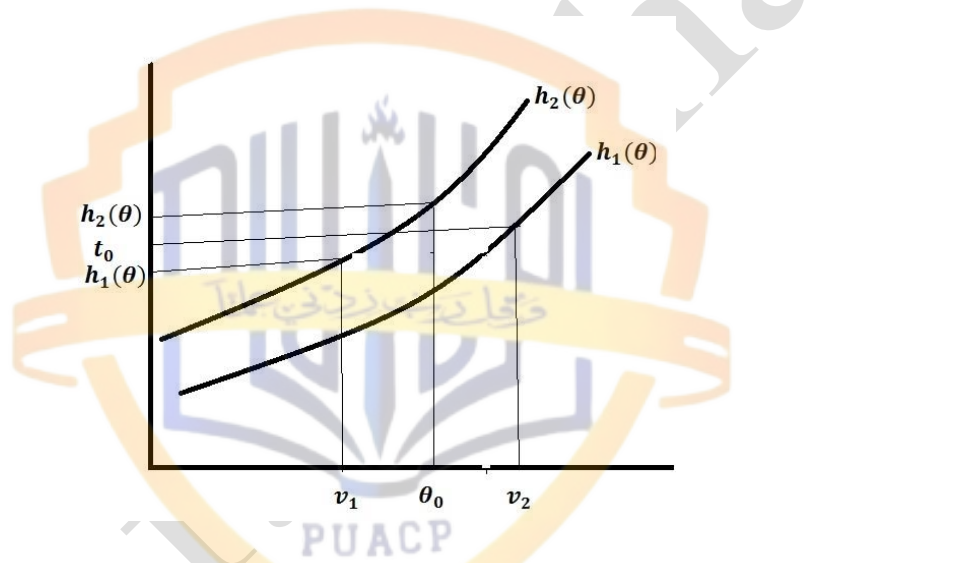
$T = t(X_1, X_2, \dots, X_n)$ denotes a statistic. The statistic can be selected in many ways. Let $f_T(t; \theta)$ denotes the density of T . We can define two functions $h_1(\theta)$ and $h_2(\theta)$ such that

$$\int_{-\infty}^{h_1(\theta)} f_T(t; \theta) dt = p_1 \quad \text{and} \quad \int_{h_2(\theta)}^{\infty} f_T(t; \theta) dt = p_2$$

Where p_1 and p_2 are two fixed values satisfying $p_1 > 0, p_2 > 0$ and $p_1 + p_2 \leq 1$.



We can plot two functions $h_1(\theta)$ and $h_2(\theta)$ as function of θ .



We know that $h_1(\theta) < h_2(\theta)$. let t_o denotes an observed value of T. plot t_o on the vertical axis and draw perpendicular v_1 and v_2 the interval (v_1, v_2) will turn to be $100(1 - p_1 - p_2)\%$ confidence interval for θ_o . As θ_o lies between v_1 and v_2 . by changing the range of v_1 and v_2 there can exist such points for which θ_o lies outside the range of v_1 and v_2 . As $h_1(\theta_o) < t_o < h_2(\theta_o)$ iff $v_1 < \theta_o < v_2$, for any observed sample $(X_1, X_2, \dots, X_n)$. Also

$$P_{\theta_o}[h_1(\theta_o) < t_o(X_1, X_2, \dots, X_n) < h_2(\theta_o)] = 1 - p_1 - p_2$$

So,

$$P_{\theta_o}[v_1(X_1, X_2, \dots, X_n) < \theta_o < v_2(X_1, X_2, \dots, X_n)] = 1 - p_1 - p_2$$

11.3.2 Example: Construction of Confidence Interval

Let $X_1, X_2 \dots \dots X_n$ be a random sample from density

$$f(x; \theta) = \frac{1}{\theta}, 0 < x < \theta$$

Let $Y_n = \max[X_1, X_2 \dots \dots X_n]$, be a sufficient statistic also it is MLE of θ . For which

$$f_T(t; \theta) = n \left(\frac{t}{\theta}\right)^{n-1} \left(\frac{1}{\theta}\right), 0 < t < \theta$$

Find confidence interval for θ using statistical method.

Solution:-

For given p_1 and p_2 we can find $h_1(\theta)$ and $h_2(\theta)$.

$$p_1 = \int_0^{h_1(\theta)} f_T(t; \theta) dt = \int_0^{h_1(\theta)} n \left(\frac{t}{\theta}\right)^{n-1} \left(\frac{1}{\theta}\right) dt = \frac{n}{\theta^n} \int_0^{h_1(\theta)} t^{n-1} dt$$

$$p_1 = \frac{n}{\theta^n} \left[\frac{t^n}{n} \right]_0^{h_1(\theta)} = \frac{[h_1(\theta)]^n}{\theta^n}$$

$$\theta^n p_1 = [h_1(\theta)]^n \Rightarrow h_1(\theta) = \theta p_1^{\frac{1}{n}} \dots \dots (1)$$

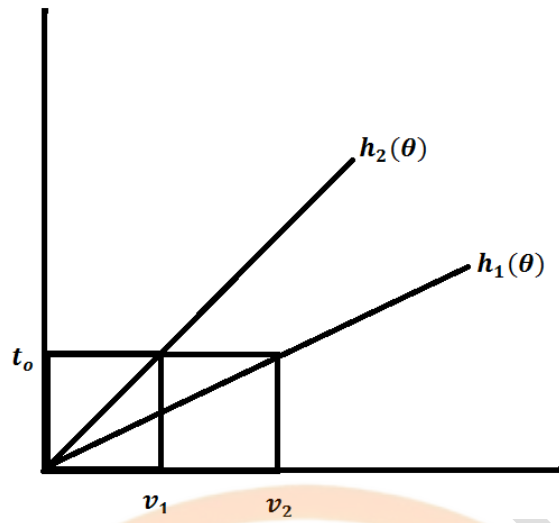
$$p_2 = \int_{h_2(\theta)}^{\theta} f_T(t; \theta) dt = \int_{h_2(\theta)}^{\theta} n \left(\frac{t}{\theta}\right)^{n-1} \left(\frac{1}{\theta}\right) dt = \frac{n}{\theta^n} \int_{h_2(\theta)}^{\theta} t^{n-1} dt$$

$$p_2 = \frac{n}{\theta^n} \left[\frac{t^n}{n} \right]_{h_2(\theta)}^{\theta} = \frac{[\theta^n - h_2(\theta)]^n}{\theta^n}$$

$$\theta^n p_2 = \theta^n - [h_2(\theta)]^n \Rightarrow [h_2(\theta)]^n = \theta^n [1 - p_2]$$

$$h_2(\theta) = \theta (1 - p_2)^{\frac{1}{n}} \dots \dots \dots (2)$$

For $t_0 = \max (X_1, X_2 \dots \dots X_n)$



$$t_0 = v_2 p_1^{\frac{1}{n}}$$

$$y_n p_1^{-\frac{1}{n}} = v_2$$

Similarly, from equation 2

$$t_0 = v_1 (1 - p_2)^{\frac{1}{n}}$$

$$y_n (1 - p_2)^{-\frac{1}{n}} = v_1$$

Hence,

$$(v_1, v_2) = \left[y_n (1 - p_2)^{-\frac{1}{n}}, y_n p_1^{-\frac{1}{n}} \right]$$

Is the required confidence interval for θ .

11.4 Confidence interval for Mean (σ^2 unknown)

Let $X_1, X_2, \dots, X_n$ be a random sample of size n from normal distribution with mean μ and variance σ^2 . The confidence interval for mean μ when σ^2 is known can easily be defined by using pivotal quantity method. we need a pivotal quantity say

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$$

But

$$q_1 < \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < q_2$$

Since, σ is unknown so this interval cannot be used to construct $t_1(X_1, X_2, \dots, X_n) < \mu < t_2(X_1, X_2, \dots, X_n)$, for any two statistics t_1 and t_2 . We need a pivotal quantity which only involve μ . We know by definition

$$t = \frac{Z}{\sqrt{\frac{\chi^2}{d.f}}} = \frac{\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}}{\sqrt{\frac{\sum(X - \bar{X})^2}{(n-1)\sigma^2}}} = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \sim t_{(n-1)}$$

This is t-distribution with $(n-1)$ degree of freedom and independent of parameter σ^2 . Hence, it is a pivotal quantity

$$q_1 < \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} < q_2$$

$$\frac{s}{\sqrt{n}} q_1 < \bar{X} - \mu < \frac{s}{\sqrt{n}} q_2$$

$$\frac{s}{\sqrt{n}} q_1 - \bar{X} < -\mu < \frac{s}{\sqrt{n}} q_2 - \bar{X}$$

$$\bar{X} - \frac{s}{\sqrt{n}} q_1 < \mu < \bar{X} - \frac{s}{\sqrt{n}} q_2$$

Where q_1 and q_2 are such that

$$P\left(q_1 < \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} < q_2\right) = \gamma = 1 - \alpha$$

Hence, $100(1 - \alpha)\%$ confidence interval for μ when σ^2 is unknown is given as

$$\left[\bar{X} - \frac{s}{\sqrt{n}}q_1, \bar{X} - \frac{s}{\sqrt{n}}q_2\right]$$

Length of confidence interval

The length of confidence interval is given as

$$L = \bar{X} - \frac{s}{\sqrt{n}}q_2 - \left(\bar{X} - \frac{s}{\sqrt{n}}q_1\right) = \frac{s}{\sqrt{n}}(q_2 - q_1)$$

The length will be minimum if q_1 and q_2 are selected in such a way that $(q_2 - q_1)$ is minimum.

We have to minimize

$$L = \frac{s}{\sqrt{n}}(q_2 - q_1) \dots \dots \dots (1)$$

Subject to

$$\int_{q_1}^{q_2} f(t)dt = \gamma \dots \dots \dots (2)$$

Where, $f(t)$ is t-distribution with $(n-1)$ degree of freedom. Differentiate equation 2 w.r.t q_1

$$\phi(q_2) \frac{\partial q_2}{\partial q_1} - \phi(q_1) = 0 \Rightarrow \frac{\partial q_2}{\partial q_1} = \frac{\phi(q_1)}{\phi(q_2)} \dots \dots \dots (3)$$

To minimize the length, we set

$$\frac{\partial L}{\partial q_1} = 0$$

$$\frac{s}{\sqrt{n}} \left[\frac{\partial q_2}{\partial q_1} - 1 \right] = 0$$

From (3)

$$\frac{s}{\sqrt{n}} \left[\frac{\phi(q_1)}{\phi(q_2)} - 1 \right] = 0$$

This equation will hold for iff.

$$\phi(q_1) = \phi(q_2)$$

In this case

$$\int_{q_1}^{q_2} f(t) dt \neq \gamma$$

Alternatively, $q_1 = -q_2$ or $q_2 = -q_1$ is the desired solution from a table of t-distribution

11.5 Confidence interval for Mean σ^2 (μ is Unknown)

Let $X_1, X_2, \dots, X_n$ be a random sample of size n from normal distribution with mean μ and variance σ^2 . The confidence interval for σ^2 when μ is unknown can easily be defined by using pivotal quantity method. We need a pivotal quantity which can be inverted

$$Q = \frac{\sum (X - \bar{X})^2}{\sigma^2} = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

Now, we have

$$q_1 < Q < q_2$$

$$q_1 < \frac{(n-1)s^2}{\sigma^2} < q_2$$

$$\frac{1}{q_1} > \frac{\sigma^2}{(n-1)s^2} < \frac{1}{q_2}$$

$$\frac{(n-1)s^2}{q_1} > \sigma^2 > \frac{(n-1)s^2}{q_2}$$

$$\frac{(n-1)s^2}{q_2} < \sigma^2 < \frac{(n-1)s^2}{q_1}$$

Is the $100(1 - \alpha)\%$ confidence interval for σ^2 when, μ is unknown, Where,

$$P(q_1 < \sigma^2 < q_2) = \gamma = 1 - \alpha$$

Here q_1 and q_2 are selected in such a way that

$$P(Q < q_1) = P(Q > q_2) = 1 - \frac{\alpha}{2}$$

q_1 and q_2 can be selected from the table of χ^2 distribution.

Length of confidence interval

The length of confidence interval is

$$L = \frac{(n-1)s^2}{q_1} - \frac{(n-1)s^2}{q_2} = (n-1)s^2 \left[\frac{1}{q_1} - \frac{1}{q_2} \right] \dots \dots (1)$$

We want to select the value of q_1 and q_2 so that the length will minimize Subject to

$$\int_{q_1}^{q_2} f(t) dt = \gamma \dots \dots (2)$$

Where, $f(t)$ is χ^2 distribution with $(n-1)$ degree of freedom. Differentiate equation 2 w.r.t q_1

$$\phi(q_2) \frac{\partial q_2}{\partial q_1} - \phi(q_1) = 0 \Rightarrow \frac{\partial q_2}{\partial q_1} = \frac{\phi(q_1)}{\phi(q_2)} \dots \dots (3)$$

To minimize the length, we set

$$\frac{\partial L}{\partial q_1} = 0$$

$$(n-1)s^2 \left[\frac{-1}{q_1^2} + \frac{1}{q_2^2} \frac{\partial q_2}{\partial q_1} \right] = 0$$

From equation (3)

$$(n-1)s^2 \left[\frac{-1}{q_1^2} + \frac{1}{q_2^2} \frac{\phi(q_1)}{\phi(q_2)} \right] = 0$$

$$(n-1)s^2 \left[\frac{-q_2^2 \phi(q_2) + q_1^2 \phi(q_1)}{q_1^2 q_2^2 \phi(q_2)} \right] = 0$$

The length of confidence interval will be minimum if

$$-q_2^2 \phi(q_2) + q_1^2 \phi(q_1) = 0$$

$$q_1^2 \phi(q_1) = q_2^2 \phi(q_2)$$

or

$$\phi(q_2) = q_1^2, \phi(q_1) = q_2^2$$

A solution for q_1 and q_2 can be obtained by trial and error or by numerical integration.

Note that the $100(1 - \alpha)\%$ for σ

$$\frac{\sqrt{(n-1)s^2}}{q_2} < \sigma < \frac{\sqrt{(n-1)s^2}}{q_1}$$



12 Testing

There is always some contentions about the value(s) of a parameter or the relationship between parameters. When parametric values are unknown, we estimate them through sample values. If the sample value is exactly the same as per our contention, there is no hitch in accepting it. And if it is far from our contention, there is no reason to accept it. But the problem arises when the sample provides a value which is neither exactly equal to the parametric value, nor too far. In this situation one has to develop some procedures(s) which enables one to decide whether to accept a contended (hypothetical) value or not on the basis of sample values. Such a procedure is known as testing of hypothesis. The theory of testing parametric statistical hypotheses was originally set forth by J.Neyman in 1928 and Karl Pearson in 1933.

12.1 Test of Statistical Hypothesis

A statistical hypothesis H is a rule or procedure which makes one to decide about the acceptance or rejection of the hypothesis H .

12.2 Principal Steps in Statistical Test

- A statistical test mainly involves four steps:-
- Evolving a test statistic.
- To know the sampling distribution of the test statistic.
- Setting of hypotheses testing conventions.
- Establishing a decision rule that lead to an inductive inference about the probable truth.

12.3 Two types of Error in Testing of Hypotheses

There is a probability of committing an error in making a decision about the hypothesis. Hence, two types of error can occur.

12.3.1 Type-I Error

The type-I error is also known as rejection error. Its probability is denoted by α . it is given as

$$P(\text{reject } H_0 | H_0 \text{ is true}) = \alpha$$

From definition it is apparent if H_0 is rejected only type-I error is committed.

12.3.2 Type-II Error

The type-II error is also known as acceptance error. Its probability is denoted by β . it is given as

$$P(\text{accept } H_0 | H_0 \text{ is false or } H_1 \text{ is true}) = \beta$$

12.3.3 Practical Example of Type-I and Type-II Errors

let us consider a practical problem, if a medicine is administered to a few patients of particular disease to cure them and the medicine is curing the disease, but it is claimed that it has no effect or has adverse effect, and hence, it is discontinued. this is type-I error. On the contrary, the medicine has adverse effect and is claimed to have good effect and the treatment is continued. This is type-II error.

12.3.4 Which Error is More Serious

If we delve into the consequences of both types of errors than we find type-ii error is more serious than type-I error. Hence, α may be relatively larger than β . in view of these facts, effort is made to minimize β even for certain risk of type-I error usually 0.01 ,0.05 or more

12.3.5 Level of Significance

The level of significance may be defined as the probability of type-I error which we already tolerate in making a decision about H_0 . It is our endeavor to carry out a test which minimizes both type of errors. Unfortunately, for a given set of observations, both errors cannot be controlled simultaneously. Hence, it is a general practice to assign a bound to type-I error and to minimize type-II error. Thus, one chooses a value of α between 0 and 1 which is known as the level of significance.

While choosing a level of significance, one should also consider the power of a test against various alternatives. If the power of test is too low, one should choose a higher value of α say 0.1,0.2,0.5 etc.

12.4 Critical Region

Let Ω denotes the sample space of observations or the potential set of data i.e. $\Omega = \{x_1, x_2, \dots, x_n\}$ where, $\{x_1, x_2, \dots, x_n\}$ is a possible value of $X_1, X_2, \dots, X_n$. then the sample space is specified by the test statistic under null hypothesis H_0 . The sample space Ω is divided into two types of regions, ω and $(\Omega - \omega)$. The level of significance is the size of critical region ω . The critical region is also known as region of rejection or rejection region. The region of rejection may be situated on the both tails or only one tail depending on the alternative hypothesis. The region $(\Omega - \omega)$ is called acceptance region. If the value of test statistic lies in the acceptance region the null hypothesis is accepted and if it lies in critical region the null hypothesis is rejected.

12.4.1 Best Critical Region (BCR)

A critical region of size α which minimizes β among all critical regions which do not exceed α is called Best Critical Region (BCR).

Consider the test of the simple null hypothesis $H_0: \theta = \theta_0$ against the alternative $H_1: \theta = \theta_1$. Let “C” be the critical region of size α . Such that

$$P(C, \theta_0) = \alpha$$

Let “D” be any other critical region of size α . Such as

$$P(D, \theta_0) = \alpha$$

Now if

$$P(C, \theta_1) \geq P(D, \theta_1)$$

Which means that if $H_1: \theta = \theta_1$, is true, the probability of rejecting $H_0: \theta = \theta_0$ using the critical region “C” is at least as great as the corresponding probability using any other critical region “D” of size α . Hence, best critical region is the one which has the greatest power among all other critical regions of size α .

12.4.2 Neyman-Pearson Lemma Sufficient Condition

Let $X_1, X_2, \dots, X_n$ be a random sample of size “n” from a distribution with the p.d.f $f(x; \theta)$ and θ_0 and θ_1 are two possible values of θ . The likelihood function

$$L(\theta) = L(x_1, x_2, \dots, x_n) = f(x_1, \theta)f(x_2, \theta) \dots \dots \dots f(x_n, \theta)$$

If there exist a positive constant “k” and a subset “C” of sample space such that

$$P(X_1, X_2, \dots, X_n \in C; \theta_0) = \alpha$$

$$\frac{L(\theta_0)}{L(\theta_1)} \leq k \text{ for all } (X_1, X_2, \dots, X_n) \in C$$

$$\frac{L(\theta_0)}{L(\theta_1)} \geq k \text{ for all } (X_1, X_2, \dots, X_n) \in C'$$

12.4.3 Use of Neyman-Pearson Lemma

- It helps to determine the size of type-I and type-II errors for given range of x .
- It helps to find out the power and power function of a test in case of testing a simple null hypothesis against simple alternative.

12.5 Power of Test and Power function

For testing the hypothesis H_0 against H_1 the test with probabilities α and β of type-I and type-II errors respectively, the quantity $(1 - \beta)$ is called power of test. The power of the test depends upon

The difference between the parameter value specified by H_1 and the actual value of the parameter. $(1 - \beta)$ can express as a function of the true parameter say, θ . If T is a test of H_0 the power function of T is defined as the probability of rejecting of H_0 when the distribution from which the sample is drawn, is parameterized by θ and denoted by $P_T(\theta)$. In terms of type-II error the function $1 - \beta(\theta)$ is known as the power function. Whereas $\beta(\theta)$ is called operating characteristic function or OC function.

12.6 Randomized and Non Randomized Test

12.6.1 Non-Randomized Test

A test T of a hypothesis H is said to be non-randomized test if the decision about the rejection or acceptance of H is based on a test statistic. H is rejected if the test statistic lies in the critical region, otherwise accepted.

12.6.2 Randomized Test

The test which uses no test statistic. The decision about H is taken on the basis of some predicted criteria. For instance, it is decided that H will be accepted if on tossing the coin head appears and will be rejected on the occurrence of tail. The randomized tests are seldom used.

12.7 Optimal Test

A test is said to be optimal if it minimizes both the errors α and β . Unfortunately, no such test is available as the reduction in one type of error causes the increase in the other. Hence, in normal practice a test which minimizes β or maximizes $1 - \beta$ for a desired low level of α is considered to be an optimum test or best test.

12.8 Properties of Test

For comparing different tests of hypothesis H , it becomes necessary to look into the properties of the tests. Main properties of the statistical non-randomized tests are

- Most Powerful Test(MPT)
- Uniformly Most Powerful Test (UMPT)
- Unbiased
- Uniformly Most Powerful Unbiased (UMPU)
- Minmax
- Admissible

12.8.1 Most Powerful Test (MPT)

The test using the best critical region is called most powerful test because it has the greatest value of the power function at $\theta = \theta_1$, when compared with that of the other tests of significance level α . If H_1 is a composite hypothesis, the power of test depends on each simple alternative in H_1

Let Ω be the sample space and ω the critical region and $\bar{\omega} = \Omega - \omega$, the acceptance region. A test T of $H_0: \theta = \theta_0$ v/s $H_1: \theta = \theta_1$ (simple v/s simple).based upon the sample values $x(x_1, x_2 \dots x_n)$ is said to be a most powerful test if and only if

$$P(x \in \omega | H_0) = \int_{\omega} L_0 dx = \alpha$$

And for any other test T_1 of the same size α

$$P_T(x \in \omega | H_1) \geq P_{T_1}(x \in \omega_1 | H_1))$$

Where, ω_1 , satisfy

$$\int_{\omega_1} L_0 dx = \alpha$$

The critical region α satisfying the above conditions is known as most powerful critical region of size α .

12.8.2 Uniformly Most Powerful Test (UMPT)

Let Ω be the sample space and ω the critical region and $\bar{\omega} = \Omega - \omega$, the acceptance region. A test T of $H_0: \theta = \theta_0$ v/s $H_1: \theta \neq \theta_0$.based upon the sample values $x(x_1, x_2 \dots x_n)$ is said to be a uniformly most powerful test if and only if

$$P(x \in \omega | H_0) = \int_{\omega} L_0 dx = \alpha$$

And for any other test T_1 of the same size α

$$P_T(x \in \omega | H_1) \geq P_{T_1}(x \in \omega_1 | H_1)) \text{ for } \theta \neq \theta_0$$

Where, ω_1 , satisfy

$$\int_{\omega_1} L_0 dx = \alpha$$

The critical region α satisfying the above conditions is known as uniformly most powerful critical region of size α .

Hence, a test defined by a critical region “C” of size α is a uniformly most powerful test if it is most power test against each simple alternative in H_1 .

12.8.3 Unbiased Test

A test T of the null hypothesis $H_0: \theta \in \Theta_0$ vs $H_1: \theta \in \Theta_1$ is said to be unbiased test if the probability of rejecting H_0 when it is false is at least as much as the probability of rejecting H_0 when it is true, i.e.

$$\sup_{\theta \in \Theta_0} P_T(\theta) \leq \inf_{\theta \in \Theta_1} P_T(\theta) \text{ where } \Theta_0 \cup \Theta_1 = \Theta \text{ and } \Theta_0 \cap \Theta_1 = \phi$$

12.8.4 Uniform Most Powerful Unbiased Test(UMPUT)

If among the class of unbiased tests, there exists, a test, which is uniformly most powerful is known as Uniformly Most Powerful Unbiased Test (UMPUT). The advantage of UMPUT is that it may exist even if UMPT do not exists.

12.8.5 Minmax Test

A test T of $H_0: \theta = \theta_0$ v/s $H_1: \theta = \theta_1$ is said to be minimax test if for any other test T_1 the following inequality holds

$$\max[R(T, \theta_0), R(T, \theta_1)] \leq \max[R(T_1, \theta_0), R(T_1, \theta_1)]$$

Where $R(T, \theta)$ is the risk in using the test T when θ is the true value of the parameter and

$R(T_1, \theta)$ is the risk in using the test T_1 when θ is the true value of the parameter θ .

12.8.6 Admissible Test

A test procedure T is said to be admissible if there exists no other procedure T_1 whose risk for testing a hypothesis about a parameter θ is less than the risk of T for testing the same hypothesis about θ . Notationally, test procedure T is admissible if

$$R(\theta, T) \leq R(\theta, T_1) \text{ for all } \theta$$

$$R(\theta, T) < R(\theta, T_1) \text{ for some } \theta$$

12.9 Methods of Testing Hypothesis

The different method of testing of hypothesis are

- Likelihood Ratio Test.
- Student's t-test.
- Normal Deviate or Z-test.
- Chi-square test.
- F-test.
- Bayes Test.
- Sequential Probability Ratio test (SPRT).

12.10 Sequential Analysis

- Sequential analysis is a procedure which leads to a statistical inference
- The number of observations to be made is not pre-determined before the experiment.
- The procedure indicates, sufficient observations have been gathered to make our decision with the risks we have chosen.
- On the average fewer observations will be required for this procedure.
- Use of this procedure will not increase the risk of α and β .
- For some cases, only half of the observations will be required for the sequential procedure.in comparison with the number required if the sample size is fixed in advance.

12.10.1 Average Sample Number

• The expected number of observations required to reach the decision in sequential analysis is called average sample number.

• In sequential procedures for statistical estimation, hypothesis testing or decision theory, the number of observations taken is not pre-determined, but depends on the observations themselves. The expected value of a sample size is called Average Sample Number(ASN).

• ASN depends upon the underlying distribution of the observations, which is often unknown.

• If the distribution of observation is determined by the value of some parameter, then ASN becomes a function of that parameter.

• ASN can be computed according to two conditions:-

1-When H_0 is true.

$$E(N) = E(N|H_0 \text{ is True})$$

$$E(N) = \frac{\alpha_a \ln k_0 + (1 - \alpha_a) \ln k_1}{E(z_i|H_0 \text{ is True})}$$

$$E(N) = \frac{\alpha_a \ln \left[\frac{\alpha_a}{1 - \beta_a} \right] + (1 - \alpha_a) \ln \left[\frac{1 - \alpha_a}{\beta_a} \right]}{E(z_i|H_0 \text{ is True})}$$

Where α_a and β_a are the values of type-I and type-II errors respectively and

$$z_i = \ln \left[\frac{f_0(x_i)}{f_1(x_i)} \right]$$

2- When H_1 is true.

$$E(N) = \frac{(1 - \beta_a) \ln k_0 + \beta_a \ln k_1}{E(z_i|H_1 \text{ is True})}$$

$$E(N) = \frac{(1 - \beta_a) \ln \left[\frac{\alpha_a}{1 - \beta_a} \right] + \beta_a \ln \left[\frac{1 - \alpha_a}{\beta_a} \right]}{E(z_i|H_1 \text{ is True})}$$

Where α_a and β_a are the values of type-I and type-II errors respectively and

$$z_i = \ln \left[\frac{f_0(x_i)}{f_1(x_i)} \right]$$

12.11 Neyman-Pearson Lemma

Statement:-

Let $X_1, X_2, \dots, X_n$, where “n” is fixed positive number denotes a random sample of size n from density $f(x; \theta)$ and if the joint p.d.f of $X_1, X_2, \dots, X_n$ is

$$L(\theta; x_1, x_2, \dots, x_n) = f(x_1, \theta)f(x_2, \theta) \dots \dots f(x_n, \theta)$$

Let θ' and θ'' be distinct fixed values of θ so that $\Omega = \{\theta: \theta = \theta', \theta''\}$. Let “k” be any positive number and let “C” be the subset of the sample space such that

(a)

$$\frac{L(\theta'; x_1, x_2, \dots, x_n)}{L(\theta''; x_1, x_2, \dots, x_n)} \leq k$$

For each point $(x_1, x_2, \dots, x_n) \in C^{CR}$

(b)

$$\frac{L(\theta'; x_1, x_2, \dots, x_n)}{L(\theta''; x_1, x_2, \dots, x_n)} \geq k$$

For each point $(x_1, x_2, \dots, x_n) \in C^*$

$$C = \alpha = P(x_1, x_2, \dots, x_n \in C, H_0)$$

Where, C is a BCR of size α , then the test γ^* corresponding to the CR “C” is most powerful test for testing $H_0: \theta = \theta'$ against a simple alternative $H_1: \theta = \theta''$.

Proof:-

Assuming the variables are of continuous type . if “C” is the only CR of size α then then theorem is proved.

If there is another CR of size α which is denoted by A then n total integrals

$$\iint \dots \dots \int L(\theta'; x_1, x_2, \dots, x_n) dx_1 dx_2 \dots dx_n = \int_R L(\theta)$$

we have to prove

$$\int_C L(\theta'') \text{ as } \int_C L(\theta'') - \int_A L(\theta'') \geq 0$$

Since, C is the union of disjoint sets $C \cap A$ and $C \cap A^*$

$$C = (C \cap A) \cup (C \cap A^*)$$

And also, “A” is the union of disjoint sets $A \cap C$ and $A \cap C^*$

$$A = (A \cap C) \cup (A \cap C^*)$$

Then

$$\int_C L(\theta'') - \int_A L(\theta'') \dots \dots \dots (1)$$

$$\int_C L(\theta'') - \int_A L(\theta'') = \left[\int_{C \cap A} L(\theta'') + \int_{C \cap A^*} L(\theta'') \right] - \left[\int_{A \cap C} L(\theta'') + \int_{A \cap C^*} L(\theta'') \right]$$

$$\int_C L(\theta'') - \int_A L(\theta'') = \int_{C \cap A^*} L(\theta'') - \int_{A \cap C^*} L(\theta'') \dots \dots \dots (2)$$

Now, from the condition (a) stated in the theorem

$$\frac{L(\theta'')}{L(\theta')} \geq \frac{1}{k}$$

$$L(\theta'') \geq \frac{1}{k} L(\theta')$$

At each point of C and hence each point of $C \cap A^*$

$$\int_{C \cap A^*} L(\theta'') \geq \frac{1}{k} \int_{C \cap A^*} L(\theta') \dots \dots \dots (3)$$

From the condition (b) stated in the theorem

$$\frac{L(\theta'')}{L(\theta')} \geq \frac{1}{k}$$

$$L(\theta'') \geq \frac{1}{k} L(\theta')$$

At each point of C^* and hence each point of $A \cap C^*$

$$\int_{A \cap C^*} L(\theta'') \geq \frac{1}{k} \int_{A \cap C^*} L(\theta'') \dots \dots \dots (4)$$

Inequalities (3) and (4) implies that

$$\int_{C \cap A^*} L(\theta'') - \int_{A \cap C^*} L(\theta'') \geq \frac{1}{k} \left[\int_{C \cap A^*} L(\theta'') - \int_{A \cap C^*} L(\theta'') \right] \dots \dots (5)$$

the L.H.S of (5) is equal to (2)

$$\int_C L(\theta'') - \int_A L(\theta'') \geq \frac{1}{k} \left[\int_{C \cap A^*} L(\theta'') - \int_{A \cap C^*} L(\theta'') \right] \dots \dots (6)$$

Take

$$\begin{aligned} \int_{C \cap A^*} L(\theta') - \int_{A \cap C^*} L(\theta') &= \int_{C \cap A^*} L(\theta') + \int_{C \cap A} L(\theta') - \int_{A \cap C} L(\theta') - \int_{A \cap C^*} L(\theta') \\ \int_{C \cap A^*} L(\theta') - \int_{A \cap C^*} L(\theta') &= \int_C L(\theta') - \int_A L(\theta') = \alpha - \alpha = 0 \end{aligned}$$

Putting in (6)

$$\int_C L(\theta'') - \int_A L(\theta'') \geq \frac{1}{k} (0)$$

$$\int_C L(\theta'') - \int_A L(\theta'') \geq 0$$

13 Questions Related to Testing

13.1 Example 1: Finding Type-I and II Errors and Power function

Question1: Suppose that $H_0: \theta = \theta_0, H_1: \theta = 2\theta_0$. We have to test H_0 v/s H_1 based on the single random observation y , from the probability $\theta e^{-\theta x}, \theta > 0, 0 \leq x < \infty$. If the hypothesis taken as $y \geq \frac{2}{\theta_0}$, then find type-I and type-II error and power function.

Solution:-

Type-I Error

$$\alpha = P(\text{Reject } H_0 | H_0 \text{ is True})$$

$$\alpha = \int_{\frac{2}{\theta_0}}^{\infty} \theta_0 e^{-\theta_0 y} dy = \theta_0 \left[\frac{e^{-\theta_0 y}}{-\theta_0} \right] \Big|_{\frac{2}{\theta_0}}^{\infty}$$

$$\alpha = -[e^{-\infty} - e^{-2}] = e^{-2} = 0.1352$$

Type-II Error

We have to find $\beta(2\theta_0)$, but first we find

$$\beta(\theta) = \int_{\frac{2}{\theta}}^{\infty} \theta e^{-\theta y} dy = \theta \left[\frac{e^{-\theta y}}{-\theta} \right] \Big|_{\frac{2}{\theta}}^{\infty}$$

$$\beta(\theta) = -\left[e^{-\infty} - e^{-\frac{2\theta}{\theta_0}} \right] = e^{-\frac{2\theta}{\theta_0}}$$

Hence

$$\beta(\theta) = e^{-\frac{2\theta}{\theta_0}}$$

Now, for $\theta = 2\theta_0$,

$$\beta = P(\text{Accept } H_0 | H_0 \text{ is False}) = \beta(\theta) = e^{-\frac{2(2\theta_0)}{\theta_0}} = e^{-4} = 0.0183$$

Power Function

$$\text{power Function} = 1 - \beta(\theta) = 1 - e^{-\frac{2\theta}{\theta_0}}$$

If $\theta > \theta_0$, then the power increases for increasing deviation of θ from θ_0 .

13.2 Finding BCR

Question 2: Let $X_1, X_2, \dots, X_n$ denote a random sample from a distribution which has the p.d.f $f(x_i)$, that is **positive on only non-negative integers**. It is desired to test the simple hypothesis $H_0: f(x) = \frac{e^{-1}}{x!}, x = 0, 1, 2, \dots, H_1: f(x) = \left(\frac{1}{2}\right)^{x+1}; x = 0, 1, 2, \dots$. Derive the expression for BCR consider the case $n=1$ and $k=1$, k being only +ve integer in the expression

$$\frac{L(\theta', x_1, x_2, \dots, x_n)}{L(\theta'', x_1, x_2, \dots, x_n)} \leq k$$

Where, $H_0: \theta = \theta'$ and $H_1: \theta = \theta''$. Find power of the test when (i) H_0 is True (ii) H_1 is True.

Solution:-

Given that

$$\frac{L(\theta', x_1, x_2, \dots, x_n)}{L(\theta'', x_1, x_2, \dots, x_n)} \leq k$$

$$\frac{\prod \left(\frac{e^{-1}}{x!} \right)}{\prod \left(\frac{1}{2} \right)^{x+1}} \leq k$$

$$\frac{e^{-n}}{x_1! \cdot x_2! \cdot \dots \cdot x_n!} \leq k \cdot \left(\frac{1}{2} \right)^n \left(\frac{1}{2} \right)^{\sum x}$$

$$\frac{(2e^{-1})^n 2^{\sum x_i}}{\prod x_i!} \leq k \dots \dots \dots (1)$$

Taking ln both sides

$$n \ln 2 - n + \sum x_i \ln 2 - \ln \prod x_i! \leq \ln k$$

$$\sum x_i \ln 2 - \ln \prod x_i! \leq \ln k - n \ln 2 + n$$

This inequality determines the BCR.

Now, consider the case $n=1$ and $k=1$. Putting in (1)

$$\frac{(2e^{-1})^1 2^x}{x!} \leq 1$$

$$\frac{2^x}{x!} \leq \frac{e}{2} \dots \dots (2)$$

This inequality, hold for all the points in the set

$$C = \{x; x = 0, 3, 4, 5 \dots\}$$

Because, at $x = 1, 2$ the inequality (2) does not hold . i.e.

$$C^* = \{x; x = 1, 2\}$$

Is the acceptance region.

(i) Power of Test When H_0 is True

Now, power of test when H_0 is True at $n=1$, $k=1$ is

$$P[X \in C; H_0 \text{ is True}] = P[X = 0, 3, 4, 5 \dots \dots, H_0 \text{ True}] = 1 - P(X = 1, 2; H_0)$$

$$P[X \in C; H_0 \text{ is True}] = 1 - [P(X = 1) + P(X = 2)] = 1 - \left[\frac{e^{-1}}{1!} + \frac{e^{-1}}{2!} \right]$$

$$P[X \in C; H_0 \text{ is True}] = 1 - [0.368 + 0.184] = 0.448$$

(ii) Power of Test When H_1 is True

$$P[X \in C; H_1 \text{ is True}] = P[X = 0, 3, 4, 5 \dots \dots, H_1 \text{ True}] = 1 - P(X = 1, 2; H_1)$$

$$P[X \in C; H_1 \text{ is True}] = 1 - [P(X = 1) + P(X = 2)] = 1 - \left[\left(\frac{1}{2}\right)^{1+1} + \left(\frac{1}{2}\right)^{2+1} \right]$$

$$P[X \in C; H_0 \text{ is True}] = 1 - \left[\frac{1}{4} + \frac{1}{8} \right] = 0.625$$

13.3 Example 1: SPRT (Bernoulli Distribution)

Question 3: Let X has the p.d.f $f(x; \theta) = \theta^x(1 - \theta)^{1-x}$, $x = 0, 1$. Perform SPRT to test $H_0: \theta = \frac{1}{3}$ v/s $H_1: \theta = \frac{2}{3}$.

Solution:-

The p.d.f

$$f(x; \theta) = \theta^x(1 - \theta)^{1-x}$$

The likelihood function

$$L(\theta) = \prod_{i=1}^n \theta^{x_i}(1 - \theta)^{1-x_i} = \theta^{\sum x_i}(1 - \theta)^{n - \sum x_i}$$

$$\frac{L\left(\frac{1}{3}; \theta\right)}{L\left(\frac{2}{3}; \theta\right)} = \frac{\left(\frac{1}{3}\right)^{\sum x_i} \left(1 - \frac{1}{3}\right)^{n - \sum x_i}}{\left(\frac{2}{3}\right)^{\sum x_i} \left(1 - \frac{2}{3}\right)^{n - \sum x_i}} = \frac{\left(\frac{1}{3}\right)^{\sum x_i} \left(\frac{2}{3}\right)^{n - \sum x_i}}{\left(\frac{2}{3}\right)^{\sum x_i} \left(\frac{1}{3}\right)^{n - \sum x_i}} = \left(\frac{1}{3}\right)^{2\sum x_i - n} \left(\frac{2}{3}\right)^{n - 2\sum x_i}$$

$$\frac{L\left(\frac{1}{3}; \theta\right)}{L\left(\frac{2}{3}; \theta\right)} = 2^{n - 2\sum x_i}$$

We will continue to observe the sample items as long as

$$K_0 < 2^{n - 2\sum x_i} < K_1$$

Taking log to the base 2

$$\log_2 K_0 < n - 2\sum x_i < \log_2 K_1$$

$$\log_2 K_0 - n < -2\sum x_i < \log_2 K_1 - n$$

Divide by -2

$$\frac{n}{2} - \frac{1}{2} \log_2 K_0 > \sum x_i > \frac{n}{2} - \frac{1}{2} \log_2 K_1$$

$$\frac{n}{2} - \frac{1}{2} \log_2 K_1 < \sum x_i < \frac{n}{2} - \frac{1}{2} \log_2 K_0$$

$$C_0(n) < \sum x_i < C_1(n)$$

Thus, we will continue to observe the sample as long as

$$C_0(n) < \sum x_i < C_1(n)$$

Accept H_0 when $C_1(n) \leq \sum x_i$,

Reject H_0 when $C_0(n) \geq \sum x_i$

13.4 Example 2: SPRT (Bernoulli Distribution)

Question 4: It is desired that $H_0: \theta = 0.5$ v/s $H_1: \theta = 0.2$ in Bernoulli population. Perform SPRT with $\alpha_a = 0.1$, $\beta_a = 0.2$. If the sequence of observations turn out to be T, H, H, T, H, T, H, H, T, T, At which observation the sampling will be stop and what will be your decision.

Solution:-

Assume we concentrate on the occurrence of Head.

$$K_0' = \frac{\alpha_a}{1 - \beta_a} = \frac{0.1}{1 - 0.2} = \frac{1}{8} = 0.125$$

$$K_1' = \frac{1 - \alpha_a}{\beta_a} = \frac{1 - 0.1}{0.2} = \frac{9}{2} = 4.5$$

The likelihood function

$$L(\theta) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^{\sum x_i} (1 - \theta)^{n - \sum x_i}$$

The likelihood ratio for Bernoulli distribution

$$\frac{L(0.5; n)}{L(0.2; n)} = \frac{(0.5)^{\sum x_i} (1 - 0.5)^{n - \sum x_i}}{(0.2)^{\sum x_i} (1 - 0.2)^{n - \sum x_i}} = \frac{(0.5)^{\sum x_i} (0.5)^{n - \sum x_i}}{(0.2)^{\sum x_i} (0.8)^{n - \sum x_i}}$$

$$\frac{L(0.5; n)}{L(0.2; n)} = \frac{(0.5)^n}{(0.2)^{\sum x_i} (0.8)^{n - \sum x_i}}$$

S.N	Outcome	$\sum x_i$	L.R
1	T	0	0.625
2	H	1	1.5625
3	H	2	3.91
4	T	2	2.44
5	H	3	6.1
6	T	3	3.82
7	H	4	
8	H	5	
9	T	5	
10	T	5	

K_1'

Thus, at 5th observation, H_0 is accepted and sampling will be stopped because $6.10 >$

13.5 Example 3: SPRT (Normal Distribution)

Question 5: Let $X \sim N(\theta, 100)$, perform SPRT to test $H_0: \theta = 75$ v/s $H_1: \theta = 78$. Such that α and β are approximately equal to 0.10.

Solution;-

Given that $\alpha_a = \beta_a = 0.10$

$$K_0 = \frac{\alpha_a}{1 - \beta_a} = \frac{0.1}{1 - 0.1} = \frac{1}{9} = 0.111$$

$$K_1 = \frac{1 - \alpha_a}{\beta_a} = \frac{1 - 0.1}{0.1} = \frac{9}{1} = 9$$

The p.d.f for $X \sim N(\theta, 100)$

$$f(x) = \frac{1}{\sqrt{2\pi(100)}} e^{-\frac{1}{2}\left(\frac{x-\theta}{10}\right)^2}, -\infty < X < \infty$$

The likelihood function

$$L = \left(\frac{1}{200\pi}\right)^{\frac{n}{2}} e^{-\frac{1}{200}\sum(x_i - \theta)^2}$$

The likelihood ratio

$$\frac{L(75; n)}{L(78; n)} = \frac{\left(\frac{1}{200\pi}\right)^{\frac{n}{2}} e^{-\frac{1}{200}\sum(x_i - 75)^2}}{\left(\frac{1}{200\pi}\right)^{\frac{n}{2}} e^{-\frac{1}{200}\sum(x_i - 78)^2}}$$

$$\frac{L(75; n)}{L(78; n)} = e^{-\frac{1}{200}[\sum(x_i - 75)^2 - \sum(x_i - 78)^2]}$$

$$\frac{L(75; n)}{L(78; n)} = e^{-\frac{1}{200}[\sum(x^2 + 5625 - 150x - x^2 - 6084 + 156x)]} = e^{-\frac{1}{200}\sum(6x - 459)}$$

$$\frac{L(75; n)}{L(78; n)} = e^{-\frac{1}{200}(6\sum x - 459n)}$$

The decision will be based upon the expression

$$K_0 < \frac{L(75; n)}{L(78; n)} < K_1$$

$$\frac{1}{9} < e^{-\frac{1}{200}(6\sum x - 459n)} < 9$$

Taking ln both sides

$$-\ln 9 < -\frac{1}{200}(6\sum x - 459n) < \ln 9$$

$$200\ln 9 > 6\sum x - 459n > -200\ln 9$$

$$-200\ln 9 < 6\sum x - 459n < 200\ln 9$$

$$459n - 200\ln 9 < 6\sum x < 459n + 200\ln 9$$

$$\frac{459n}{6} - \frac{200\ln 9}{6} < \sum x < \frac{459n}{6} + \frac{200\ln 9}{6}$$

$$\frac{153n}{2} - \frac{100\ln 9}{3} < \sum x < \frac{153n}{2} + \frac{100\ln 9}{3}$$

We continue sampling if $\sum x$ lies between the above limit.

Accept H_0 : If $\sum x$ is increase

Reject H_0 : If $\sum x$ is decrease

13.6 Example 4: SPRT Equality of Means (Normal Distribution)

Question 6: Let $\bar{X}$ and $\bar{Y}$ be sample means of random samples each of size n drawn independently from $N(\mu_1, 400)$ and $N(\mu_2, 225)$.

$H_0: \theta = 0$ v/s $H_1: \theta > 0$, is rejected iff $\bar{X} - \bar{Y} \geq K$, where $\theta = \mu_1 - \mu_2$. Find n and K so that $\pi(0) = 0.05$, $\pi(10) = 0.90$, where $\pi(\theta)$ is power function.

Solution:-

$$E(\bar{X} - \bar{Y}) = E(\bar{X}) - E(\bar{Y}) = \mu_1 - \mu_2 = 0 \text{ (Under } H_0\text{)}$$

Also

$$\sigma_{\bar{X}-\bar{Y}}^2 = V(\bar{X} - \bar{Y}) = V(\bar{X}) + V(\bar{Y}) = \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n} = \frac{400}{n} + \frac{225}{n} = \frac{625}{n}$$

Hence,

$$\sigma_{\bar{X}-\bar{Y}} = \frac{25}{\sqrt{n}}$$

As given

$$\alpha = P_{\theta=0}(\bar{X} - \bar{Y} \geq K) = 0.05$$

$$\alpha = \int_K^{\infty} \frac{1}{\sqrt{2\pi} \frac{25}{\sqrt{n}}} e^{-\frac{1}{2} \left[\frac{\bar{X}-\bar{Y}-0}{\frac{25}{\sqrt{n}}} \right]^2} d_{\bar{X}-\bar{Y}}$$

Put

$$t = \bar{X} - \bar{Y} \Rightarrow dt = d_{\bar{X}-\bar{Y}}$$

$$\alpha = \int_K^{\infty} \frac{\sqrt{n}}{25\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{\sqrt{n}}{25} t \right)^2} dt$$

Put

$$z = \frac{\sqrt{n}}{25} t \Rightarrow \frac{25}{\sqrt{n}} dz = dt$$

When $t = k$, then $z = \frac{\sqrt{n}}{25} K$, when $t = \infty$, $z = \infty$

$$\alpha = \int_{\frac{\sqrt{n}}{25} K}^{\infty} \frac{\sqrt{n}}{25\sqrt{2\pi}} e^{-\frac{z^2}{2}} \left(\frac{25}{\sqrt{n}} dz \right)$$

$$\alpha = \int_{\frac{\sqrt{n}}{25}K}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$z = \frac{\sqrt{n}}{25}K \text{ at } \alpha = 0.05 \text{ is } 1.645$$

$$\frac{\sqrt{n}}{25}K = 1.645 \dots \dots \dots (1)$$

Now,

$$1 - \beta = P_{\theta=10}(\bar{X} - \bar{Y} \geq K) = P_{\theta=10}(t \geq K)$$

$$0.90 = \int_K^{\infty} \frac{\sqrt{n}}{25\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{t-10}{\frac{25}{\sqrt{n}}}\right)^2} dt$$

put

$$z = \frac{\sqrt{n}}{25}(t - 10) \Rightarrow \frac{25}{\sqrt{n}} dz = dt$$

$$\text{When } t = k, \text{ then } z = \frac{\sqrt{n}}{25}(K - 10), \text{ when } t = \infty, z = \infty$$

$$0.90 = \int_{\frac{\sqrt{n}}{25}(K-10)}^{\infty} \frac{\sqrt{n}}{25\sqrt{2\pi}} e^{-\frac{z^2}{2}} \left(\frac{25}{\sqrt{n}} dz\right)$$

$$0.90 = \int_{\frac{\sqrt{n}}{25}(K-10)}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$z = \frac{\sqrt{n}}{25}(K - 10) \text{ at } \alpha = 0.90 \text{ is } -1.282$$

$$\frac{\sqrt{n}}{25}(K - 10) = -1.282 \dots \dots \dots (2)$$

In eq(2)

$$\frac{\sqrt{n}}{25}K - \frac{\sqrt{n}}{25}(10) = -1.282$$

From eq(1)

$$1.645 - \sqrt{n}(0.4) = -1.282$$

$$-\sqrt{n}(0.4) = -1.282 - 1.645$$

$$\sqrt{n} = \frac{2.927}{0.4} = 7.3175$$

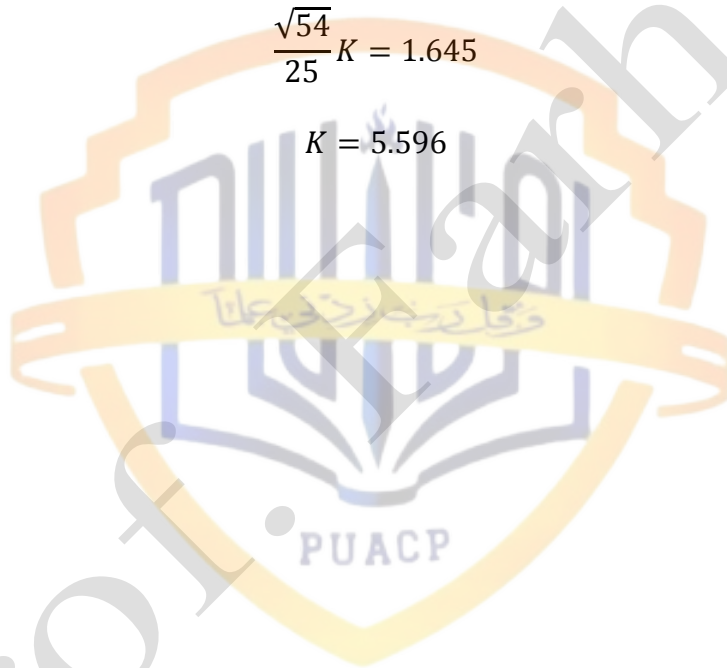
Squaring and rounding up to next integer we get

$$n = 54$$

Put in eq(1)

$$\frac{\sqrt{54}}{25} K = 1.645$$

$$K = 5.596$$



13.7 Example 5:SPRT (Appropriate Values of K_0 and K_1)

Question 7: Determination of appropriate value of K_0 and K_1 in SPRT.

or

for SPRT show that

$$A \cong \frac{1-\beta}{\alpha} \text{ and } B \cong \frac{\beta}{1-\alpha}$$

Where α and β are respectively error sizes for testing $H_0: \theta = \theta_0$ v/s $H_1: \theta = \theta_1$

Solution:-

if we denote the power of test when H_0 by the symbol α and the power of the test when H_1 is true by the symbol $1 - \beta$, with the sets:-

C_n Critical Region and A_n : Acceptance Region

$$P(\text{Reject } H_0 | H_0 \text{ is true}) = \alpha = \int_{C_n} L(\theta'; n)$$

$$P(\text{Reject } H_0 | H_1 \text{ is true}) = 1 - \beta = \int_{C_n} L(\theta''; n)$$

We Also have

$$1 - \alpha = \int_{A_n} L(\theta'; n) \text{ and } \beta = \int_{A_n} L(\theta''; n)$$

If $X_1, X_2, \dots, X_n$ belong to C_n ; we have

$$\frac{L(\theta'; n)}{L(\theta''; n)} \leq K_0$$

$$L(\theta'; n) \leq K_0 L(\theta''; n)$$

Integrate under Critical Region

$$\int_{C_n} L(\theta'; n) \leq K_0 \int_{C_n} L(\theta''; n)$$

$$\alpha \leq K_0(1 - \beta) \dots \dots (A)$$

If $X_1, X_2, \dots, X_n$ belong to A_n ; we have

$$\frac{L(\theta'; n)}{L(\theta''; n)} \geq K_1$$

$$L(\theta'; n) \geq K_1 L(\theta''; n)$$

Integrate under acceptance Region

$$\int_{A_n} L(\theta'; n) \geq K_1 \int_{A_n} L(\theta''; n)$$

$$1 - \alpha \geq K_1 \beta \dots \dots \dots (b)$$

From (A) and (B) we get

$$\frac{\alpha}{1 - \beta} \leq K_0 \text{ and } \frac{1 - \alpha}{\beta} \geq K_1$$

Provided, $\beta \neq 0$ and $1 - \beta \neq 0$

$$K_0 = \frac{\alpha_a}{1 - \beta_a} \text{ and } K_1 = \frac{1 - \alpha_a}{\beta_a}$$

Where α_a and β_a are the assumed values of α and β and K_0 and K_1 are determine values of K_0 and K_1 .

Note: For proving $A \cong \frac{1 - \beta}{\alpha}$ and $B \cong \frac{\beta}{1 - \alpha}$, we need to replace $K_0 = A$ and $K_1 = B$

13.8 Example 6 SPRT (Relationship Between Error Sizes)

Question 8: let α' and β' are the error sizes of SPRT defined by K_0 and K_1 . Where $K_0 = \frac{\alpha}{1 - \beta}$ and $K_1 = \frac{1 - \alpha}{\beta}$, then show that

$$\alpha' + \beta' \leq \alpha + \beta$$

Solution:

if we denote the power of test when H_0 by the symbol α' and the power of the test when H_1 is true by the symbol $1 - \beta'$, with the sets:-

C_n Critical Region and A_n : Acceptance Region

$$P(\text{Reject } H_0 | H_0 \text{ is true}) = \alpha' = \int_{C_n'} L(\theta'; n)$$

$$P(\text{Reject } H_0 | H_1 \text{ is true}) = 1 - \beta' = \int_{C_n'} L(\theta''; n)$$

We Also have

$$1 - \alpha' = \int_{A_n'} L(\theta'; n) \text{ and } \beta' = \int_{A_n'} L(\theta''; n)$$

If $X_1, X_2, \dots, X_n$ belong to C_n' ; we have

$$\frac{L(\theta'; n)}{L(\theta''; n)} \leq K_0'$$

$$L(\theta'; n) \leq K_0' L(\theta''; n)$$

Integrate under Critical Region

$$\int_{C_n'} L(\theta'; n) \leq K_0' \int_{C_n'} L(\theta''; n)$$

$$\alpha' \leq K_0' (1 - \beta')$$

$$\frac{\alpha'}{1 - \beta'} \leq K_0'$$

Given that

$$K_0' = \frac{\alpha}{1 - \beta}$$

$$\frac{\alpha'}{1 - \beta'} \leq \frac{\alpha}{1 - \beta}$$

$$\alpha' (1 - \beta) \leq \alpha (1 - \beta') \dots \dots \dots (A)$$

If $X_1, X_2, \dots, X_n$ belong to A_n' ; we have

$$\frac{L(\theta'; n)}{L(\theta''; n)} \geq K_1'$$

$$L(\theta'; n) \geq K_1' L(\theta''; n)$$

Integrate under acceptance Region

$$\int_{A_n'} L(\theta'; n) \geq K_1' \int_{A_n'} L(\theta''; n)$$

$$1 - \alpha' \geq K_1' \beta'$$

$$\frac{1 - \alpha'}{\beta'} \geq K_1'$$

As given

$$K_1' = \frac{1 - \alpha}{\beta}$$

$$\frac{1 - \alpha'}{\beta'} \geq \frac{1 - \alpha}{\beta}$$

$$(1 - \alpha')\beta \geq (1 - \alpha)\beta' \dots \dots \dots (B)$$

Adding (A) and (B)

$$\alpha'(1 - \beta) + (1 - \alpha)\beta' \leq (1 - \alpha')\beta + \alpha(1 - \beta')$$

$$\alpha' - \alpha'\beta + \beta' - \alpha\beta' \leq \beta - \alpha'\beta + \alpha + \alpha\beta'$$

$$\alpha' + \beta' \leq \alpha + \beta$$